Diffusion-Driven and Divisor-Sensitive Integer Factorization

Exact Kernels, Certified Collision Detection, Thermodynamic Interfaces and an Industrial Evaluation Dossier

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Abstract

We develop a detailed mathematical and engineering study of integer factorization through programmed diffusion, tagged exponent sampling and exact arithmetic certification. The dyadic averaging schedule has return value $1/r + s(r-s)/(rQ^2)$, where $s = Q \bmod r$, and admits logarithmic algebraic depth with explicit precision costs. The main extension exploits collisions modulo an unknown prime factor, not only equality modulo the entire input. For a product N of two distinct odd primes, a batch of $2\lceil N^{1/4} \rceil$ four-wise-independent exponent samples with $Q \geq 128N$ returns a verified factor with probability at least $3/10$ when processed by a complete combined detector. We prove this using a local-order dichotomy and finite second-moment bounds. A derivative product/remainder-tree construction detects partial congruences over $\mathbb{Z}/N\mathbb{Z}$, including saturated-product recovery. Fixed-base windows reduce endpoint work, while a distinct-key isolation theorem improves constant-register receiver capture. A proposed stochastic arithmetic interface supplies quotient–remainder certificates with quantified checking and repair. The package includes reproducible software, logical circuit instances, resource ledgers, an Extropic evaluation plan and a preliminary invention-disclosure annex. The sampling improvement is relative to the earlier equality-only architecture; no new best factoring exponent, measured hardware advantage or patent novelty is asserted.

Executive assessment and reading guide

This revised monograph develops two related research directions: exact programmed diffusion as an order/cardinality primitive, and tagged sampling with certified arithmetic detection as a possible hardware workload. The mathematical identity for the dyadic return kernel is retained with its full proof. The main extension is a divisor-sensitive detector and a finite semiprime success theorem that no longer waits exclusively for equality modulo the entire input.

The strongest new result. For an odd semiprime $N = pq$ with distinct primes, a batch of $2\lceil N^{1/4} \rceil$ four-wise-independent dyadic exponent samples, with exact endpoint arithmetic and a complete two-branch detector, returns a proper factor with probability at least $3/10$. The proof uses a dichotomy between different local odd orders and equal local odd orders. A monic product/remainder-tree detector computes $\gcd(F'(z_i), N)$ and resolves saturated products; full equalities feed the existing relation-extraction theorem. Neither factors nor orders are inputs to the algorithm.

What this result does not assert. The quarter-power sample scale is an improvement to this manuscript's former equality-only route when its group order is large. It is not a new best exponent for integer factorization and does not outperform established classical algorithms on the evidence supplied. In the small reproducible software comparison, Pollard–Brent remains faster. A prototype must pay for exact arithmetic, polynomial operations, memory and communication as well as stochastic sampling.

Further improvements. Shared fixed-base window tables reduce repeated endpoint multiplication. A distinct-key isolation argument raises the guaranteed capture probability of a one-record-per-receiver bank to at least three quarters of the useful-collision probability when it has four times as many receivers as records. It is a capture guarantee, not a logarithmic transport-time theorem. A quantitative seed-error statement and a checked quotient–remainder proposal interface connect imperfect stochastic hardware to exact output verification.

Industrial proposal. Section 25 specifies the modules, messages, measurements and decision gates for an Extropic evaluation. Section 21 distinguishes a simple seed-source experiment from a more demanding conditional arithmetic experiment. Section 26 organizes an invention disclosure and names relevant prior art. No patent novelty, affiliation, fabricated device, measured energy advantage or guaranteed chip availability is asserted. The title-page author is preserved from the supplied source; legal inventorship and ownership are separate matters.

How to read the full treatment. The foundations establish graph diffusion, the CRT reduction and the fixed walk. The programmed-kernel part supplies the exact schedule, relation theorem and finite sampling bounds. The divisor-sensitive part gives the principal extension and its detector. The network part retains the previous detailed routing theory with its stated conditions. The physical part treats RC limitations, stochastic arithmetic, experiments and the evaluation dossier. Readers assessing the new contribution should begin with Theorem 14.4, Proposition 15.1 and the measured comparison.

Evidence convention. A *theorem* is proved under its written assumptions. A *reference result* is a finite software computation with saved inputs and outputs. A *design specification* is an unmeasured proposed module. A *hardware result* would require fabrication or execution on an identified device; there are no such new results here. Historical projections based on unspecified

billion-cell devices and nanosecond trials are excluded from the active case and retained only in the source archive.

Notation and resource conventions

Symbol	Meaning
N, n	Input integer; $n = \lceil \log_2 N \rceil$ bit-size parameter in the new analysis
a, b	Original unit base; oddified base $b = a^{2^n}$ for odd semiprimes
M	$\lfloor \log_2 N \rfloor + 1$ in the original fixed-walk sections; equals n for the odd inputs used in factoring
r	Multiplicative order of the base under discussion; later the order of b
ℓ, L	Fixed-walk iteration count; programmed-layer count
$Q = 2^K$	Exponent interval length; the kernel section also writes $Q = 2^L$ for its own schedule
T	Number of exponent samples or tagged records, with each section declaring its use
m	Number of prime-power factors in CRT sections; half-batch size in the explicitly restricted semiprime theorem
C, H, B	Legacy sample budget, receiver count or order bound as declared locally; never an automatic transistor count
C_0	Electrical capacitance in the RC model
$\mathsf{M}_R(s)$	Cost of degree-s polynomial multiplication over $R = \mathbb{Z}/N\mathbb{Z}$

A modular product operates on n-bit residues; a finite-field product acts on K-bit field representations. A compare–exchange moves whole records. None of these is a unit-cost physical bit operation. A claimed logarithmic algebraic depth does not set the device's clock period, memory size, wiring area or sampling independence.

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Part I

Foundations: diffusion and the arithmetic reduction

1 Introduction

1.1 Motivation and scope

Factoring a composite integer N is a cornerstone problem in computational number theory and cryptography, intimately tied to the structure of $(\mathbb{Z}/N\mathbb{Z})^*$; see, e.g., [HW08, CP05, Me96, BS96]. Classical methods are highly nontrivial and often subexponential, but no polynomial-time algorithm in $\log_2 N$ is known in the standard digital model [CP05].

Shor's algorithm shows that, in the quantum circuit model, factoring reduces to *order finding* and order finding can be solved in polynomial time via quantum period finding/Fourier sampling [Sh94, Sh97, NC10, EJ96]. For recent advances and generalizations see [CBK22, Hh25, Ra24, Re25, XQLM23] and references therein. In broad terms: one encodes modular exponentiation into a coherent linear evolution, extracts periodicity through interference, and completes the computation by classical post-processing (continued fractions and verification) [Sh94, Sh97, KSV99, NC10].

This paper investigates a different computational primitive: *diffusion on a finite graph* as an analog mechanism for order finding. Specifically, we analyze an iterated half-lazy random walk (discrete-time heat flow) on a weighted Cayley graph attached to a cyclic subgroup of $(\mathbb{Z}/N\mathbb{Z})^*$. Our objective is not to improve on Shor's algorithm within the usual digital/RAM or quantum-circuit models. Rather, we propose a *hybrid* model in which standard digital arithmetic is augmented by a diffusion process that can be iterated and queried. In this accounting, order recovery is achieved using $\mathrm{poly}(\log_2 N)$ digital work together with $O((\log_2 N)^2)$ diffusion iterations, under explicit interface assumptions.

1.2 Quantum layers and diffusion iterations

For clarity of resource accounting, we compare both non-digital paradigms schematically as alternating applications of a linear operator and (i) an observation/projection in the quantum case, and (ii) a classical readout followed by conditioning (post-processing) in the diffusion case:

$$\text{quantum:} \quad \cdots \xrightarrow{U} \xrightarrow{\text{measure/proj}} \xrightarrow{U} \xrightarrow{\text{measure/proj}} \cdots,$$

$$\text{diffusion:} \quad \cdots \xrightarrow{W} \xrightarrow{\text{read/cond}} \xrightarrow{W} \xrightarrow{\text{read/cond}} \cdots.$$

This is bookkeeping only: U is unitary (interference-capable), while W is Markovian and dissipative [NC10, LPW09]. The point is to make explicit where work is assumed to occur in the hybrid model: theorems bound the number of invocations of the diffusion primitive by $O((\log_2 N)^2)$, while the remaining processing is standard digital arithmetic.

1.3 Physical diffusion primitive, resource scaling and locality

Our complexity bounds track two resources separately: digital steps and diffusion steps.

Digital steps. Deterministic operations on $O(\log_2 N)$-bit integers (modular arithmetic, gcd, verification, bookkeeping), all polynomial in $\log_2 N$ in the standard arithmetic/RAM sense [Kn97, BS96].

Diffusion steps. One application of a fixed local-averaging operator, the half-lazy walk operator W on a graph X, together with a prescribed readout of a small number of "temperature" coordinates of a ℓ^1-normalized state vector.

The intended interpretation of a diffusion step is hardware-based. A potential depth advantage has physical significance only if diffusion is treated as a genuine *hardware primitive*: a single *hardware-level update* that performs local averaging simultaneously at every vertex, whose wall-clock cost (the actual elapsed real time as measured by a clock) does not scale with the number of vertices, and whose required readout precision is achievable with $\mathrm{poly}(\log_2 N)$ overhead. In this model, the wall-clock cost of one diffusion step is treated as independent of the number of vertices represented, because *all vertices are updated in parallel*. If diffusion is digitally simulated by explicitly updating a length-r state vector (with $r = \mathrm{ord}_N(b)$ possibly as large as $\varphi(N)$, the Euler totient function), then the advantage disappears and the procedure becomes exponential in $\log_2 N$.

This assumption shifts cost from time to physical resources. Any literal diffuser must represent a state over a vertex set of size $\Omega(|X|)$, so device area/energy scales (at least) linearly with $|X|$. Our results are therefore conceptual and model-based: they show that, given such a diffusion primitive with sufficient readout precision, the order r is determined from a single heat-kernel value after $O((\log_2 N)^2)$ diffusion iterations.

A further operational point is *locality*. Computing the labels of neighbors is an arithmetic-local operation, not evidence of geometrically local wires. Evolving a small explored subgraph generally does not preserve the full heat kernel. Section 20 explains the all-residue construction and the cost of representing the unknown orbit without supplying its order.

1.4 Main results

Fix an odd composite integer

$$N = \prod_{i=1}^{m} p_i^{e_i} \qquad (m \geq 2,\ p_i \text{ distinct odd primes}). \tag{1.1}$$

Let $b \in (\mathbb{Z}/N\mathbb{Z})^*$ have *odd* order $r = \mathrm{ord}_N(b)$. Without knowing r in advance, we consider a weighted Cayley graph on $\langle b \rangle$ generated by the moves $b^{\pm 2^t}$ for $0 \leq t \leq M = \lfloor \log_2 N \rfloor + 1$. We analyze the associated half-lazy walk operator W and its discrete-time heat kernel. We prove that for

$$\ell_0 = 4(M+1)\big(\log_2 N + 2\big) = O((\log_2 N)^2),$$

the single heat-kernel value at the identity, $p_{\ell_0}(e)$, determines r uniquely by rounding $1/p_{\ell_0}(e)$.

There exist deterministic algorithms which ascertain if N is either a prime or the power of a prime; see, for example, [AKS04], [Be07], or [Ra80]. In these two problems, meaning the determination of N is prime or a prime power, the best known algorithms have (classical) complexity of order $O((\log N)^a)$ for some constant a. With this, we do not view the assumption that N is neither a prime nor a prime power as being restrictive, at least from the point of view of theoretical computability.

1.5 Organization

The paper is organized as follows. Section 2 develops the diffusion/heat-kernel framework on weighted graphs and Cayley graphs. Section 3 collects the number-theoretic preliminaries used in the reduction from factoring to order finding. Section 4 proves the main diffusion order-finding theorem. Section 5 gives pseudocode for the diffusion-assisted factoring algorithm and for numerical simulation of the diffusion primitive (e.g. in Python). Section 6 presents an

accelerated, collision-based relation-finding mechanism based on detecting closed cycles and using gcd stabilization of the resulting loop lengths. Section 7 analyzes collision statistics for the walk on G, including scale estimates. Finally, Section 8 provides illustrative numerical examples.

2 Diffusion preliminaries

2.1 Weighted graphs

For the preliminary simple-graph discussion, graphs are finite, undirected and connected, without self-loops. Later Cayley operators are defined directly by signed shift occurrences; coincident shifts and identity occurrences are retained. A weighted graph $X = (V, E, w)$ consists of a finite vertex set V, an edge set E of 2-element subsets of V, and a weight function $w\colon V \times V \to \mathbb{R}$ with the following properties.

- **Symmetry:** $w(x, y) = w(y, x)$ for all $x, y \in V$.
- **Non-negativity:** $w(x, y) \geq 0$ for all $x, y \in V$.
- **Support on edges:** $w(x, y) > 0$ iff $\{x, y\} \in E$.

Fix an ordering of V with $|V| = k$. The adjacency operator A_X is the operator on the space of functions f on V such that $(A_X f)(x) = \sum_{y \in V} w(x, y)\, f(y)$.
The degree of a vertex $x \in V$ is

$$d(x) = \sum_{y \in V} w(x, y).$$

We call X *regular of degree* d if $d(x) = d$ for all $x \in V$.

2.2 Half-lazy random walks and the discrete-time heat kernel

Assume X is d-regular. The *half-lazy walk matrix* is

$$W = \frac{1}{2}\left(I + \frac{1}{d}A_X\right). \tag{2.1}$$

Starting from an initial probability distribution $p_0 \in \mathbb{R}^k$ ($p_0 \geq 0$, $\|p_0\|_1 = 1$), define

$$p_\ell = W^\ell p_0.$$

Write $w_\ell(x, y)$ for the (x, y)-entry of W^ℓ. This is the probability that a walk started at y is at x after ℓ steps. We refer to p_ℓ as the *discrete-time heat kernel*.
Namely, the discrete-time derivative $\partial_\ell p_\ell = p_{\ell+1} - p_\ell$ and the discrete (probabilistic) Laplacian $\Delta = W - I$ satisfy

$$\partial_\ell p_\ell = \Delta p_\ell, \tag{2.2}$$

the discrete heat equation on X. An explicit expression for the discrete time heat kernel p_ℓ on a regular graph is derived in [CHJSV23].

2.3 Spectral expansion and a uniform bound

Since the graph is d-regular, the random-walk matrix is $P = \frac{1}{d}A_X$. Then P is real symmetric and

$$\operatorname{spec}(P) \subset [-1, 1] \qquad \text{(see e.g. [Ni18, Thm. 7.5])}.$$

Recall that $W = \frac{1}{2}(I + P)$. If $\theta \in \operatorname{spec}(P)$, then $\frac{1}{2}(1+\theta) \in \operatorname{spec}(W)$. Hence

$$\operatorname{spec}(W) = \left\{ \tfrac{1}{2}(1+\theta) : \theta \in \operatorname{spec}(P) \right\} \subset [0,1].$$

In particular, W is symmetric and all its eigenvalues are real and nonnegative. Let

$$1 = \lambda_0 > \lambda_1 \geq \cdots \geq \lambda_{k-1} \geq 0 \tag{2.3}$$

be the eigenvalues of W, with an orthonormal eigenbasis $\{\psi_0, \psi_1, \ldots, \psi_{k-1}\}$ satisfying $W\psi_j = \lambda_j \psi_j$.

Moreover, $P\mathbf{1} = \mathbf{1}$ (and hence $W\mathbf{1} = \mathbf{1}$), so we may take $\psi_0 = \frac{1}{\sqrt{k}}\mathbf{1}$. Therefore, for any initial distribution p_0 we have the spectral expansion

$$p_\ell = W^\ell p_0 = \sum_{j=0}^{k-1} \langle \psi_j, p_0 \rangle \, \lambda_j^\ell \, \psi_j,$$

where $\langle \psi, \phi \rangle$ denotes the usual inner product of vectors $\psi, \phi \in \mathbb{R}^k$.

Proposition 2.1. *Let λ_1 be the largest eigenvalue of W strictly less than 1. If p_0 is a probability distribution, then for every vertex $x \in V$ and every $\ell \geq 0$,*

$$\left| p_\ell(x) - \frac{1}{k} \right| \leq \lambda_1^\ell.$$

Proof. Let $\psi_0 = \frac{1}{\sqrt{k}}\mathbf{1}$ and extend this vector to an orthonormal eigenbasis $\{\psi_0, \psi_1, \ldots, \psi_{k-1}\}$ of W: $W\psi_j = \lambda_j \psi_j$, with $1 = \lambda_0 > \lambda_1 \geq \lambda_2 \geq \cdots$. Since p_0 is a probability distribution,

$$\langle \psi_0, p_0 \rangle = \frac{1}{\sqrt{k}} \sum_{x \in V} p_0(x) = \frac{1}{\sqrt{k}}.$$

Therefore

$$p_\ell = W^\ell p_0 = \sum_{j=0}^{k-1} \langle \psi_j, p_0 \rangle \, \lambda_j^\ell \, \psi_j = \frac{1}{k}\mathbf{1} + \sum_{j=1}^{k-1} \langle \psi_j, p_0 \rangle \, \lambda_j^\ell \, \psi_j.$$

Set $u_\ell = \sum_{j=1}^{k-1} \langle \psi_j, p_0 \rangle \, \lambda_j^\ell \, \psi_j$, so $p_\ell = \frac{1}{k}\mathbf{1} + u_\ell$ and u_ℓ is orthogonal to $\mathbf{1}$.

Fix $x \in V$ and set $a_j = \langle \psi_j, p_0 \rangle$ and $b_j = \lambda_j^\ell \psi_j(x)$ for $1 \leq j \leq k-1$. Then $u_\ell(x) = \sum_{j=1}^{k-1} a_j b_j$, and by Cauchy–Schwarz,

$$|u_\ell(x)| \leq \Big(\sum_{j=1}^{k-1} |a_j|^2 \Big)^{1/2} \Big(\sum_{j=1}^{k-1} |b_j|^2 \Big)^{1/2} = \Big(\sum_{j=1}^{k-1} |\langle \psi_j, p_0 \rangle|^2 \Big)^{1/2} \Big(\sum_{j=1}^{k-1} \lambda_j^{2\ell} \psi_j(x)^2 \Big)^{1/2}.$$

By Parseval's identity,

$$\|p_0\|_2^2 = \sum_{j=0}^{k-1} |\langle \psi_j, p_0 \rangle|^2, \qquad \text{hence} \qquad \sum_{j=1}^{k-1} |\langle \psi_j, p_0 \rangle|^2 \leq \|p_0\|_2^2.$$

Since $\lambda_j \in [0, \lambda_1]$ for $j \geq 1$,

$$\sum_{j=1}^{k-1} \lambda_j^{2\ell} \psi_j(x)^2 \leq \lambda_1^{2\ell} \sum_{j=1}^{k-1} \psi_j(x)^2 \leq \lambda_1^{2\ell} \sum_{j=0}^{k-1} \psi_j(x)^2.$$

Finally, using $\langle \psi_j, \delta_x \rangle = \psi_j(x)$ and Parseval's identity for δ_x, we see that

$$\sum_{j=0}^{k-1} \psi_j(x)^2 = \sum_{j=0}^{k-1} |\langle \psi_j, \delta_x \rangle|^2 = \|\delta_x\|_2^2 = 1.$$

Combining those bounds yields $|u_\ell(x)| \leq \lambda_1^\ell \|p_0\|_2$. Since $p_0 \geq 0$ and $\|p_0\|_1 = 1$, we have $\|p_0\|_2 \leq \|p_0\|_1 = 1$, hence $|p_\ell(x) - \frac{1}{k}| = |u_\ell(x)| \leq \lambda_1^\ell$. □

2.4 Weighted Cayley graphs of finite abelian groups

Let G be a finite abelian group written additively, $S \subseteq G$ a symmetric generating set (so $s \in S \Rightarrow -s \in S$), and $\alpha\colon S \to \mathbb{R}_{>0}$ with $\alpha(s) = \alpha(-s)$. The weighted Cayley graph $X = \mathrm{Cay}(G, S, \alpha)$ has vertex set G and edge weights

$$w(x, y) = \alpha(x - y) \quad \text{if } x - y \in S, \qquad w(x, y) = 0 \text{ otherwise.}$$

It is regular of degree

$$d = \sum_{s \in S} \alpha(s).$$

Characters diagonalize the adjacency operator A_X; see [CR62] for background and [Ba79, Cor. 3.2] for the Cayley-graph spectral formula. If χ is a character of G, then it is an eigenvector of A_X with eigenvalue

$$\eta(\chi) = \sum_{s \in S} \alpha(s)\chi(s),$$

hence an eigenvector of W with eigenvalue $\lambda(\chi) = \frac{1}{2}\,(1 + \eta(\chi)/d)$.

3 Number-theoretic preliminaries

This section records the elementary number theory used in the factoring reduction. All statements below are standard; we include short proofs in the form and level of generality needed later. General references include [HW08, Chs. 1–3], [CP05, Chs. 2–4], and [BS96, Chs. 2–4]; see also [Me96, §2–5] for standard cryptographic formulations of the order-finding to factoring reduction.

Let N be a positive odd integer which we write as a product

$$N = \prod_{i=1}^{m} p_i^{e_i}, \tag{3.1}$$

where $m \geq 2$, the primes $p_1, \ldots, p_m$ are distinct and odd, and the exponents satisfy $e_i > 0$. In particular, N is neither prime nor a prime power.

Let $\mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z}$ denote the residue ring modulo N, and let $\mathbb{Z}_N^*$ denote its group of units. There is a natural mapping

$$\mathbb{Z}_N \longrightarrow \mathbb{Z}_{p_1^{e_1}} \times \cdots \times \mathbb{Z}_{p_m^{e_m}}, \qquad a \longmapsto (a \bmod p_1^{e_1}, \ldots, a \bmod p_m^{e_m}). \tag{3.2}$$

By the Chinese Remainder Theorem (CRT), (3.2) is an isomorphism of rings. Restricting to units yields an isomorphism of groups

$$g_N : \mathbb{Z}_N^* \xrightarrow{\sim} \mathbb{Z}_{p_1^{e_1}}^* \times \cdots \times \mathbb{Z}_{p_m^{e_m}}^*. \tag{3.3}$$

In a slight abuse of notation, we occasionally use x to denote either an element of $\mathbb{Z}_N^*$ or its image $g_N(x)$.

For each i, the group $\mathbb{Z}_{p_i^{e_i}}^*$ is cyclic under multiplication. Fix a generator u_i. Its order is

$$\mathrm{ord}_{p_i^{e_i}}(u_i) = \varphi(p_i^{e_i}) = p_i^{e_i - 1}(p_i - 1) = 2^{c_i} p_i',$$

where $c_i > 0$ and p_i' is odd.

Without loss of generality we assume that the primes are ordered so that

$$c_1 \geq c_2 \geq \cdots \geq c_m. \tag{3.4}$$

The following proposition shows that, assuming a parity condition on the exponents in the CRT decomposition, one can produce a nontrivial square root of 1 modulo N.

Proposition 3.1. *Let $a \in \mathbb{Z}_N^*$. Let $s \geq 0$ be the* least *integer such that*

$$a^{2^s q} \equiv 1 \pmod{N}$$

for some odd integer q. Write

$$g_N(a) = (u_1^{d_1}, \ldots, u_m^{d_m}).$$

If there exist indices $i < j$ such that d_i is odd and d_j is even, then $s > 0$. Moreover, setting

$$x = a^{2^{s-1} q},$$

one has that

$$x^2 \equiv 1 \pmod{N} \qquad \textit{and} \qquad x \not\equiv \pm 1 \pmod{N}. \tag{3.5}$$

Proof. Assume d_i is odd and d_j is even.
Since $a^{2^s q} \equiv 1 \pmod{N}$, it follows that

$$u_i^{2^s q d_i} \equiv 1 \pmod{p_i^{e_i}}.$$

Thus the order $\mathrm{ord}_{p_i^{e_i}}(u_i) = 2^{c_i} p_i'$ divides $2^s q d_i$. Because q and d_i are odd, the 2-part forces $2^{c_i} \mid 2^s$, hence $s \geq c_i \geq 1$.
Set

$$x = a^{2^{s-1} q} = (u_1^{2^{s-1} q d_1}, \ldots, u_m^{2^{s-1} q d_m}).$$

Then

$$x^2 = a^{2^s q} \equiv 1 \pmod{N}.$$

By minimality of s, we have $x \not\equiv 1 \pmod{N}$.
It remains to show $x \not\equiv -1 \pmod{N}$. If $d_j = 0$, then $x \equiv 1 \pmod{p_j^{e_j}}$, so this follows because p_j is odd. Otherwise write $d_j = 2^v d_j'$ with $v > 0$ and d_j' odd. Now

$$x \equiv u_j^{2^{s-1} q d_j} = u_j^{2^{s-1} q \cdot 2^v d_j'} = u_j^{2^{s+v-1} q d_j'} \pmod{p_j^{e_j}}.$$

We claim that x is 1 modulo $p_j^{e_j}$. Indeed, since $a^{2^s q} \equiv 1 \pmod{p_j^{e_j}}$ we have

$$u_j^{2^s q d_j} = u_j^{2^{s+v} q d_j'} \equiv 1 \pmod{p_j^{e_j}}.$$

Thus the order $2^{c_j} p_j'$ divides $2^{s+v} q d_j'$. As $q d_j'$ is odd, this implies

$$s + v \geq c_j \quad \text{and} \quad p_j' \mid q d_j'.$$

Since $s \geq c_i \geq c_j$ (by (3.4)) and $v > 0$, we have $s + v - 1 \geq c_j$, hence $2^{c_j} p_j'$ divides $2^{s+v-1} q d_j'$, which is exactly the condition that

$$u_j^{2^{s+v-1} q d_j'} \equiv 1 \pmod{p_j^{e_j}}.$$

Therefore $x \equiv 1 \pmod{p_j^{e_j}}$.
But this implies that $x \not\equiv -1 \pmod{N}$, for otherwise $x \equiv -1 \pmod{p_j^{e_j}}$ also holds. Subtracting the two congruences would give $2 \equiv 0 \pmod{p_j^{e_j}}$, a contradiction because p_j is odd.

□

Lemma 3.2 (Early repeated signed powers)**.** *Write the generator occurrences as pairs* (ϵ, t)*, with* $\epsilon \in \{-1, 1\}$ *and* $0 \le t \le M$*. If two distinct occurrences have equal residues* $a^{\epsilon 2^t} = a^{\epsilon' 2^{t'}}$*, their nonzero integer difference* $D = \epsilon 2^t - \epsilon' 2^{t'}$ *satisfies* $a^D = 1$*. Remove the power of two from* $|D|$ *to obtain an odd* q *and use repeated squaring to find the least* s *with* $a^{2^s q} = 1$*. The two-primary parity pattern of Proposition 3.1 is a sufficient condition for successful extraction whenever this relation has been supplied. For a uniform unit, that sufficient pattern has probability* $p(m) = 1 - (m+1)/2^m$*.*

Proof. Distinct signed powers of two are distinct integers, including the case $t = t'$ with opposite signs. Equality modulo N gives the relation. Writing $|D| = 2^v q$ with q odd, at most v squarings of a^q find the first one. The bit length of D is $O(\log N)$; all modular costs are charged at their actual word width. For the probability assertion, CRT makes the local exponent parities independent fair bits. The absence of an odd parity followed by an even one describes exactly the $m+1$ nondecreasing binary vectors. This counts a sufficient event; its complement is not the full failure event of factor extraction. □

Remark 3.3. For $m \ge 2$, the sufficient parity event has probability at least $1/4$. If a valid relation is supplied for every fresh uniform-unit trial, this gives failure at most $(3/4)^k$ after k trials. Conditioning instead on an early repeated-power collision does not preserve a uniform base, so that conditional claim is not made. The exact success law, proved in Theorem 11.2, is stronger: at least $1/2$ for semiprimes and at least $1 - 2^{1-m}$ in general.

Lemma 3.4. *Let* G *be a finite cyclic group of even order* $\ell = 2^c m$ *with* $c > 0$ *and* m *odd, and let* u *be a generator. Then for* $1 \le d \le \ell$ *one has*

$$\operatorname{ord}_G(u^d) = \frac{\ell}{\gcd(\ell, d)}.$$

Proof. This is standard since u^d generates the subgroup of index $\gcd(\ell, d)$. □

Lemma 3.5. *For any* $a \in \mathbb{Z}_N^*$*, the exponent* $v_2(\operatorname{ord}_N(a))$ *is strictly less than* $\log_2 N$*. In particular, if* $M = \lfloor \log_2 N \rfloor + 1$ *and* $b = a^{2^M}$*, then* $\operatorname{ord}_N(b)$ *is odd.*

Proof. Write $g_N(a) = (a_1, \ldots, a_m)$ with $a_i \in \mathbb{Z}^*_{p_i^{e_i}}$. Then

$$\operatorname{ord}_N(a) = \operatorname{lcm}\big(\operatorname{ord}_{p_1^{e_1}}(a_1), \ldots, \operatorname{ord}_{p_m^{e_m}}(a_m)\big).$$

Each $\operatorname{ord}_{p_i^{e_i}}(a_i)$ divides $\varphi(p_i^{e_i}) = 2^{c_i} p_i'$, hence the 2-adic valuation of $\operatorname{ord}_N(a)$ is at most $\max_i c_i$. Finally, $2^{c_i} \mid \varphi(p_i^{e_i}) < p_i^{e_i} \le N$, so $c_i < \log_2 N$ for all i, hence $\max_i c_i < \log_2 N$. Taking $M = \lfloor \log_2 N \rfloor + 1$ makes 2^M divisible by the entire 2-part of $\operatorname{ord}_N(a)$, so $b = a^{2^M}$ has odd order. □

Proposition 3.6. *Set* $M = \lfloor \log_2 N \rfloor + 1$*. For any* $a \in \mathbb{Z}_N^*$*, let* $b = a^{2^M}$ *and write*

$$r_b = \operatorname{ord}_N(b),$$

which is odd by Lemma 3.5. If r_b *is known, then* $\operatorname{ord}_N(a)$ *can be computed in at most* $O(\log_2 N)$ *deterministic steps.*

Moreover, if a *is chosen uniformly at random from* $\mathbb{Z}_N^*$*, then with probability at least* $p(m) = 1 - (m+1)/2^m$ *the order* $r_a = \operatorname{ord}_N(a)$ *is even and the element* $x = a^{r_a/2}$ *satisfies* (3.5)*.*

Proof. Since $b = a^{2^M}$ and r_b is odd, the order of a has the form $r_a = 2^k r_b$ for some $0 \le k \le M$. One finds the least such k by testing $a^{2^k r_b} \equiv 1 \pmod N$ for $k = 0, 1, \dots, M$, which costs $O(\log_2 N)$ deterministic steps (repeated squaring / modular exponentiation).

For the probability statement, apply Lemma 3.2 and Proposition 3.1. □

4 Diffusion order finding

This section contains the analytic core: an order–finding primitive based on iterating a *half–lazy walk operator* on the Cayley graph of the cyclic subgroup generated by b. The transition has a succinct arithmetic description; this does not by itself avoid the cost of representing the full state or implementing its connections. A single application of the walk operator at a point $x \in \langle b \rangle$ only requires evaluating the neighbors

$$x \longmapsto x\, b^{\pm 2^t} \pmod N \qquad (0 \le t \le M),$$

which can be executed by modular multiplication (and modular inversion).

The underlying Cayley graph and the random walk matrix. Let

$$M = \lfloor \log_2 N \rfloor + 1, \qquad G = \langle b \rangle \subseteq (\mathbb{Z}/N\mathbb{Z})^*.$$

Consider the (unweighted) Cayley graph $X = X_{N,b} = \mathrm{Cay}(G, S)$ with generating multiset

$$S = \{\, b^{\pm 2^t} : 0 \le t \le M \,\}.$$

Equivalently, X is the weighted graph with vertex set G and weights

$$w(x, y) = \sum_{t=0}^{M} (\mathbf{1}_{\{y = x b^{2^t}\}} + \mathbf{1}_{\{y = x b^{-2^t}\}}).$$

This counts signed occurrences separately, including loops; the corresponding transition matrix has constant row sum $d = 2(M + 1)$, with adjacency matrix A_X given by

$$(A_X p)(x) = \sum_{y \in G} w(x, y)\, p(y).$$

As in Section 2.2, the (simple) random-walk matrix on X is

$$P = \frac{1}{d} A_X.$$

In particular, for functions $p \colon G \to \mathbb{R}$ we have the concrete formula

$$(Pp)(x) = \frac{1}{2(M+1)} \sum_{t=0}^{M} \Big(p(x\, b^{2^t}) + p(x\, b^{-2^t}) \Big), \qquad x \in G. \tag{4.1}$$

Fix an ordering $G = \{x_1, \dots, x_{|G|}\}$ and identify $p : G \to \mathbb{R}$ with the column vector

$$(p(x_1), \dots, p(x_{|G|}))^{\mathsf{T}} \in \mathbb{R}^{|G|}.$$

As before, define the associated half-lazy walk operator

$$W = W_{N,b} = \frac{1}{2}(I + P).$$

4.1 The diffusion theorem

Theorem 4.1 (Diffusion order finding)**.** *Let $N > 1$ be an integer. Let $b \in (\mathbb{Z}/N\mathbb{Z})^*$ have multiplicative order $r = \mathrm{ord}_N(b)$. Set*

$$M = \lfloor \log_2 N \rfloor + 1, \qquad G = \langle b \rangle \subseteq (\mathbb{Z}/N\mathbb{Z})^*, \qquad |G| = r.$$

Let $\{p_\ell\}_{\ell \geq 0}$ be a half-lazy walk on G with transition matrix W and the initial state being the delta function at the identity, meaning

$$p_0 = \delta_e, \qquad p_\ell = W^\ell p_0, \qquad (\ell \geq 0).$$

Then, for every

$$\ell \;\geq\; 4(M+1)\big(\log_2 N + 2\big), \tag{4.2}$$

the single readout value $p_\ell(e)$ determines r uniquely as the unique integer in $(0, N]$ whose reciprocal lies within $1/(4N^2)$ of $p_\ell(e)$.

Theorem 4.1 will be proved in subsection 4.2.

Corollary 4.2. *Under the hypotheses of Theorem 4.1, if $|p_\ell(e) - 1/r| \leq 1/(4N^2)$ then*

$$\mathrm{round}\big(1/p_\ell(e)\big) = r.$$

Proof. Since $r \leq N$, we have

$$p_\ell(e) \geq \frac{1}{r} - \frac{1}{4N^2} \geq \frac{1}{N} - \frac{1}{4N^2} = \frac{4N-1}{4N^2} > \frac{3}{4N}.$$

Therefore

$$\left|\frac{1}{p_\ell(e)} - r\right| = \left|\frac{r(1/r - p_\ell(e))}{p_\ell(e)}\right| \leq \frac{N \cdot (1/(4N^2))}{3/(4N)} = \frac{1}{3} < \frac{1}{2},$$

which forces rounding to equal r. □

Remark 4.3. Theorem 4.1 is stated in full generality. It only concerns diffusion on the cyclic subgroup $G = \langle b \rangle$ of order $r = \mathrm{ord}_N(b)$, and its proof does not use any special arithmetic hypotheses on N beyond the explicit inequalities stated. In the remainder of this paper, however, we will invoke Theorem 4.1 only in the factoring setting, where N is assumed to be an odd integer that is neither prime nor a prime power, and where we choose b so that $r = \mathrm{ord}_N(b)$ is odd.

Next, we will state and prove a *doubling lemma* we will need in the proof of Theorem 4.1 in the next section. Conceptually, this is a lacunary-series phenomenon: the doubling sequence 2^t spreads residues quickly enough that one must encounter an interval where the phase has nonpositive cosine value, reminiscent of the mechanisms exploited in Korobov-type bounds for lacunary exponential sums, see e.g. [KM12] or [Va19].

Lemma 4.4 (Doubling Lemma)**.** *Let r be an integer and let $M \geq 0$ be such that $2^M > r$. For every integer k with $1 \leq k \leq r-1$, there exists $t \in \{0, 1, \ldots, M\}$ such that*

$$k2^t \bmod r \in \Big[\frac{r}{4}, \frac{3r}{4}\Big].$$

Proof. For each $t \geq 0$, let $a_t \in \{0, 1, \ldots, r-1\}$ be the least residue of $k2^t$ modulo r. Assume for contradiction that for all $t \in \{0, \ldots, M\}$,

$$a_t \in \Big[0, \frac{r}{4}\Big) \;\cup\; \Big(\frac{3r}{4}, r\Big).$$

Set

$$b_t = \min(a_t,\ r - a_t).$$

Initially $b_0 > 0$. The doubling identity proved next propagates positivity as long as the middle interval is avoided, so $0 < b_t < r/4$ for the indices in question, also when r is even.

We claim that $b_{t+1} = 2b_t$ for $t = 0, \dots, M-1$. If $a_t < r/4$, then $2a_t < r/2 < r$, so $a_{t+1} = 2a_t$ and hence $b_{t+1} = 2b_t$. If $a_t > 3r/4$, then $2a_t \in (3r/2, 2r)$, so $a_{t+1} = 2a_t - r \in (r/2, r)$ and therefore

$$b_{t+1} = r - a_{t+1} = r - (2a_t - r) = 2(r - a_t) = 2b_t.$$

Thus $b_t = 2^t b_0$ for all $t \le M$. Since $b_0 = \min(k, r-k) \ge 1$, we have $b_M \ge 2^M$.

By hypothesis $2^M > r$, hence $b_M \ge 2^M > r/4$, contradicting $b_M < r/4$. This contradiction proves the lemma. □

Recall that for $k \in \{0, 1, \dots, r-1\}$, if

$$\chi_k(j) = \exp\left(\frac{2\pi i k j}{r}\right), \qquad j \in \mathbb{Z}/r\mathbb{Z},$$

then

$$\frac{1}{r}\sum_{k=0}^{r-1} \chi_k(j) = \begin{cases} 1, & j \equiv 0 \pmod r, \\ 0, & j \not\equiv 0 \pmod r. \end{cases}$$

4.2 Proof of Theorem 4.1

Proof. If $r = 1$, the return value is identically one and the conclusion is immediate. Assume $r > 1$. We change to additive notation. The subgroup $G = \langle b \rangle$ is cyclic of order r. Hence,

$$\phi : \mathbb{Z}/r\mathbb{Z} \longrightarrow G, \qquad \phi(j) = b^j$$

is an isomorphism. Using ϕ we transport functions $p : G \to \mathbb{R}$ to functions $\widetilde{p} : \mathbb{Z}/r\mathbb{Z} \to \mathbb{R}$ by $\widetilde{p}(j) = p(b^j)$. In particular $\widetilde{p}(0) = p(e)$. Under this identification, right multiplication by $b^{\pm 2^t}$ becomes translation by $\pm 2^t$ modulo r. Hence the random walk operator P becomes the translation-invariant walk

$$(P\widetilde{p})(j) = \frac{1}{2(M+1)} \sum_{t=0}^{M} \Big(\widetilde{p}(j + 2^t) + \widetilde{p}(j - 2^t)\Big), \qquad j \in \mathbb{Z}/r\mathbb{Z}.$$

Now the characters are eigenfunctions of P. Since $\chi_k(j \pm a) = \chi_k(j)\chi_k(\pm a)$ one has

$$(P\chi_k)(j) = \chi_k(j) \cdot \frac{1}{2(M+1)} \sum_{t=0}^{M} \left(e^{2\pi i k 2^t/r} + e^{-2\pi i k 2^t/r}\right) = \chi_k(j) \cdot \frac{1}{M+1} \sum_{t=0}^{M} \cos\left(\frac{2\pi k 2^t}{r}\right).$$

Thus $P\chi_k = \mu_k \chi_k$ with

$$\mu_k = \frac{1}{M+1} \sum_{t=0}^{M} \cos\left(\frac{2\pi k 2^t}{r}\right),$$

and therefore $W\chi_k = \lambda_k \chi_k$ where $\lambda_k = (1 + \mu_k)/2$. As noted before (2.3), $\lambda_0 = 1$ and $0 \le \lambda_k \le 1$ for all k.

Let δ_0 be the delta function at $0 \in \mathbb{Z}/r\mathbb{Z}$. By orthogonality,

$$\delta_0(j) = \frac{1}{r} \sum_{k=0}^{r-1} \chi_k(j).$$

Applying W^ℓ and evaluating at 0 gives

$$(W^\ell \delta_0)(0) = \frac{1}{r}\sum_{k=0}^{r-1} \lambda_k^\ell = \frac{1}{r} + \frac{1}{r}\sum_{k=1}^{r-1} \lambda_k^\ell.$$

Transporting back to G, the left-hand side is exactly $p_\ell(e)$, hence

$$\left| p_\ell(e) - \frac{1}{r} \right| \le \max_{1\le k\le r-1} \lambda_k^\ell. \tag{4.3}$$

Fix $k \in \{1, \dots, r-1\}$. By Lemma 4.4 there exists $t \in \{0, \dots, M\}$ such that

$$k2^t \bmod r \in \left[\frac{r}{4}, \frac{3r}{4}\right], \qquad \text{hence} \qquad \cos\left(\frac{2\pi k 2^t}{r}\right) \le 0.$$

All other cosine terms are ≤ 1, so

$$\mu_k \le \frac{M}{M+1} = 1 - \frac{1}{M+1}, \qquad \text{and therefore} \qquad \lambda_k = \frac{1+\mu_k}{2} \le 1 - \frac{1}{2(M+1)}.$$

Combining with (4.3) and using $1 - x \le \mathrm{e}^{-x}$,

$$\left| p_\ell(e) - \frac{1}{r} \right| \le \left(1 - \frac{1}{2(M+1)}\right)^\ell \le \exp\left(-\frac{\ell}{2(M+1)}\right).$$

If $\ell \ge 4(M+1)(\log_2 N + 2)$ then

$$\exp\left(-\frac{\ell}{2(M+1)}\right) \le \exp\bigl(-2(\log_2 N + 2)\bigr) = \mathrm{e}^{-4}\,\mathrm{e}^{-2\log_2 N}.$$

Since $\log_2 N = \ln N / \ln 2 \ge \ln N$, we have $\mathrm{e}^{-2\log_2 N} \le \mathrm{e}^{-2\ln N} = N^{-2}$, so the right-hand side is $\le \mathrm{e}^{-4} N^{-2} < \frac{1}{4N^2}$. Hence

$$\left| p_\ell(e) - \frac{1}{r} \right| \le \frac{1}{4N^2}.$$

Uniqueness of r. Since $r = \mathrm{ord}_N(b)$ divides $|\mathbb{Z}_N^*| = \varphi(N)$ and $N > 1$, we have $\varphi(N) < N$, hence $r < N$. (In particular $2^M > N > r$, for $M = \lfloor \log_2 N \rfloor + 1$.) For distinct integers $a \ne b$ in $\{1, \dots, N\}$,

$$\left| \frac{1}{a} - \frac{1}{b} \right| = \frac{|a-b|}{ab} \ge \frac{1}{N^2}.$$

Therefore there is at most one integer $r' \in (0, N]$ such that $\left| p_\ell(e) - \frac{1}{r'} \right| \le \frac{1}{4N^2}$. The preceding estimate shows that r satisfies this inequality, hence r is uniquely determined by $p_\ell(e)$. □

5 A diffusion-assisted factoring algorithm

In this section, we describe diffusion assisted factorization algorithm. It receives as an input a positive integer N with $m \ge 2$ odd prime factors. The algorithm returns a divisor d of N, with probability at least $p(m) = 1 - 2^{-m}(m+1)$ for each choice of a random integer $a \in \{1, 2, \dots, N-1\}$.

Before we proceed with the pseudocode for the algorithm, let us state the following two remarks.

Remark 5.1. Algorithm 1 is probabilistic in the same sense as Shor's factoring algorithm: a single *trial* (i.e. one choice of a and the subsequent steps) may end in a restart (`continue` in Algorithm 1; equivalently, FAIL for that particular a), for instance because the extracted square root satisfies $x \equiv \pm 1 \pmod{N}$ and hence yields only a trivial gcd.

However, for composite N with at least two distinct prime factors, each independent trial has a nonzero success probability depending only on the CRT parity pattern (cf. Proposition 3.6 and Lemma 3.2). Consequently, repeating the procedure independently drives the overall failure probability very close to 0.

We will leave for elsewhere the problem of optimizing the probability $p(m)$ of success. In that regard, the methodology of [Za13] seems applicable.

Remark 5.2. In practice one can efficiently exclude the cases "N is prime" and "N is a prime power" before invoking Algorithm 1. For primality testing there are fast randomized tests (e.g. Miller–Rabin), deterministic polynomial-time algorithms (AKS), and practical certificate-based methods (e.g. ECPP). To test whether N is a prime power, one may first perform a *perfect-power test* (decide whether $N = u^k$ with $k \geq 2$) and, if so, apply a primality test to the base u. See, for example, [CP05].

Input: odd composite N with at least two distinct prime factors. **Output:** a verified proper divisor.

1. Choose a uniformly from $\{1, \ldots, N-1\}$ and compute $d = \gcd(a, N)$. If $1 < d < N$, return d.
2. Set $M = \lfloor \log_2 N \rfloor + 1$. Generate the signed powers $a^{\pm 2^t} \bmod N$ for $0 \leq t \leq M$ by repeated squaring and inversion.
3. If two signed occurrences repeat, form their nonzero exponent difference, remove its power of two, and find the first squaring of a^q that equals one. Test the preceding value with $\gcd(x-1, N)$. Return a proper divisor if one is obtained.
4. Compute $b = a^{2^M} \bmod N$. Use the diffusion primitive, with its stated readout precision, to obtain the odd order r_b.
5. Find the least $k \geq 0$ with $a^{2^k r_b} = 1 \pmod{N}$. If $k > 0$, test $d = \gcd(a^{2^{k-1} r_b} - 1, N)$ and return it if proper.
6. If no factor was returned, restart with a fresh independent base.

The ideal diffusion oracle and its costs are those of Theorem 4.1. The later divisor-sensitive algorithm replaces this readout primitive with tagged samples and arithmetic detection.

Algorithm 1: Diffusion-assisted factoring: the fixed-walk baseline

6 Relation finding via collisions and gcd stabilization

This section discusses a complementary mechanism for recovering the order $r = \mathrm{ord}_N(b)$ from explicit relations produced by collisions while exploring the Cayley graph of

$$G = \langle b \rangle \subseteq (\mathbb{Z}/N\mathbb{Z})^*.$$

We explore G from the identity using short words in the dyadic generating multiset

$$S = \{\, b^{\pm 2^t} : \ 0 \leq t \leq M \,\}, \qquad M = \lfloor \log_2 N \rfloor + 1.$$

Whenever two sampled words land at the same group element, we obtain an arithmetic relation in the exponent. Repeating this yields many such relations, and taking running gcds often stabilizes to r (or a small multiple of r). In Section 7 we complement this with a quantitative analysis of how frequently such collisions occur for the random walk on G.

6.1 Exponent words and loop relations

As in the proof of Theorem 4.1 (and throughout the paper), we work in exponent coordinates. Assume r is odd and identify

$$G = \langle b \rangle \simeq \mathbb{Z}/r\mathbb{Z}, \qquad j \longmapsto b^j.$$

A (non-lazy) word of length L is a sequence

$$w = [(\varepsilon_1, t_1), \ldots, (\varepsilon_L, t_L)], \qquad \varepsilon_i \in \{\pm 1\}, \quad t_i \in \{0, \ldots, M\}.$$

Its integer exponent and endpoint are

$$E(w) = \sum_{i=1}^{L} \varepsilon_i 2^{t_i} \in \mathbb{Z}, \qquad x(w) = b^{E(w)} \in G.$$

Clearly,

$$|E(w)| \le L\, 2^M. \tag{6.1}$$

Definition 6.1 (Word collision and loop relation)**.** Let w, w' be words of length at most L. We say that (w, w') is a *word collision* if

$$x(w) = x(w') \quad \text{in } G.$$

The associated *loop difference* (or *loop exponent*) is

$$D(w, w') = E(w) - E(w').$$

A word collision is called *nontrivial* if $E(w) \neq E(w')$, equivalently if $D(w, w') \neq 0$.

Proposition 6.2. *Let w, w' be words of length at most L forming a* nontrivial *word collision (Definition 6.1). Then*

$$D(w, w') \neq 0, \qquad r \mid D(w, w'), \qquad \textit{and} \qquad 0 < |D(w, w')| \le 2L\, 2^M.$$

Equivalently, there exists $q \in \mathbb{Z} \setminus \{0\}$ such that

$$D(w, w') = q\, r \qquad \textit{and} \qquad |q\, r| \le 2L\, 2^M.$$

Proof. The collision $x(w) = x(w')$ means

$$b^{E(w)} \equiv b^{E(w')} \pmod{N},$$

hence

$$b^{E(w) - E(w')} \equiv 1 \pmod{N}.$$

By definition of $r = \operatorname{ord}_N(b)$, this implies $r \mid D(w, w')$.

If $E(w) = E(w')$, then $D(w, w') = 0$ and the collision yields no relation; by assumption we are in the nontrivial case, so $D(w, w') \neq 0$.

Finally, by (6.1),

$$|D(w, w')| \le |E(w)| + |E(w')| \le 2L\, 2^M. \qquad \square$$

Remark 6.3 (Running gcd is always a multiple of r). If $D_1, \ldots, D_s$ are nonzero loop differences coming from nontrivial word collisions, then each D_i is divisible by r, hence

$$r \mid g_s = \gcd(|D_1|, \ldots, |D_s|).$$

Thus the running gcd can only decrease as more relations are collected, and it always remains a multiple of the true order.

6.2 Gcd stabilization and the zeta heuristic

Each nontrivial collision yields, by Proposition 6.2, a relation

$$D_i = q_i r \neq 0, \qquad i = 1, \dots, s,$$

and we form the running gcd

$$g_s = \gcd(|D_1|, \dots, |D_s|).$$

Then

$$g_s = r \cdot \gcd(|q_1|, \dots, |q_s|).$$

In particular, $g_s = r$ if and only if $\gcd(q_1, \dots, q_s) = 1$.

Remark 6.4. In favorable long explorations, distinct word collisions arise from essentially unrelated pairs, and the multipliers $q_i = D_i/r$ behave like approximately independent nonzero integers in a comparable magnitude range. Under this heuristic, order recovery reduces to understanding how quickly $\gcd(q_1, \dots, q_s)$ drops to 1.

The following theorem is standard (see, e.g., [Nym72, Leh00]).

Theorem 6.5 (Riemann's zeta law for gcds)**.** *Fix $s \geq 2$. Let $U_1, \dots, U_s$ be independent random integers, each uniform on $\{1, 2, \dots, Q\}$. Then as $Q \to \infty$,*

$$\Pr(\gcd(U_1, \dots, U_s) = 1) = \frac{1}{\zeta(s)} + O\left(\frac{\delta_{s=2} \log Q + 1}{Q}\right),$$

where $\delta_{s=2} = 1$ if $s = 2$ and $\delta_{s=2} = 0$ otherwise.

Remark 6.6.

$$\frac{1}{\zeta(2)} = \frac{6}{\pi^2} \approx 0.608, \qquad \frac{1}{\zeta(3)} \approx 0.832, \qquad \frac{1}{\zeta(4)} = \frac{90}{\pi^4} \approx 0.924.$$

This motivates a heuristic under the explicitly stated independence model. It does not certify the exact order for the collision process. The extraction theorem below avoids any such assumption.

7 Collision bounds for the walk on G

We now quantify how often endpoint collisions occur for our (half–lazy) random walk on the cyclic group

$$G = \langle b \rangle, \qquad |G| = r,$$

with transition operator W as in the previous sections. For $x, y \in G$ and $t \geq 0$, define the t–step transition probabilities by

$$p(y, x; t) = (W^t \delta_y)(x).$$

In particular, the endpoint distribution at time t for a walk started at the identity e is

$$p_t(x) = p(e, x; t) = (W^t \delta_e)(x), \qquad x \in G.$$

To avoid dependence between samples taken along a single trajectory, we generate endpoints by *independent restarts*: we run the walk for t steps starting from e, record the endpoint, restart at e, and repeat this procedure T times.

7.1 Independent restarts, collisions, and the collision parameter

Definition 7.1 (Independent restarts and collisions)**.** Fix $t \geq 0$ and an integer $T \geq 1$. Let $(X_1, \ldots, X_T)$ be G–valued random variables.

1. We say that $(X_1, \ldots, X_T)$ are sampled by *independent restarts at time t* if the random variables are independent and have common distribution p_t, i.e.
$$\Pr(X_i = x) = p(e, x; t) \qquad (x \in G,\ i = 1, \ldots, T).$$
2. For $1 \leq i < j \leq T$, we say that the pair (i, j) *collides* if $X_i = X_j$. Let
$$Z = \#\{(i, j) : 1 \leq i < j \leq T,\ X_i = X_j\}$$
be the number of colliding pairs. We say that *a collision occurs* in the sample if $Z \geq 1$.
3. The *collision parameter* at time t is
$$s_2(t) = \sum_{x \in G} p_t(x)^2.$$

Theorem 7.2. *Assume $(X_1, \ldots, X_T)$ are sampled by independent restarts at time t (Definition 7.1). Let Z be the number of colliding pairs. Then*
$$\mathbb{E}[Z] = \binom{T}{2} s_2(t).$$
In particular, if t is large enough that p_t is close to uniform on G (so $p_t(x) \approx 1/r$ for all $x \in G$), then
$$\mathbb{E}[Z] \ \approx \ \frac{T(T-1)}{2r}. \tag{7.1}$$
Moreover, the probability of at least one collision satisfies the general bound
$$\Pr(Z \geq 1) \ \leq \ \mathbb{E}[Z].$$

Proof. For each pair $1 \leq i < j \leq T$, define the event
$$A_{ij} := \{X_i = X_j\}.$$
Recall that Z is the number of colliding pairs, i.e.
$$Z = \#\{(i, j) : 1 \leq i < j \leq T,\ A_{ij} \text{ occurs}\}.$$
Taking expectations and using linearity of expectation for finite sums gives
$$\mathbb{E}[Z] = \sum_{1 \leq i < j \leq T} \Pr(A_{ij}) = \sum_{1 \leq i < j \leq T} \Pr(X_i = X_j).$$
Fix $i < j$. By independence and the common law p_t,
$$\Pr(X_i = X_j) = \sum_{x \in G} \Pr(X_i = x,\ X_j = x) = \sum_{x \in G} \Pr(X_i = x) \Pr(X_j = x) = \sum_{x \in G} p_t(x)^2 = s_2(t).$$
There are $\binom{T}{2}$ pairs (i, j), hence $\mathbb{E}[Z] = \binom{T}{2} s_2(t)$.
If $p_t(x) \approx 1/r$ for all x, then
$$s_2(t) = \sum_{x \in G} p_t(x)^2 \approx \sum_{x \in G} \frac{1}{r^2} = \frac{1}{r},$$

which yields (7.1).

Finally, the event $\{Z \geq 1\}$ is the same as the union $\bigcup_{i<j} A_{ij}$, so by the union bound

$$\Pr(Z \geq 1) = \Pr\left(\bigcup_{1 \leq i < j \leq T} A_{ij}\right) \leq \sum_{1 \leq i < j \leq T} \Pr(A_{ij}) = \mathbb{E}[Z]. \qquad \square$$

Remark 7.3 (Heuristic collision threshold). Heuristically, the first collision among T independent restarts becomes plausible once

$$\mathbb{E}[Z] = \binom{T}{2} s_2(t) \approx 1, \qquad \text{equivalently} \qquad T \approx \frac{1}{\sqrt{s_2(t)}}.$$

In particular, if p_t is close to uniform so that $s_2(t) \approx 1/r$, then

$$T \approx \sqrt{r}.$$

Remark 7.4 (Digital collision search versus diffusion order-finding). The scale $T \approx \sqrt{r}$ governs a *purely digital* collision search: if $r = \mathrm{ord}_N(b)$ is large (often comparable to $\varphi(N)$), then the number of independent restarts needed to see a collision is itself of order $\sqrt{r}$, consistent with the classical "birthday paradox".

If instead the chosen base has *small order* $r \ll \varphi(N)$, then collisions appear after only $T \approx \sqrt{r}$ restarts, and the gcd-stabilization mechanism of Section 6 often recovers r quickly from a small number of relations.

This collision-based scale should be distinguished from what governs performance in the diffusion-based order-finding model analyzed earlier: when the diffusion primitive is available, the order is recovered from a controlled number of diffusion iterations and a single readout, rather than by waiting for endpoint collisions.

8 Examples

This section illustrates the diffusion–based order recovery on different composite integers as well as examples employing the digital collision strategy.

8.1 Example 1: $N = 299$

Take $N = 299 = 13 \cdot 23$ and $b = 3 \in (\mathbb{Z}/N\mathbb{Z})^*$. One checks that $\gcd(b, N) = 1$. The element b has *odd* multiplicative order modulo N:

$$r = \mathrm{ord}_{299}(3) = 33.$$

As in Section 4, we form the Cayley graph on the cyclic subgroup $\langle b \rangle \subset (\mathbb{Z}/N\mathbb{Z})^*$ using generators $b^{\pm 2^t}$ for $0 \leq t \leq M$ with $M = \lfloor \log_2 N \rfloor + 1$. See Figure 1. Then we run the half–lazy random walk started at the identity element.

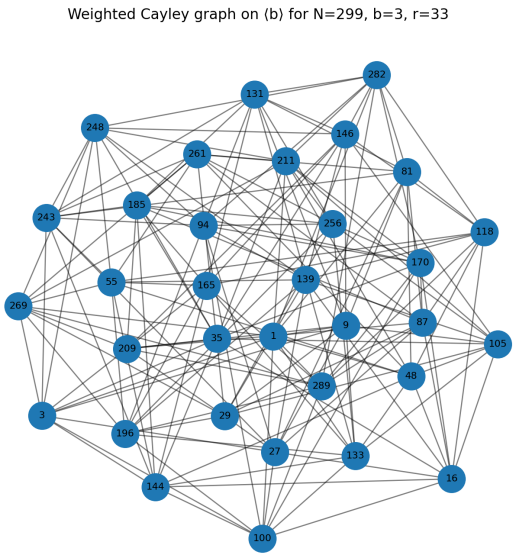


Figure 1: Cayley graph on $\langle 3\rangle \subset (\mathbb{Z}/299\mathbb{Z})^*$ with generators $b^{\pm 2^t}$. Here $N = 299$, $b = 3$, and $\mathrm{ord}_{299}(3) = 33$.

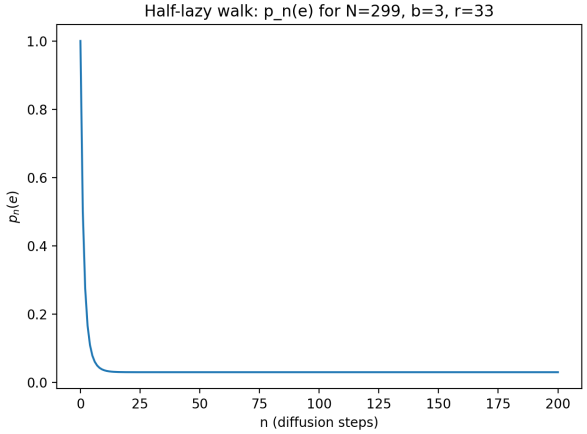


Figure 2: The identity value $p_\ell(e)$ for the half–lazy walk on $\langle 3\rangle$ (with $N = 299$).

Recovering the order by rounding. As shown in Figure 2, $p_\ell(e)$ rapidly approaches $1/r$, so that $1/p_\ell(e)$ approaches $r = 33$. Figure 3 plots $1/p_\ell(e)$ together with the target level r. Numerically, rounding stabilizes quickly: from $\ell = 17$ onward one has

$$\mathrm{round}\big(1/p_\ell(e)\big) = 33.$$

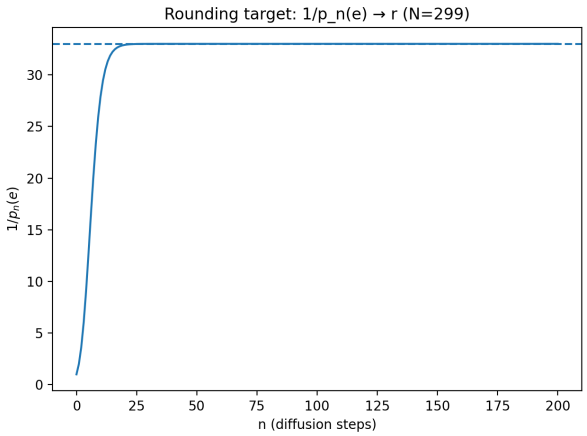


Figure 3: The sequence $1/p_\ell(e)$ converging to $r = \mathrm{ord}_{299}(3) = 33$; rounding recovers r. (The dashed line indicates $r = 33$.)

This provides a concrete "small-N" visualization of the diffusion model: the dynamics mix toward the uniform distribution on $\langle b \rangle$ (of size r), and a single scalar readout $p_\ell(e)$ determines r once it is close enough to $1/r$.

8.2 Example 2: $N = 1022117$

Now, we take $N = 1009 \cdot 1013 = 1022117$, and $b = 576 \in (\mathbb{Z}/N\mathbb{Z})^*$. One checks that $\gcd(b, N) = 1$, and the order of b modulo N is odd:

$$r = \mathrm{ord}_N(b) = 5313.$$

We run the half-lazy diffusion on the weighted Cayley graph of the cyclic subgroup $\langle b \rangle$, using generators $b^{\pm 2^t}$ for $0 \le t \le M$ with $M = \lfloor \log_2 N \rfloor + 1$. The theory predicts that after $\ell = O((\log_2 N)^2)$ diffusion steps, the single heat-kernel value at the identity determines r by rounding $1/p_\ell(e)$. See Figures 5 and 6.

Remark 8.1. The transition labels can be computed from a short rule, but a full-vector implementation must still represent the orbit. For visualization we draw only a small neighborhood of the identity (a local view), since the full graph has $r = 5313$ vertices. See Figure 4.

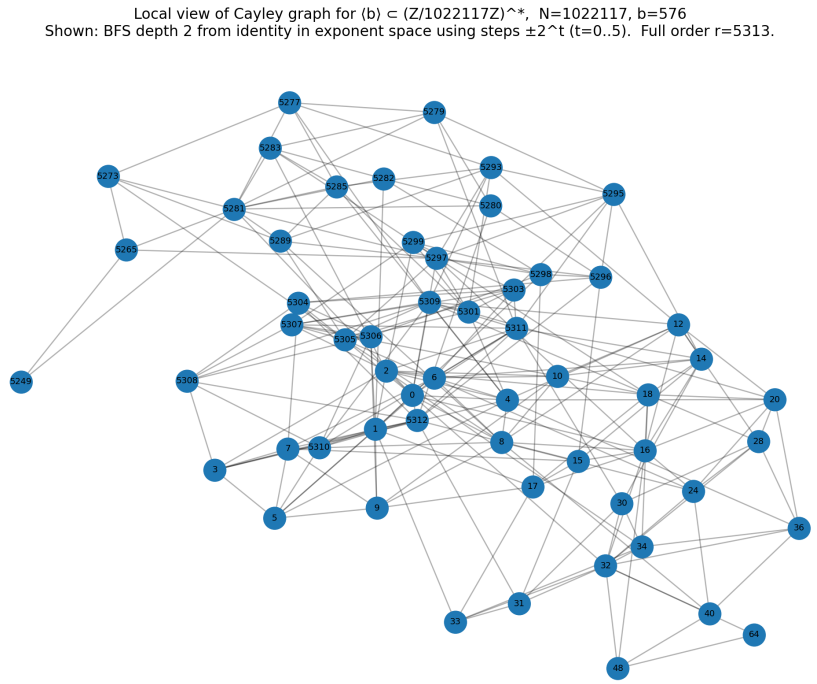


Figure 4: A *local* view of the Cayley graph near the identity (shown in exponent coordinates). This is only for illustration; the full Cayley graph has $r = 5313$ vertices.

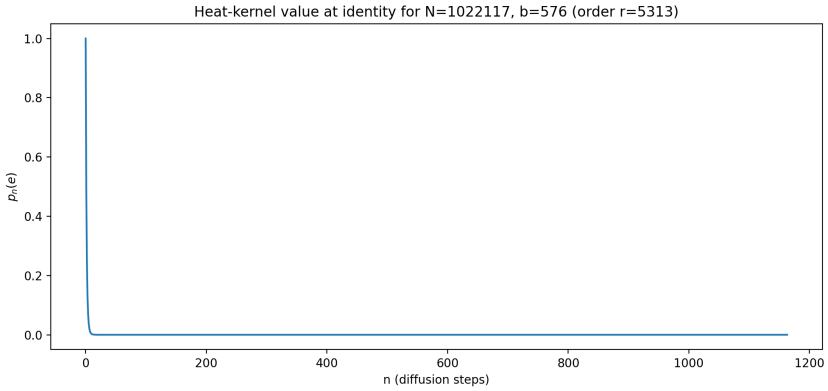


Figure 5: The identity heat-kernel value $p_\ell(e)$ as a function of the number ℓ of diffusion steps. As $\ell \to \infty$ one has $p_\ell(e) \to 1/r$.

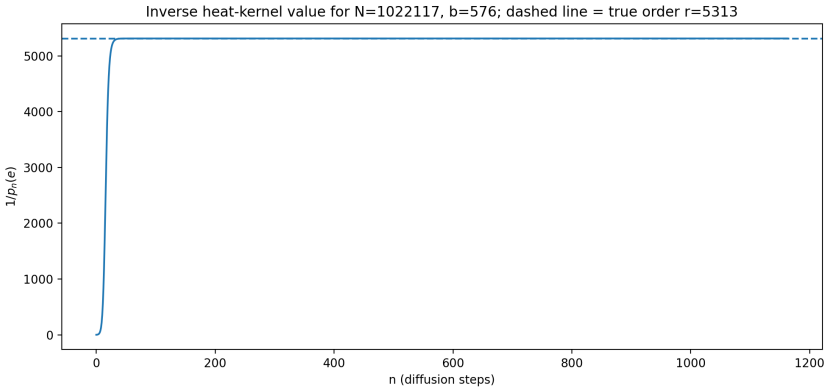


Figure 6: The inverse $1/p_\ell(e)$ converging to the true order $r = 5313$ (dashed line). In this run, rounding $1/p_\ell(e)$ recovers r after relatively few diffusion steps.

8.3 Exact certificates from the two original factoring runs

The original manuscript reported a Fermat-number run with $N = 4294967297$, $a = 3945765912$ and first relation $D = 314919552$. Exact modular arithmetic verifies $a^D = 1 \pmod N$. Removing the power of two from D and extracting a square root returns 6700417, whose complementary factor is 641. Thus

$$2^{32} + 1 = 641 \cdot 6700417.$$

The original comparison instance $N = 8219999$ has first reported relation $D = 12962750$ for $a = 7081686$; the same exact procedure yields $8219999 = 251 \cdot 32749$. Both first relations suffice. Waiting for several unchanged running gcds is unnecessary for factorization. The original wall-clock logs are preserved in the archived source, but are not used as hardware evidence or as controlled comparisons here. Current reproducible experiments are reported in Sections 22 and 24.

Part II

Programmed kernels, exact relations and finite sampling

9 Scope, conventions, and the fixed diffusion model

Let $N > 1$, let $b \in (\mathbb{Z}/N\mathbb{Z})^\times$, and write $G = \langle b \rangle$, $r = \mathrm{ord}_N(b)$. We use $e = 1 \pmod N$ for the identity. The input size is $n = \lceil \log_2 N \rceil$. A modular multiplication is an operation on $O(n)$-bit integers, not a unit-cost bit operation. We distinguish arithmetic work, parallel depth, stored state, routing, and readout precision throughout.

The original diffusion idea is to prepare δ_e, mix on G, and read a coordinate approaching $1/r$. The revised question is whether the mixing schedule and the arithmetic post-processing can be improved while exposing their physical costs. Our main result is the exact kernel identity in Theorem 10.1. The arithmetic reduction and success probability in Section 11 are classical facts proved here in the form needed for relation finding; we make no originality claim for those reductions. Binary averaging and accelerated walks have substantial antecedents, including the work of Chung, Diaconis, and Graham and of Dubail and Massoulié [1, 2]. Our explicit schedule is analyzed directly, without appealing to a new general mixing phenomenon. Its precise novelty relative to that literature requires a broader bibliographic assessment.

Discrete heat kernels on regular graphs provide a related spectral framework; explicit formulas and return-distribution applications were developed in [CHJSV23]. Here the cyclic structure permits a different, elementary finite-product formula. General background on finite-group walks and spectral mixing is available in [LPW09], and on factorization in [CP05].

Shor's algorithm works in a different resource model [Sh97]. A classical array with one scalar per group element uses r physical degrees of freedom. An $O(\log N)$-qubit register does not provide independently readable classical access to its N amplitudes. Counting the applications of a linear operator in both settings is therefore insufficient for a complexity comparison.

9.1 Shift operators and generator multiplicities

Identify G with $\mathbb{Z}/r\mathbb{Z}$ for the analysis only, via $j \mapsto b^j$, and define

$$(Tf)(j) = f(j-1).$$

Thus $T\delta_0 = \delta_1$, $T^{-1} = T^*$, and $T^r = I$. In residue labels, a shift is multiplication by a known power of b. The analysis does not require supplying the unknown r to the algorithm.

Set $M = \lfloor \log_2 N \rfloor + 1$. The fixed half-lazy walk is

$$W = \frac{1}{2} I + \frac{1}{4(M+1)} \sum_{t=0}^{M} (T^{2^t} + T^{-2^t}). \tag{9.1}$$

This operator formula counts every signed generator occurrence, including coincident shifts and identities. It is valid for even order and for $r = 1$. A simple-graph definition that excludes loops, or counts a coincident positive and negative shift only once, would not in general give (9.1).

Lemma 9.1 (Dyadic escape). *If $1 \le k < r$ and $2^M > r$, some $0 \le t \le M$ satisfies*

$$k2^t \bmod r \in [r/4, 3r/4].$$

Proof. Suppose otherwise. Let a_t be the least nonnegative residue and $d_t = \min(a_t, r - a_t)$. Initially $d_0 \geq 1$. As long as the middle interval is avoided, a direct calculation on the two intervals $[0, r/4)$ and $(3r/4, r)$ gives $d_{t+1} = 2d_t$. This also proves inductively that $d_t > 0$, including when r is even. Consequently $d_M = 2^M d_0 \geq 2^M > r$, whereas avoidance requires $d_M < r/4$, a contradiction. □

Proposition 9.2 (Fixed-walk baseline)**.** *Let $p_t = W^t \delta_e$. For $r > 1$,*

$$0 \leq p_t(e) - \frac{1}{r} \leq \exp\left(-\frac{t}{2(M+1)}\right).$$

In particular, the sufficient original bound

$$t \geq 4(M+1)(\log_2 N + 2)$$

implies $|p_t(e) - 1/r| \leq 1/(4N^2)$.

Proof. The characters $\chi_k(j) = \exp(2\pi i k j/r)$ diagonalize W with eigenvalues

$$\lambda_k = \frac{1}{2} + \frac{1}{2(M+1)} \sum_{u=0}^{M} \cos(2\pi k 2^u / r).$$

Each eigenvalue belongs to $[0, 1]$, and $\lambda_0 = 1$. Lemma 9.1 makes at least one cosine nonpositive for $k \neq 0$. Hence $\lambda_k \leq 1 - 1/[2(M+1)]$. Character orthogonality gives

$$p_t(e) = \frac{1}{r} \sum_{k=0}^{r-1} \lambda_k^t.$$

Subtract $1/r$, use nonnegativity, and bound the remaining sum by the largest summand. Finally $1 - x \leq e^{-x}$ and $\log_2 N \geq \ln N$ give the stated precision. For $r = 1$, $p_t(e) = 1$ exactly. □

Lemma 9.3 (Reciprocal rounding)**.** *If $1 \leq r \leq B$, $B \geq 1$, and $|y - 1/r| \leq 1/(4B^2)$, then $y > 0$ and* $\mathrm{round}(1/y) = r$.

Proof. We have $y \geq 1/B - 1/(4B^2) \geq 3/(4B)$. Therefore

$$\left|\frac{1}{y} - r\right| = \frac{r|y - 1/r|}{y} \leq \frac{B/(4B^2)}{3/(4B)} = \frac{1}{3} < \frac{1}{2}.$$

There is no tie in rounding. □

The fixed-walk bound is sufficient and often conservative. Reducing this bound is not the same as proving that every individual instance needs quadratically many steps. The programmed construction below changes the permitted operators and gives a sharper uniform guarantee in the enlarged model.

10 Programmed dyadic diffusion and its exact kernel

For $t \geq 0$, define

$$A_t = \frac{I + T^{2^t}}{2}, \qquad B_t = A_t A_t^* = \frac{1}{2} I + \frac{1}{4}(T^{2^t} + T^{-2^t}). \tag{10.1}$$

Each B_t is a symmetric stochastic averaging operator. A programmed run applies $B_0, B_1, \ldots, B_{L-1}$, using the repeated-squaring sequence $b, b^2, b^4, \ldots$ to select the connections. This differs from repeatedly applying their average W.

Theorem 10.1 (Exact programmed return kernel)**.** *Let $L \geq 0$, $Q = 2^L$, $s = Q \bmod r$, and*

$$h_L = \left(\prod_{t=0}^{L-1} B_t \right) \delta_e.$$

Then

$$\boxed{h_L(e) = \frac{1}{r} + \frac{s(r-s)}{rQ^2}.} \tag{10.2}$$

Moreover,

$$\max_{x \in G} |h_L(x) - 1/r| \leq h_L(e) - 1/r \leq \frac{r}{4Q^2}. \tag{10.3}$$

These statements hold for all $r \geq 1$, without any parity restriction.

Proof. All shifts commute, so

$$\prod_{t=0}^{L-1} B_t = U_Q U_Q^*, \qquad U_Q = \prod_{t=0}^{L-1} A_t.$$

The elementary polynomial identity

$$\prod_{t=0}^{L-1} (1 + z^{2^t}) = \sum_{u=0}^{Q-1} z^u$$

follows by writing each $0 \leq u < Q$ uniquely in binary. Substituting $z = T$ gives

$$U_Q = \frac{1}{Q} \sum_{u=0}^{Q-1} T^u. \tag{10.4}$$

Thus $v = U_Q \delta_e$ is the law of b^U for U uniform on $\{0, \ldots, Q-1\}$. Write $Q = ar + s$, $0 \leq s < r$. There are s residue classes with $a+1$ representatives in the interval, and $r - s$ classes with a representatives. Consequently

$$h_L(e) = \langle U_Q \delta_e, U_Q \delta_e \rangle = \frac{s(a+1)^2 + (r-s)a^2}{Q^2}.$$

Expanding the numerator and using $Q = ar + s$ yields (10.2).

For a general coordinate, put $u = v - r^{-1}\mathbf{1}$. Circular autocorrelation gives

$$h_L(x) - 1/r = \sum_{j \in \mathbb{Z}/r\mathbb{Z}} u(j) u(j - x).$$

Translation preserves the Euclidean norm. By Cauchy–Schwarz, the absolute value is at most $\sum_j u(j)^2 = h_L(e) - 1/r$. Finally $s(r-s) \leq r^2/4$. The calculation includes $r = 1$, in which case $s = 0$ and the error vanishes. □

Corollary 10.2 (Logarithmic programmed depth)**.** *If $r \leq B$ and $Q^2 \geq B^3$, then* $\operatorname{round}(1/h_L(e)) = r$. *In particular, $L = \lceil \frac{3}{2} \log_2 B \rceil$ layers suffice for exact readout. Taking $B = N$ requires $O(\log N)$ layers and $O(\log N)$ modular squarings to specify their shifts.*

Proof. Equation (10.3) gives $0 \leq h_L(e) - 1/r \leq 1/(4B^2)$. Apply Lemma 9.3. □

The formula depends on the unknown r only in the proof; the implemented sequence uses N and b. For powers of two r, it becomes exactly uniform as soon as $r \mid Q$. For odd $r > 1$, that exact divisibility never occurs, but the error still decays as specified.

10.1 A certificate that permits early stopping

Proposition 10.3 (One-sided order certificate)**.** *Suppose $z > 0$ is a certified upper bound for a return value h satisfying $h \geq 1/r$. Define $c = \lceil 1/z \rceil$. If $b^c \equiv 1 \pmod{N}$, then $c = r$.*

Proof. The inequalities $z \geq h \geq 1/r$ imply $1/z \leq r$, hence $1 \leq c \leq r$. The modular check implies $r \mid c$. Therefore $c = r$. □

Both $p_t(e)$ in Proposition 9.2 and $h_L(e)$ in Theorem 10.1 satisfy the required one-sided bound. With a measured value y and a certified error bound ϵ, use $z = y + \epsilon$. The certificate must not be applied to an unbounded floating-point or analog reading. Without the inequality $c \leq r$, the check $b^c = 1$ certifies only a multiple of the order. This distinction is also why a stabilized gcd is not an exact-order certificate.

10.2 An explicit perturbation budget

Theorem 10.4 (Layer and readout errors)**.** *Suppose the actual normalized state obeys*

$$\widetilde{h}_{t+1} = B_t \widetilde{h}_t + e_t, \quad \|e_t\|_\infty \leq \eta_t, \quad \|\widetilde{h}_0 - \delta_e\|_\infty \leq \eta_{\text{init}}.$$

Let the final readout satisfy $|y - \widetilde{h}_L(e)| \leq \rho$. If $r \leq B$, then

$$|y - 1/r| \leq \frac{B}{4Q^2} + \eta_{\text{init}} + \sum_{t=0}^{L-1} \eta_t + \rho.$$

In particular, $Q^2 \geq 2B^3$ and

$$\eta_{\text{init}} + \sum_t \eta_t + \rho \leq \frac{1}{8B^2}$$

guarantee reciprocal rounding to r.

Proof. Because each row of B_t is a probability vector, $\|B_t u\|_\infty \leq \|u\|_\infty$. Induction in the error recurrence gives

$$\|\widetilde{h}_L - h_L\|_\infty \leq \eta_{\text{init}} + \sum_t \eta_t.$$

Add the readout error and (10.3). Under the displayed budgets the total is at most $1/(4B^2)$, so Lemma 9.3 applies. □

This theorem is an absolute, deterministic error statement. It does not identify a root-mean-square noise level with a worst-case bound. In a statistical device, a confidence interval must first supply the required error bound. If the matrix itself is perturbed, its effect on the actual state must be included in e_t; small entrywise coefficient error alone is not a complete error budget.

10.3 Other radices and finite abelian subgroups

For an integer radix $d \geq 2$, replace A_t by

$$A_t^{(d)} = \frac{1}{d} \sum_{j=0}^{d-1} T^{jd^t}.$$

Unique base-d expansion gives $\prod_{t<L} A_t^{(d)} = d^{-L} \sum_{u<d^L} T^u$. Consequently Theorem 10.1 holds with $Q = d^L$ for the symmetric factors $A_t^{(d)}(A_t^{(d)})^*$. A larger radix reduces the number of layers but increases the number of taps to at most $2d - 1$ per symmetric layer. This is a depth–fan-in tradeoff, not a free acceleration.

Theorem 10.5 (Finite abelian subgroup cardinality)**.** *Let $H = \langle g_1, \dots, g_d \rangle$ be a finite abelian subgroup, let $r_i = \operatorname{ord}(g_i)$, and let k_i be the programmed kernel on $\langle g_i \rangle$ with $Q_i = 2^{L_i}$, extended by zero to H. Set $k = k_1 * \cdots * k_d$. Then*

$$\|k - |H|^{-1}\mathbf{1}_H\|_\infty \le \sum_{i=1}^{d} \frac{r_i}{4Q_i^2}.$$

If $|H| \le B$ and $Q_i^2 \ge dB^3$ for each i, reciprocal rounding of $k(e)$ recovers $|H|$. The total programmed depth is $O(d \log B + d \log d)$.

Proof. Let u_i be the uniform probability measure on $\langle g_i \rangle$. The multiplication map $\prod_i \langle g_i \rangle \to H$ is a surjective homomorphism, so all fibers have the same size. Thus $u_1 * \cdots * u_d$ is uniform on H. Convolution with a probability measure contracts the infinity norm. The telescoping identity

$$k_1 * \cdots * k_d - u_1 * \cdots * u_d = \sum_{i=1}^{d} k_1 * \cdots * k_{i-1} * (k_i - u_i) * u_{i+1} * \cdots * u_d$$

and (10.3) prove the bound. Since $r_i \le B$, the prescribed Q_i make the sum at most $1/(4B^2)$. Apply Lemma 9.3 with $r = |H|$. □

The same cyclic result applies to the unknown orbit length of an accessible permutation σ: replace multiplication by b with σ, provided powers $\sigma^{\pm 2^t}$ can actually be implemented at their charged cost. For an arbitrary black-box permutation these powers may be expensive; their availability is an assumption, not a consequence of the kernel identity.

10.4 A support bound and the meaning of the improvement

A product of L operators, each supported on at most k shifts, has support of size at most k^L when applied to a delta mass. If its distribution is within $1/(2r)$ of uniform in every coordinate, every coordinate is positive. Therefore $k^L \ge r$, and

$$L \ge \log_k r.$$

For our three-tap layers, logarithmic depth is thus optimal in order of magnitude within this bounded-support positive-averaging model when r grows comparably to N. The lower bound says nothing about the precision or hardware area required to attain it.

11 Factor extraction from a relation, without exact order recovery

For this section assume

$$N = \prod_{i=1}^{m} p_i^{e_i}, \qquad m \ge 2,$$

where the p_i are distinct odd primes. Primes, even integers, and perfect powers may be handled separately by standard arithmetic preprocessing. The factorization displayed here is used for analysis; it is not an input to any algorithm.

Choose a unit a and write

$$\operatorname{ord}_N(a) = 2^k u, \qquad u \text{ odd}.$$

With $M = \lfloor \log_2 N \rfloor + 1$ and $b = a^{2^M}$, we have $\operatorname{ord}_N(b) = u$. Indeed $2^k \le \operatorname{ord}_N(a) \le \varphi(N) < N < 2^M$, so $k < M$, and the standard formula $\operatorname{ord}(a^d) = \operatorname{ord}(a)/\gcd(\operatorname{ord}(a), d)$ applies. The correct bound is on the exponent $k = v_2(\operatorname{ord}_N(a)) < \log_2 N$; it is not the assertion $2^k < \log_2 N$.

Theorem 11.1 (Any nonzero relation is sufficient)**.** *Suppose $D \neq 0$ and $b^D \equiv 1 \pmod N$. Put $q = |D|/2^{v_2(|D|)}$. Starting with $x = a^q$, square repeatedly until the first occurrence of 1. If $a^q = 1$, declare failure for this base. Otherwise, let x denote the preceding value and compute $d = \gcd(x - 1, N)$.*

For each i, put $k_i = v_2(\mathrm{ord}_{p_i^{e_i}}(a))$. The output is a proper divisor if and only if the k_i are not all equal. In particular, for a fixed favorable base, every nonzero relation succeeds, regardless of its multiplier. No factorization of D or recovery of the exact order is necessary.

Proof. Since $u = \mathrm{ord}_N(b)$ is odd and divides D, it divides q. Write the local orders as $2^{k_i}u_i$ with u_i odd. Each $u_i \mid u$, so $u_i \mid q$. Because q is odd, a^q has local order exactly 2^{k_i}. Let $K = \max_i k_i$. If $K = 0$, $a^q = 1$ and all k_i are equal. Suppose $K > 0$. The first global value 1 occurs after K squarings. Its predecessor is $x = a^{q2^{K-1}}$. In the component modulo $p_i^{e_i}$, it equals 1 if $k_i < K$, and has order two if $k_i = K$. The unit group of an odd prime power is cyclic, so its unique element of order two is -1. Therefore

$$x \equiv \begin{cases} -1 \pmod{p_i^{e_i}}, & k_i = K, \\ 1 \pmod{p_i^{e_i}}, & k_i < K. \end{cases}$$

At least one component is -1. A component is 1 precisely when the k_i are not all equal. In that case $\gcd(x - 1, N)$ contains exactly the prime-power factors with $k_i < K$, and is a proper nontrivial divisor. If all are equal, $x = -1 \pmod N$, giving gcd equal to one. □

The exponentiation a^q costs $O(\log |D|)$ modular products by binary exponentiation, and the chain uses fewer than $\log_2 N$ further squarings. If relations have polynomial bit length, post-processing has polynomial bit complexity. Factoring the relation integer to certify its exact order could be substantially more expensive and is unnecessary here.

> **Algorithm: extraction from an oddified relation.** Input: odd N, a unit a, $b = a^{2^M} \pmod N$, and $D \neq 0$.
>
> 1. Verify $b^{|D|} = 1 \pmod N$; otherwise return Invalid.
> 2. Set $q = |D|/2^{v_2(|D|)}$ and $x = a^q \pmod N$. If $x = 1$, return UnfavorableBase.
> 3. Repeat at most M times: compute $y = x^2 \pmod N$. If $y \neq 1$, replace x by y and continue. If $y = 1$, compute $d = \gcd(x - 1, N)$; return d when $1 < d < N$, and otherwise return UnfavorableBase.
> 4. If the loop ends without reaching 1, return Invalid.

When sampling arbitrary residues, compute $\gcd(a, N)$ first: a nonunit immediately gives a factor. The probability statements below are conditional on a uniform unit. Conditioning on "an early dyadic collision occurred" does not by itself preserve that distribution or its success probability.

11.1 The exact success probability

Theorem 11.2 (Classical CRT success law)**.** *Write $\varphi(p_i^{e_i}) = 2^{c_i}u_i$ with u_i odd, let $C = \sum_i c_i$ and $c = \min_i c_i$. For a uniformly random unit a, Theorem 11.1 succeeds with probability*

$$S_N = 1 - 2^{-C}\left(1 + \frac{2^{mc} - 1}{2^m - 1}\right) \geq 1 - 2^{1-m}. \tag{11.1}$$

For two distinct prime factors the universal lower bound is $1/2$. After j independent completed trials, failure probability is at most $2^{-(m-1)j}$, provided each trial supplies a valid relation.

Proof. CRT makes the local components independent and uniform. In a cyclic group with two-primary part of order 2^{c_i}, the two-adic valuation K_i of a uniform element's order has distribution

$$\mathbb{P}(K_i = 0) = 2^{-c_i}, \qquad \mathbb{P}(K_i = t) = 2^{t-1-c_i} \quad (1 \le t \le c_i).$$

The first equality counts the identity in the two-primary component. The second counts its $\varphi(2^t) = 2^{t-1}$ elements of exact order 2^t.

Failure is the event $K_1 = \cdots = K_m$, so independence gives

$$\mathbb{P}(\text{failure}) = 2^{-C} \left(1 + \sum_{t=1}^{c} 2^{m(t-1)} \right),$$

which is the complement in (11.1). Because $C \ge mc$, this is at most

$$\frac{1}{2^m - 1} + \frac{2^m - 2}{2^m - 1} \, 2^{-mc}.$$

This expression decreases with $c \ge 1$, and its value at $c = 1$ is 2^{1-m}. Independence of completed trials proves the last assertion. □

The weaker $1 - (m+1)/2^m$ bound in the previous manuscript counts a sufficient parity pattern. Its complement is not the complete failure event. Formula (11.1) supplies the exact event and includes the cases omitted by that sufficient condition. The formula is consistent with known analyses of the classical part of quantum factoring [Za13]; it is not a quantum-specific enhancement.

11.2 Independence of the favorable pattern and the oddified base

Proposition 11.3. *For uniform a, the value $b = a^{2^M}$ is uniform on the odd-order subgroup*

$$H_{\text{odd}} \simeq \prod_{i=1}^{m} C_{u_i}.$$

It is independent of the two-primary pattern $(K_1, \ldots, K_m)$. Consequently, any randomized relation-search success event depending only on b and independent search randomness is independent of favorability.

Proof. Each local cyclic group splits as $C_{2^{c_i}} \times C_{u_i}$. A uniform element has independent uniform components. Raising to 2^M kills the first factor and is an automorphism of the second, since $\gcd(2^M, u_i) = 1$. The K_i depend only on the first factors. Applying a function of b and independent randomness preserves independence. □

For a semiprime $N = pq$, write $p - 1 = 2^\alpha u$ and $q - 1 = 2^\beta v$ with u, v odd. The exact joint mass of favorable bases with oddified order at most B is therefore

$$\frac{S_N}{uv} \sum_{\substack{d \mid u, \ f \mid v \\ \operatorname{lcm}(d,f) \le B}} \varphi(d)\varphi(f). \tag{11.2}$$

This clean separation is useful in hardware-budget analyses. It does not assume that an arbitrary cutoff test is available before finding a relation.

11.3 Retained full-order cutoff bounds

For comparison, define $\pi_B(p,q)$ as the probability that a uniform unit a is favorable and $\operatorname{ord}_N(a) \le B$, using its full order. Counting elements of exact local orders gives

$$\pi_B(p,q) = \frac{1}{(p-1)(q-1)} \sum_{\substack{d\mid p-1,\ f\mid q-1 \\ \operatorname{lcm}(d,f)\le B,\ v_2(d)\ne v_2(f)}} \varphi(d)\varphi(f). \tag{11.3}$$

The sum is an analytic formula in the unknown factors, not an efficient recipe for choosing a favorable base.

Proposition 11.4. *For distinct odd primes $p < q$, with $g = \gcd(p-1, q-1)$,*

$$\pi_{\sqrt{pq}}(p,q) \ge \frac{1}{q-1}, \qquad \pi_B(p,q) \le \min\left(1, \frac{Bg\,\tau(p-1)\tau(q-1)}{(p-1)(q-1)}\right).$$

Here τ is the positive-divisor counting function.

Proof. For each $x \in \mathbb{F}_p^\times$ of order d, choose the component $y = 1$ modulo q if d is even, and $y = -1$ if d is odd. In the first case the combined order is d; in the second it is $2d$, which divides $p-1$ because d is an odd divisor of $p-1$. Both combined orders are at most $p-1 < \sqrt{pq}$ and their two-primary valuations differ. These give $p-1$ distinct favorable units, proving the lower bound. For each summand of (11.3),

$$\varphi(d)\varphi(f) \le df = \operatorname{lcm}(d,f)\gcd(d,f) \le Bg.$$

There are at most $\tau(p-1)\tau(q-1)$ summands, proving the upper bound. □

For balanced semiprimes with $\gcd(p-1, q-1) = N^{o(1)}$, the standard divisor estimate $\tau(t) = t^{o(1)}$ yields $\pi_{N^\gamma} \le N^{\gamma-1+o(1)}$. At $\gamma = 1/2$ the lower bound matches the exponent, giving $\pi_{\sqrt{N}} = N^{-1/2+o(1)}$. Thus searching only favorable full orders below $\sqrt{N}$ incurs an expected $N^{1/2+o(1)}$ base choices. A birthday-sized table $O(N^{1/4})$ per eligible base does not remove that selection cost.

12 Useful collisions, rigorous probabilities, and faster sampling

If two recorded exponents satisfy $b^E = b^{E'}$ and $E \ne E'$, their difference $D = E - E'$ is a nonzero multiple of $r = \operatorname{ord}_N(b)$. An endpoint repeat with $E = E'$ is not a useful relation. Different words can have the same integer exponent without using any arithmetic modulo r, so this distinction cannot be omitted.

12.1 A second-moment lower bound

Proposition 12.1. *Let $X_1, \dots, X_T$ be independent with common distribution p on a finite set. Put*

$$s_j = \sum_x p(x)^j, \quad Z = \sum_{i<j} \mathbf{1}_{X_i = X_j}, \quad \mu = \binom{T}{2} s_2.$$

Then

$$\mathbb{E}Z = \mu, \qquad \mathbb{E}Z^2 = \mu + 6\binom{T}{3}s_3 + 6\binom{T}{4}s_2^2.$$

For $T \ge 2$,

$$\mathbb{P}(Z > 0) \ge \frac{\mu^2}{\mu + 6\binom{T}{3}s_3 + 6\binom{T}{4}s_2^2}. \tag{12.1}$$

Proof. Every pair agrees with probability s_2. In the expansion of Z^2, repeated pairs contribute μ. For each triple of sample indices there are three unordered pairs of pair-events sharing an index, each with probability s_3, and the cross terms occur twice. For each quadruple there are three disjoint pairings, each with probability s_2^2, again counted twice. This proves the second moment. Finally,

$$(\mathbb{E}Z)^2 = (\mathbb{E}[Z\mathbf{1}_{Z>0}])^2 \le \mathbb{E}[Z^2]\mathbb{P}(Z>0)$$

by Cauchy–Schwarz. □

An expectation of order one alone was not a lower bound on the probability of seeing a collision. Equation (12.1) supplies one. For a symmetric fixed walk $p_t = W^t\delta_e$, the useful identity

$$s_2(t) = \|W^t\delta_e\|_2^2 = \langle \delta_e, W^{2t}\delta_e\rangle = p_{2t}(e)$$

connects the collision scale with the return kernel. The second-moment bound still counts all endpoint collisions; useful relations require the additional nonzero-exponent condition.

12.2 Independent binary exponents replace long random words

Choose $K \ge 1$ and $Q = 2^K$, sample E uniformly on $\{0,\dots,Q-1\}$, and compute $X = b^E$. Precompute b^{2^j} for $0 \le j < K$. Forming one endpoint uses one modular product for each nonzero bit of E, hence at most K and exactly $K/2$ in expectation with this implementation. The setup requires $K-1$ squarings. This replaces a 2,000-increment random word by a controlled interval sample.

Theorem 12.2 (Finite-interval useful-collision guarantee)**.** *Let $E_1,\dots,E_T$ be independent and uniform on $\{0,\dots,Q-1\}$ and put $X_i = b^{E_i}$. Define*

$$\beta_T(r) = \begin{cases} 1 - \prod_{j=0}^{T-1}(1-j/r), & T \le r, \\ 1, & T > r. \end{cases}$$

The probability of a pair with $X_i = X_j$ and $E_i \ne E_j$ is at least

$$\max\left\{0,\ \beta_T(r) - \frac{Tr}{4Q} - \frac{\binom{T}{2}}{Q}\right\}. \tag{12.2}$$

For each specified pair, its exact useful-collision probability is

$$\frac{1}{r} + \frac{s(r-s)}{rQ^2} - \frac{1}{Q}, \qquad s = Q \bmod r. \tag{12.3}$$

Proof. The endpoint distribution has s probabilities $(a+1)/Q$ and $r-s$ probabilities a/Q, where $Q = ar + s$. Its total variation distance from uniform is exactly

$$\epsilon = \frac{s(r-s)}{rQ} \le \frac{r}{4Q}.$$

Couple each endpoint with an independent uniform point so that the probability of any disagreement among T samples is at most $T\epsilon$. The uniform collision probability is $\beta_T(r)$, by sequential sampling without repetitions. Therefore the actual endpoint-collision probability is at least $\beta_T(r) - T\epsilon$.

An endpoint collision without a useful relation must include a repeated integer exponent. By the union bound this event has probability at most $\binom{T}{2}/Q$. Subtraction proves (12.2). For one specified pair, endpoint equality has probability $\sum_x p(x)^2$, computed in Theorem 10.1. Equality of the integer exponents has probability $1/Q$ and is a subset of endpoint equality. Subtraction gives (12.3). □

If $r \le B$, the simpler lower bound

$$\mathbb{P}(\text{useful relation}) \ge \max\left\{0, 1 - e^{-T(T-1)/(2B)} - \frac{TB}{4Q} - \frac{\binom{T}{2}}{Q}\right\}$$

holds for all T. Taking T of order $\sqrt{B}$ and Q sufficiently larger than $B^{3/2}$ gives a fixed positive success probability with $K = O(\log B)$ bits per exponent. This is a rigorous sampling statement, not a mixing assumption. With $b = a^{2^M}$, Proposition 11.3 permits multiplication of the relation-search success probability by S_N; Theorem 11.1 then handles the first useful relation.

In particular, if a valid relation produces no proper factor, continuing to collect relations for the same base cannot repair its two-primary pattern. For an equality-only strategy a new base is appropriate. The divisor-sensitive extension in Section 14 should first complete its partial-congruence detector, which can succeed even for that base.

12.3 Running gcds: what they certify

For nonzero relations $D_i = q_i r$,

$$\gcd(|D_1|, \dots, |D_j|) = r \gcd(|q_1|, \dots, |q_j|).$$

The gcd is always an order multiple. Its stabilization proves neither that the remaining multiplier is one nor that the q_i are independent uniform integers. A $1/\zeta(j)$ law applies to independent integers in a growing interval, but is not established for the multipliers produced by the original word process. We therefore do not use that heuristic for correctness or stopping. Exact order, when specifically required, can instead be certified by Proposition 10.3; factorization can use Theorem 11.1.

12.4 Memory, batching, and parallel throughput

For independent uniform endpoints, a batch of T samples has collision probability at least $1 - e^{-T(T-1)/(2r)}$. A hash table stores endpoint labels together with their exponents, not just occupied graph vertices. A record requires at least $\lceil \log_2 N \rceil + K$ bits before control fields and table slack. This endpoint table is a different object from the r-coordinate probability vector.

With a table capacity $H \ll \sqrt{r}$ and independent discarded batches of size H, the collision probability per batch is of order H^2/r under the uniform model. Thus the expected number of generated endpoints is of order r/H, versus order $\sqrt{r}$ when a sufficiently large table can be retained. A computational budget must count generation of those endpoints as well as storage. These order-of-magnitude assertions use the stated uniform model; finite-interval errors can be bounded by Theorem 12.2.

Suppose P arithmetic engines collectively sustain $P\nu$ modular products per second, and a collision table accepts at most β records per second. For the binary implementation, its sustained endpoint rate is at most

$$\min\left\{\frac{2P\nu}{K},\ \beta\right\} \tag{12.4}$$

in expectation, excluding setup, inversion, factor extraction, and synchronization. Generation of a complete H-record batch cannot be assigned a one-nanosecond latency independently of H without specifying corresponding arithmetic and memory bandwidth.

Shared distinguished-point searches can reduce stored records and permit efficient cooperation between processors [5]. Applying such a strategy here requires a concrete transition function, exponent bookkeeping, and an analysis of its cycles; the independent-sample theorem does not automatically extend to that correlated walk. This is a reasonable implementation direction rather than an already-proved improvement for this manuscript.

For generic order collisions the relevant domain has size r. Pollard's factoring method instead detects coincidences modulo an unknown prime divisor through gcds [3, 4]. Its usual heuristic scale is $\sqrt{p}$ for the least prime factor p, which is about $N^{1/4}$ for balanced semiprimes. A method requiring $\sqrt{r}$ endpoints with r near N does not thereby improve the standard factoring scale. Number-field-sieve and elliptic-curve methods are additional essential baselines for a serious performance study.

13 Distributed collision detection with four-wise independent dyadic exponents

Let $N = pq$ with p, q distinct odd primes. Choose a uniformly from $(\mathbb{Z}/N\mathbb{Z})^\times$, put $M = \lfloor \log_2 N \rfloor + 1$, $b = a^{2^M} \bmod N$, and write $r = \mathrm{ord}_N(b)$. The factor-extraction theorem of Section 11 states that any verified relation $b^D = 1 \pmod N$ with $D \neq 0$ yields a proper factor whenever the two local two-primary order exponents of a differ. This event has probability at least $1/2$ and is independent of b.

The following construction produces T endpoint samples with a random seed of size $4K$ bits rather than retaining all T exponents. It does *not* remove the need for endpoint labels, modular arithmetic, or communication between physical processors.

Let $Q = 2^K$ and let $\mathbb{F}_Q$ be a finite field. Choose $T \leq Q$ distinct, publicly specified elements $\xi_1, \ldots, \xi_T \in \mathbb{F}_Q$. Choose $A_0, A_1, A_2, A_3 \in \mathbb{F}_Q$ independently and uniformly, and define

$$F(z) = A_0 + A_1 z + A_2 z^2 + A_3 z^3.$$

Fix an arbitrary bijection $\iota : \mathbb{F}_Q \longrightarrow \{0, \ldots, Q-1\}$ and put

$$E_i = \iota(F(\xi_i)), \qquad X_i = b^{E_i} \bmod N.$$

Each E_i is uniform, and any four distinct exponents are jointly independent and uniform: this is the invertibility of the corresponding Vandermonde matrix over $\mathbb{F}_Q$. In a collision, the two exponents can be regenerated from the saved indices and the global coefficients $A_0, \ldots, A_3$.

Theorem 13.1 (Four-wise independent birthday bound)**.** *Let $T \geq 4$, $r \leq B$, and let the E_i, X_i be as above. Put*

$$A = \binom{T}{2}, \qquad U = 6\binom{T}{3}, \qquad V = 6\binom{T}{4}.$$

Then

$$\mathbb{P}\big(\exists i < j : \ X_i = X_j, \ E_i \neq E_j\big) \geq \frac{1}{\dfrac{B}{A} + \dfrac{U+V}{A^2} + \dfrac{UB}{A^2 Q}} - \frac{A}{Q}. \tag{13.1}$$

In particular, if $T = C$, $B = C^2$, and $Q \geq 100C^3$, then

$$\mathbb{P}(\textit{nonzero relation}) \geq \frac{1}{3} - O(C^{-1}).$$

The hidden constant is absolute; the displayed finite bound is valid without asymptotic notation.

Proof. Let $\pi(x) = \mathbb{P}(X_i = x)$, and define $s_2 = \sum_x \pi(x)^2$ and $s_3 = \sum_x \pi(x)^3$. The Q consecutive exponents are as evenly distributed modulo r as possible. Therefore $s_2 \geq 1/r \geq 1/B$, and $\max_x \pi(x) \leq 1/r + 1/Q \leq s_2 + 1/Q$, which yields

$$s_3 \leq (s_2 + Q^{-1}) s_2.$$

Put $Z = \sum_{i<j} \mathbf{1}_{\{X_i = X_j\}}$. Since all moments in $\mathbb{E}(Z^2)$ involve at most four sample indices, four-wise independence suffices to give exactly the same moment formulas as full independence:

$$\mathbb{E}Z = As_2, \qquad \mathbb{E}Z^2 = As_2 + Us_3 + Vs_2^2.$$

By Cauchy–Schwarz,

$$\mathbb{P}(Z > 0) \geq \frac{(As_2)^2}{As_2 + Us_3 + Vs_2^2} \geq \frac{1}{\dfrac{1}{As_2} + \dfrac{U+V}{A^2} + \dfrac{U}{A^2 Q s_2}}.$$

Use $s_2 \geq B^{-1}$ to obtain the first term on the right-hand side of (13.1). A collision is unhelpful only if some $E_i = E_j$. Pairwise independence and the union bound give $\mathbb{P}(\exists i < j : E_i = E_j) \leq A/Q$. Subtract this event. For $T = C, B = C^2$, the three denominator terms tend respectively to $2, 1, 0$, and $A/Q = O(C^{-1})$ under the stated condition. □

Corollary 13.2 (Factorization scale of an ideal parallel batch)**.** *Suppose $N = pq$ with p, q distinct odd primes, and that a device can produce, retain, and compare C tagged endpoints in a batch, including all endpoint equalities, without omissions. If*

$$N \leq 4C^2 \qquad \text{and} \qquad 2^K \geq 100C^3,$$

then, for a uniform unit a and independent four-wise sampling seed, its probability of obtaining a nontrivial factor in one batch is at least

$$\frac{1}{2}\left[\frac{1}{\dfrac{C^2}{\binom{C}{2}} + \dfrac{6\binom{C}{3} + 6\binom{C}{4}}{\binom{C}{2}^2} + \dfrac{6\binom{C}{3} C^2}{\binom{C}{2}^2 2^K}} - \frac{\binom{C}{2}}{2^K}\right] = \frac{1}{6} - O(C^{-1}), \tag{13.2}$$

for $C \geq 4$. If an initially chosen a is not a unit, the nontrivial gcd already provides a factor.

Proof. Write $p - 1 = 2^\alpha u$ and $q - 1 = 2^\beta v$, where u, v are odd and $\alpha, \beta \geq 1$. Since b has odd order,

$$r \mid \operatorname{lcm}(u, v) \leq uv \leq \frac{(p-1)(q-1)}{4} < \frac{N}{4} \leq C^2.$$

Theorem 13.1 thus applies to every unit a. The two-primary favorability event for a is independent of b and of the independent exponent-sampling seed, and its probability is at least $1/2$, by Proposition 11.3 and Theorem 11.2. A useful relation is valid for factor extraction by Theorem 11.1. Multiply the two bounds. □

13.1 Local rendezvous versus global communication

A distributed collision detector may hash each numeric endpoint X_i to a destination bucket $h(X_i)$, retaining the pair (X_i, i). Numerically equal endpoints always reach the same bucket. Unequal endpoints that happen to share a bucket must be distinguished by an exact comparison of their full labels. A universal hash family can make the probability that distinct values share a bucket at most $1/H$, where H is the number of buckets. Thus, conditional on any set of endpoints, if Z denotes the number of equal-label pairs and Y the number of pairs sent to the same bucket,

$$\mathbb{E}_h(Y \mid X_1, \ldots, X_T) \leq Z + \frac{\binom{T}{2} - Z}{H}.$$

For $H \asymp T \asymp C$ and $r \asymp C^2$, only $O(C)$ bucket-local comparisons are expected. This replaces a *central* collision table by a *distributed* collision table; it does not remove the global movement of labeled packets to their buckets. Buffering and bucket overflow must be analyzed separately.

Proposition 13.3 (A two-dimensional nearest-neighbor routing barrier)**.** *Let $C = m^2$ physical processors form an $m \times m$ mesh, each initially holding one packet. Suppose the packets' hash destinations are independently uniform over all C processors. If each link crossing the vertical bisection transports at most κ packets in each direction per communication round, then the expected number of rounds required to route all packets satisfies*

$$\mathbb{E}R \geq \frac{m}{4\kappa}.$$

This statement applies to nearest-neighbor transport and does not exclude nonlocal interconnects, hierarchical switching fabrics, or other architectures whose physical communication resources are accounted for separately.

Proof. Of the $C/2$ initial packets in the left half, each has probability $1/2$ of requiring a destination in the right half. The expected number crossing left-to-right is therefore $C/4$. There are m edges through the bisection, each with directional capacity κ packets per round. Every routing realization requires at least the number of left-to-right packets divided by $m\kappa$ rounds. Taking expectations gives $\mathbb{E}R \geq C/(4m\kappa) = m/(4\kappa)$. □

Remark 13.4 (Interpretation)*.* The programmed diffusion identity proves $O(\log N)$ *algebraic* layers, and the present theorem proves a *birthday-scale* probability in an ideal batch. Neither establishes $O(\log N)$ physical time. Building endpoints requires $O(K)$ modular operations per record by the binary method, while routing a large distributed collision table is a separate communication problem. A purely analog three-tap heat cell, or one cell with only a single latch bit, cannot be counted as a modular-arithmetic engine or as an entire tagged bucket without additional circuitry. For context, parallel collision search and distinguished-point techniques predate this construction (see van Oorschot–Wiener, *J. Cryptology* 12 (1999), 1–28).

Part III

Divisor-sensitive detection and improved sampling architecture

14 Divisor-sensitive collisions: reducing the relevant sampling scale

14.1 The arithmetic information lost by an equality-only detector

The preceding collision networks recognize $X_i = X_j$ as integers modulo N. A factorization algorithm can exploit a strictly weaker event:

$$X_i \equiv X_j \pmod{p}, \qquad X_i \not\equiv X_j \pmod{q}, \qquad N = pq.$$

Then $\gcd(X_i - X_j, N) = p$, even though a numeric equality comparator reports two different keys. This observation is classical in factoring, not a claim of a new principle [3, 4, 13]. Its integration with the programmed exponent distribution changes the resource target of this project. The algorithm need not learn either p or q to test the gcd.

In this section p, q are distinct odd primes, $p < q$; the factorization is used only in the proof. Choose a unit a, set

$$n = \lceil \log_2 N \rceil, \quad b = a^{2^n} \bmod N, \quad d_p = \operatorname{ord}_p(b), \quad d_q = \operatorname{ord}_q(b), \quad r = \operatorname{ord}_N(b) = \operatorname{lcm}(d_p, d_q).$$

All three orders are odd. In particular

$$d_p \le (p-1)/2, \qquad d_q \le (q-1)/2, \qquad r < N/4. \tag{14.1}$$

We draw $2m$ four-wise independent exponents in $\{0, \dots, Q-1\}$, using the degree-three finite-field construction already proved above. Divide their indices into two groups of size m for the probability analysis. The actual detector may use every pair.

Lemma 14.1 (The collision dichotomy)**.** *For a fixed b, one of the following alternatives holds.*

(i) *If $d_p \neq d_q$, let $d = \min(d_p, d_q)$. For exponents uniform modulo r, a specified pair yields a proper gcd associated with the component of order d with probability*

$$\theta = \frac{1}{d} - \frac{1}{r} \ge \frac{1}{2d}.$$

(ii) *If $d_p = d_q$, then $r = d_p = d_q \le (p-1)/2$. Full endpoint collisions occur on this smaller group; any nonzero exponent difference can be used in Theorem 11.1.*

Proof. Assume, for example, $d_p = d < d_q$. The congruence modulo p is equivalent to $d \mid E - E'$, whereas simultaneous congruence modulo p and q is equivalent to $r \mid E - E'$. A uniform difference modulo r has the former property with probability $1/d$ and the latter with probability $1/r$. The events are nested. Since r/d is an integer greater than one, subtraction gives the stated bound. In fact r/d is odd and at least three here; the weaker factor $1/2$ suffices. The argument is unchanged after exchanging p and q. If the orders are equal, their least common multiple is their common value and the last bound follows from (14.1). □

14.2 A finite second-moment theorem for translation-invariant events

The next proof supplies a quantitative guarantee without a random-walk mixing assumption, without knowing an order, and with only four-wise independence.

Lemma 14.2 (Bipartite collision moments)**.** *Let $S \subseteq \mathbb{Z}/r\mathbb{Z}$ and $\theta = |S|/r$. Consider the two lists*

$$U_1, \dots, U_m, \qquad V_1, \dots, V_m.$$

Assume their $2m$ members are four-wise independent, each uniform on $\mathbb{Z}/r\mathbb{Z}$. Set

$$Z_S = \sum_{i,j=1}^{m} \mathbf{1}_{\{U_i - V_j \in S\}}, \qquad A = m^2, \qquad \mu = A\theta.$$

Then

$$\mathbb{E}Z_S = \mu, \qquad \mathbb{E}Z_S^2 = \mu + A(A-1)\theta^2.$$

Consequently $\mathbb{P}(Z_S > 0) \geq \mu/(1+\mu)$ when $\mu > 0$.

Proof. Every difference is uniform, giving probability θ for one pair. For two disjoint pairs, four-wise independence gives probability θ^2. If two pairs share U_i, condition on that value. Each of the two remaining independent variables belongs to a translate of S with probability θ, independently of the value of U_i. Three-wise independence is enough for this calculation. Pairs sharing a V_j are identical in this respect. Thus all distinct pair indicators are pairwise independent. Expanding the square gives the two formulas. Finally Cauchy–Schwarz yields

$$\mathbb{P}(Z_S > 0) \geq \frac{\mu^2}{\mu + A(A-1)\theta^2} \geq \frac{\mu}{1+\mu}.$$

□

Theorem 14.3 (Finite-interval version)**.** *Replace the uniform residues in the preceding lemma by residues of four-wise independent uniform integers in $\{0, \dots, Q-1\}$, where $Q \geq r$, and write $\alpha = r/Q$. For $\mu = m^2|S|/r > 0$,*

$$\mathbb{P}(Z_S > 0) \geq G_\alpha(\mu) := \frac{(1-\alpha)^4\mu}{(1+\alpha)^2 + (1+\alpha)^4\mu}. \tag{14.2}$$

For full endpoint collisions, the probability that some cross-pair has different integer exponents *is at least*

$$G_\alpha(m^2/r) - m^2/Q. \tag{14.3}$$

A negative lower bound is interpreted as zero.

Proof. Each residue has probability between $(1-\alpha)/r$ and $(1+\alpha)/r$: its number of representatives is either $\lfloor Q/r \rfloor$ or $\lceil Q/r \rceil$. For any event involving $k \leq 4$ distinct sample indices, its probability is therefore between $(1-\alpha)^k$ and $(1+\alpha)^k$ times its probability under independent uniform residues. This is a pointwise bound on the product probabilities, summed over the event. Four-wise independence of the integer exponents justifies that product formula.

Let $Z = Z_S$. The preceding lemma now gives

$$\mathbb{E}Z \geq (1-\alpha)^2\mu, \qquad \mathbb{E}Z^2 \leq (1+\alpha)^2\mu + (1+\alpha)^4\mu^2.$$

For overlapping pairs there are only three indices, so the fourth power remains a valid upper bound. Applying Cauchy–Schwarz proves (14.2). Take $S = \{0\}$ for full endpoint equality. A specified pair has equal integer exponents with probability $1/Q$; a union bound over the m^2 cross-pairs proves (14.3). □

The function $G_\alpha(\mu)$ increases with $\mu > 0$ and decreases with $0 \le \alpha < 1$. The first assertion follows by writing it as $c\mu/(d + e\mu)$ with positive constants; for the second, the numerator decreases and the denominator increases.

14.3 An order-oblivious quarter-power guarantee

Theorem 14.4 (Divisor-sensitive semiprime batch)**.** *Let $N = pq$ be a product of two distinct odd primes. Put*

$$\boxed{m = \lceil N^{1/4} \rceil, \qquad n = \lceil \log_2 N \rceil, \qquad Q = 2^{n+7}.}$$

Choose a uniformly from $\{1, \dots, N-1\}$. First test $\gcd(a, N)$; if it is one, form $b = a^{2^n} \bmod N$. Generate $2m$ four-wise independent uniform exponents in $\{0, \dots, Q-1\}$, independently of a, and their exact endpoints $X_i = b^{E_i} \bmod N$.

Use a detector with both of the following capabilities:

(a) *it finds a proper $\gcd(X_i - X_j, N)$ whenever any pair has one;*

(b) *it finds a full equality with $E_i \ne E_j$ whenever such a pair exists, and applies Theorem 11.1.*

Then a batch returns a verified proper factor with probability at least $3/10$. The algorithm is supplied neither the factors nor their orders. Independent repetitions have failure probability at most $(7/10)^k$; 13 batches suffice for success probability at least 0.99.

Proof. A nonunit base already returns a proper divisor. Conditional on being a unit, a is uniform on the unit group, so it suffices to handle this case. Fix the oddified value b. We have $\alpha = r/Q < 1/128$ by $r < N$ and $Q \ge 128N$. Also $m^2 \ge \sqrt{N}$ and $m^2 \le N$ for $N \ge 15$. For the latter, $m \le N^{1/4} + 1 \le \sqrt{N}$ when $N \ge 15$ (or directly check the least case); this deliberately loose bound is sufficient.

If $d_p \ne d_q$, the smaller order d is at most $d_p \le (p-1)/2 < \sqrt{N}/2$. Hence

$$\mu = m^2 \left(\frac{1}{d} - \frac{1}{r} \right) \ge \frac{m^2}{2d} > 1.$$

The proper-gcd event therefore has probability at least

$$G_{1/128}(1) = \frac{260144641}{549569025} > 0.473 > 3/10.$$

No favorable two-primary pattern is needed for this branch.

If $d_p = d_q = r$, then $r < \sqrt{N}/2$, so $m^2/r > 2$. A useful full collision has probability at least

$$G_{1/128}(2) - \frac{1}{128}.$$

By Proposition 11.3, conditional on any fixed b, the favorable two-primary pattern of a still has probability $S_N \ge 1/2$. The exponent seed is independent of it. Thus in this case the conditional probability of obtaining a factor is at least

$$\frac{1}{2} \left(G_{1/128}(2) - \frac{1}{128} \right) = \frac{32885268095}{105790963968} > 0.3108 > 3/10.$$

Both alternatives have the asserted lower bound for each fixed b, with the second averaged only over the independent two-primary part. Averaging over b and reinstating nonunit bases proves the theorem. Independent trials multiply failure probabilities, and $(7/10)^{13} < 0.01$. □

Remark 14.5 (What improves, and what is classical). The guarantee replaces the earlier uniform target of $\Theta(\sqrt{r})$ samples for equality modulo N by $2\lceil N^{1/4} \rceil$ samples for the combined detector. When r is comparable to N, the change is substantial. It is *not* a new general complexity exponent for factoring: Pollard-type searches and the Pollard–Strassen tradition already exploit factors of size about $\sqrt{N}$ [3, 4, 13]. The result here is a self-contained finite guarantee for this particular four-wise dyadic sampler and two-branch detector. Priority and patent novelty of that combination have not been established.

14.4 Examples explaining both branches

For $N = 77$ and $b = 4$, the local orders are $\mathrm{ord}_7(4) = 3$ and $\mathrm{ord}_{11}(4) = 5$, so $r = 15$. The endpoints for $E = 0$ and $E' = 3$ are 1 and 64. They are different modulo 77, but

$$\gcd(64 - 1, 77) = 7.$$

Thus no full-group collision is necessary. In the opposite case, $N = 91$ and $b = 9$ give local orders three at both primes. Here all local collisions are full collisions, so the relation branch matters. For the original base $a = 3$, oddification with $n = 7$ gives $b = 3^{128} \equiv 9 \pmod{91}$. A relation $D = 3$ gives $a^3 = 27$, whose square is one modulo 91, and $\gcd(27 - 1, 91) = 13$. Both local orders of b are three, but the local orders of a have different two-primary parts.

14.5 Limited memory and unknown least-factor size

If a hardware budget permits $2h$ samples rather than $2m$, the same proof applies with $\mu = h^2\theta$ and the exact finite bounds above. There is no abrupt size limit: smaller batches reduce success probability. For differing local orders, at small h^2/d the lower bound is of order h^2/d; independent discarded batches then consume order d/h samples in expectation within this bound. The cost of recomputing the product tree and exponent tables must be included.

One may try $h = 2, 4, 8, \ldots, \lceil N^{1/4} \rceil$ before repeating at the final size. This is a practical schedule that requires no order oracle. It should not be described as a uniform constant success bound at every small size. The theorem certifies the final budget; earlier successes are additional savings. Trial division, prime-power preprocessing, ECM and other conventional stages may be placed before this pipeline without affecting its conditional proof.

15 An exact product-tree detector over a composite residue ring

15.1 From all pairwise gcds to one derivative evaluation problem

The sample bound of Theorem 14.4 would be of limited value if detection required all $\Theta(m^2)$ pairwise gcds. We now specify a detector based on standard polynomial product and remainder trees. Its operations are valid over $R = \mathbb{Z}/N\mathbb{Z}$; a field is not assumed. Product/remainder trees are classical computational tools, including in factorization [13, 14]. The role of the derivative below is to encode all pair differences while removing each self-pair.

First group the records by their full numeric endpoint X. Within a block of equal endpoints, compare each exponent with the first one. If any differs, pass its difference to the exact relation extractor. If this returns a factor, stop. Otherwise continue: an unfavorable base for *full relations* may still supply a proper partial congruence. Retain one representative of every distinct endpoint, obtaining

$$z_1, \ldots, z_s \in \{0, \ldots, N - 1\}, \qquad z_i \neq z_j \ (i \neq j).$$

The original two groups used in the probability proof no longer matter. Deduplication preserves every possible proper gcd between different numeric endpoints.

Set

$$F(X) = \prod_{j=1}^{s} (X - z_j) \in R[X].$$

The formal derivative is well defined over any commutative ring, and

$$F'(z_i) = \prod_{j \neq i} (z_i - z_j) \pmod{N}. \tag{15.1}$$

Indeed, differentiate the product before substituting $X = z_i$: all terms except the ith contain a zero factor. No division by $F(z_i)$ occurs.

Proposition 15.1 (Proper-factor detection)**.** *For $N = pq$ with distinct primes, if some pair of distinct numeric keys has a proper gcd with N, then some $g_i = \gcd(F'(z_i), N)$ is greater than one. If $1 < g_i < N$ it is already a factor. If $g_i = N$, a balanced subproduct tree deterministically recovers a proper factor, using $O(s)$ additional ring operations and $O(\log s)$ gcd calls for this one selected index. Thus the detector is complete for partial congruences as required by Theorem 14.4.*

Proof. A prime dividing $z_i - z_j$ divides (15.1), so its gcd with N is nonunit. Consider a selected index with $g_i = N$. The root polynomial contains the self-factor $X - z_i$, so evaluating the root at z_i would be uninformative. Follow the root-to-leaf path of that self-factor. The sibling subtrees on this path form a disjoint partition of all other factors. Their evaluations at z_i multiply to $F'(z_i)$.

Evaluate those sibling polynomials and take their gcds with N. If every gcd were one, the product would be a unit, a contradiction. A proper gcd finishes the computation. Otherwise choose a sibling subtree with gcd N. At its next split, the two child evaluations multiply to the parent evaluation. Test the left child. If its gcd is proper, stop; if it is N, descend left; if it is one, the right child must have gcd N, so descend right. Continue in this way.

A leaf in this subtree is $X - z_j$ with $j \neq i$. Since the numeric representatives are distinct and lie in $[0, N)$, its value $z_i - z_j$ is a nonzero integer of absolute value less than N. It cannot have gcd N. Thus the process must encounter a proper gcd before reaching a saturated leaf. This proves completeness.

The sibling degrees sum to $s - 1$. Horner evaluation therefore takes $O(s)$ ring operations for all siblings. Along the later balanced descent, the evaluated degrees form a geometrically decreasing sequence, again of total $O(s)$. There are $O(\log s)$ siblings and descent levels, so the gcd count has the stated bound. □

Example 15.2 (Saturation is information, not automatic failure)**.** For $N = 15$ and distinct keys $1, 4, 6$, the derivative at the first key is

$$(1 - 4)(1 - 6) = 15 \equiv 0 \pmod{15}.$$

The aggregate gcd is 15, although $\gcd(1 - 4, 15) = 3$ and $\gcd(1 - 6, 15) = 5$. Splitting the product recovers a proper divisor. Discarding every aggregate gcd equal to N would lose this batch.

15.2 Why no modular inverse is needed

A product tree has leaves $X - z_i$; every internal node is the product of its children. All these polynomials are monic. To evaluate F', reduce it modulo the root polynomial, then recursively reduce each remainder modulo the child polynomials. At leaf $X - z_i$, the remainder is the constant $F'(z_i)$.

Lemma 15.3 (Monic division over an arbitrary commutative ring)**.** *If $g \in R[X]$ is monic and $f \in R[X]$, there are unique $q, h \in R[X]$ with $f = qg + h$ and $\deg h < \deg g$. Long division uses only ring addition and multiplication. Fast division can be obtained by reversing g and inverting the resulting power series, whose constant coefficient is one.*

Proof. Subtract the leading coefficient of the current dividend times an appropriate power of X times g. Since the leading coefficient of g is one, the degree decreases without dividing by a possibly noninvertible element. This yields existence. If two divisions existed, $(q - q')g = h' - h$; the left has degree at least $\deg g$ when $q \neq q'$ because g is monic, whereas the right does not. Thus both differences vanish. For fast division, a series g^* with constant term one has a unique inverse obtained recursively; Newton doubling $u \leftarrow u(2 - g^*u)$ increases the precision using ring operations only. It is valid in all characteristics, including over composite residue rings. □

Repeated endpoint roots do not make monic division invalid, but we remove them first to avoid trivially zero derivative products and to obtain the saturation-splitting guarantee. Interpolation, which might require inverting differences between evaluation points, is unnecessary.

Theorem 15.4 (Arithmetic and storage ledger)**.** *Let $\mathsf{M}_R(s)$ bound multiplication of polynomials of degree less than s over R, with $\mathsf{M}_R(s)/s$ nondecreasing and the usual constant-factor doubling regularity. Once exact endpoints are available, the detector requires*

$$O(\mathsf{M}_R(s) \log s)$$

ring operations, $O(s)$ gcd calls and $O(s \log s)$ residue coefficients in a straightforward stored-tree implementation, in addition to sorting and the relation checks. Recovering a factor from a selected saturated evaluation adds only the cost stated in Proposition 15.1.

Proof. At any product-tree level, the sum of degrees is s. The assumptions on M_R bound the work at that level by $O(\mathsf{M}_R(s))$. There are $O(\log s)$ levels. Differentiation takes $O(s)$ ring operations. The same argument applies to fast monic remainders in the evaluation tree. One gcd is taken for each leaf, and one saturated index suffices if no proper gcd has already been found. Keeping all levels uses $O(s \log s)$ coefficients. Sorting $2m$ word records and regenerating tagged exponents are additional, separately charged tasks. □

For $s = O(N^{1/4})$, fast polynomial arithmetic makes the detector quasilinear in the number of records, up to the integer-arithmetic and logarithmic factors. With schoolbook polynomials it can instead cost $O(s^2)$ ring operations. The supplied Python reference intentionally uses the transparent schoolbook version, so its timings do not establish the fast bound. A production backend must explicitly implement and benchmark fast multiplication and division; an FFT routine over a field cannot simply be applied modulo an arbitrary composite N without explaining its ring arithmetic or exact reconstruction.

15.3 An explicit exact backend through integer packing

The fast detector need not assume a convenient root of unity in $\mathbb{Z}/N\mathbb{Z}$. There is a direct reduction to ordinary integer multiplication. Take two degree-less-than-s polynomials with coefficients represented in $\{0, \ldots, N-1\}$. Choose a power of two B_0 with

$$B_0 > s(N-1)^2,$$

and encode them as nonnegative integers

$$A = \sum_{i=0}^{s-1} a_i B_0^i, \qquad B = \sum_{j=0}^{s-1} b_j B_0^j.$$

Every coefficient $c_k = \sum_{i+j=k} a_i b_j$ of the unreduced product is less than B_0. Consequently no carry crosses a coefficient slot: the base-B_0 digits of AB are exactly the c_k. Reducing those digits modulo N produces the desired polynomial product over R.

A slot width of $2n+\lceil \log_2 s \rceil + 1$ bits is sufficient. The packed integers therefore have $O(s(n+\log s))$ bits. If $\mathsf{M}_{\mathrm{int}}(u)$ denotes exact integer-multiplication bit cost, a polynomial product is reduced to

$$O(\mathsf{M}_{\mathrm{int}}(s(n+\log s)) + s\,\mathsf{R}(n+\log s, n))$$

bit operations, where $\mathsf{R}(u,n)$ is the cost of reducing a u-bit coefficient modulo an n-bit modulus. The actual unreduced coefficient width is $O(n+\log s)$ up to a constant factor; that is why it appears in the reduction term. Extraction of binary slots is linear in the packed length. Fast exact integer algorithms make this quasilinear up to logarithmic factors, with all word widths accounted for. Schoolbook integer multiplication does not have that property.

The same construction can serve as a test oracle for a hardware polynomial multiplier. An NTT/CRT implementation is another option, but its auxiliary moduli must support the transform length and their product must exceed the coefficient bound before exact reconstruction. Numerical floating-point convolution without a proved rounding bound is not an exact substitute. Each tree node should reduce coefficients back to $[0, N)$ so that the next multiplication has the same certified coefficient bound.

For an FPGA study, the first meaningful comparison is thus between three fully counted backends: schoolbook residue arithmetic; a packed integer implementation with an exact host backend; and a dedicated transform/reconstruction engine. Repacking, memory layout, transform parameters and coefficient reduction belong in the benchmark. This specifies a concrete route toward fast detection, while leaving the choice to measured arithmetic and bandwidth costs.

15.4 Complete algorithm and stopping semantics

Input: odd semiprime N and independent random seed. **Output:** verified proper factor or RETRY.

1. Set $n = \lceil \log_2 N \rceil$, $m = \lceil N^{1/4} \rceil$, $K = n+7$. Choose a uniformly in $\{1, \ldots, N-1\}$; compute $d = \gcd(a, N)$. If $1 < d < N$, return d.
2. Compute $b = a^{2^n} \bmod N$ by n squarings. Generate $2m$ four-wise independent K-bit exponents and exact endpoints $X_i = b^{E_i}$.
3. Group equal numeric keys. Test a nonzero relation if present and return its factor if proper. Otherwise retain distinct keys $z_1, \ldots, z_s$. If $s < 2$, return RETRY.
4. Build $F(X) = \prod_i (X - z_i)$ and evaluate $F'(z_i)$ using monic remainder trees. Compute $g_i = \gcd(F'(z_i), N)$.
5. Return any proper g_i. If some $g_i = N$, split its sibling products using Proposition 15.1 and return the recovered factor.
6. If every gcd is one and no relation succeeded, return RETRY.

Every released output is independently checked to divide N properly.

Algorithm 2: Divisor-sensitive dyadic batch

The relation stage need only process one useful equality: once it establishes that the two-primary pattern is unfavorable, other full relations cannot help, although the partial-gcd stage must still run. The implementation may check additional equalities for diagnostics; this does not improve that branch's mathematical success probability. Every externally reported divisor passes $1 < d < N$ and $N \bmod d = 0$. The entire computation is Las Vegas in its answer semantics: a retry or timeout is possible, but an unverified answer is never returned. The geometric running-time guarantee requires the ideal independence and complete-detector hypotheses already stated.

15.5 Hardware implications

A hash rendezvous network sends equal *numeric* keys together. It cannot route unknown congruence classes modulo p merely by changing the hash function. The new detector therefore replaces that part of the pipeline with polynomial arithmetic, coefficient memory and a gcd engine. A balanced tree gives a hierarchy of arithmetic tasks suitable for banked memory and tiled processing. It does not provide unlimited memory bandwidth or remove the need for exact multipliers.

For a modulus of roughly 40 decimal digits, the conservative theorem uses about 2×10^{10} records. Even at only 133 residue bits per record, that is about 332.5 GB before tags, polynomial coefficients and buffering. The quarter-power theorem is a mathematical improvement to the previous sampling scale, not a declaration that a 40-digit factorization machine is economical. Smaller experimental instances are the immediate target; comparison with optimized digital factoring remains mandatory.

16 Shared exponent tables and a stronger constant-register receiver

16.1 Fixed-base windows: reducing repeated modular work

All endpoints in a batch use the same base b. Computing each with an independent binary product misses reusable work. Fixed-base window exponentiation is a standard arithmetic optimization [Me96]; here its costs can be matched explicitly to the sampling budget.

Write a K-bit exponent in windows of width $w \geq 1$:

$$E = \sum_{j=0}^{h-1} e_j 2^{jw}, \qquad h = \lceil K/w \rceil, \qquad 0 \leq e_j < 2^{w_j},$$

where $w_j = \min(w, K - jw)$. Precompute

$$H_j(d) = b^{d2^{jw}} \bmod N, \quad 0 \leq d < 2^{w_j}.$$

Then $b^E = \prod_j H_j(e_j)$. The distribution of E is unchanged: this is an exact arithmetic reorganization, valid for independent, four-wise independent, or even deterministic exponents.

Proposition 16.1 (Explicit work for a batch)**.** *For T exponents, all endpoints can be generated with at most*

$$\sum_{j=0}^{h-1} (2^{w_j} - 2) + (h-1) + Th \leq h(2^w + T)$$

modular products, after b is supplied. The tables contain $\sum_j 2^{w_j} \leq h2^w$ residues. In a sequential generator, an endpoint uses at most h products; a balanced multiplication tree gives depth $O(\log h)$ if enough multipliers and table ports are available.

Proof. At a window base $c = b^{2^{jw}}$, store $H_j(0) = 1$ and $H_j(1) = c$. Successive multiplication by c fills the remaining table using $2^{w_j} - 2$ products. If a further window is needed, $H_j(2^{w_j} - 1)c = c^{2^{w_j}}$ gives its base in one more product. The T outputs require at most h products each, even if multiplication by one is not omitted. Sum these counts. The product tree for one endpoint has logarithmic depth in its number of factors. □

Taking $w = \lfloor \log_2 T \rfloor$ for $T \geq 2$ yields $O(T\lceil K/\log_2 T \rceil)$ modular work and the same order of table entries. In Theorem 14.4, $K = \Theta(\log N)$ and $T = \Theta(N^{1/4})$, so this is $O(T)$ modular products

and $O(T)$ table entries asymptotically. The four-wise finite-field generator still costs three field products per record and its field setup; that work has not vanished.

Large tables may destroy the locality or energy benefits of a smaller arithmetic engine. The FPGA should sweep $w = 2, 4, 6, 8$ rather than choose $w = \lfloor \log_2 T \rfloor$ blindly. The end-to-end objective includes table fill, memory reads, fanout and reuse across batches. Tables may be reused for the same b; changing the base requires new tables. Their construction cannot be hidden in a benchmark's untimed setup when the reported task includes random-base selection.

16.2 One stored record can capture a useful collision more often

The isolated-pair bound in Section 19 required exactly two packets at a receiver. That condition is stronger than necessary. A receiver succeeds if it receives only one distinct key, possibly many times, and at least two of its exponent tags differ.

Theorem 16.2 (Distinct-key isolation, with an explicit capture factor)**.** *Fix any T endpoint–exponent records, with no independence assumption on them. Suppose some key occurs with two different exponents. Choose an independent pairwise-universal hash into H receivers, so distinct keys collide with probability at most $1/H$. Deliver every record. Each receiver retains its first record and compares every later arrival with it, checking differing exponents when the keys agree. If the list contains s distinct keys, the probability that this receiver bank reports at least one useful relation is at least*

$$\max\{0, 1 - (s-1)/H\}.$$

For a random record list whose useful-collision probability is p_T,

$$\mathbb{P}(\textit{capture}) \geq \max\{0, 1 - (T-1)/H\}\, p_T. \tag{16.1}$$

In particular $H = 4T$ gives at least $3p_T/4$.

Proof. Conditional on the whole record list, select a key x_* with two different exponent values by a deterministic rule. For each of the other $s-1$ distinct keys, the chance of sharing its receiver is at most $1/H$. The union bound implies that no other key reaches $h(x_*)$ with probability at least $1 - (s-1)/H$.

On that event, the receiver's first record has key x_*. Because there are at least two exponent values, some later record differs from the first in its exponent, irrespective of arrival order. The receiver then reports a useful relation. Additional copies of the same key do not hurt. Average over record lists having a useful collision, and use $s \leq T$, to obtain (16.1). □

This strengthens the *capture* component of the previous isolated pair analysis and removes its need for a lane near $\sqrt{r}$ when all packets can be delivered. It does not strengthen the congestion theorem: a very frequent key can still create a long queue at its destination. The old multiscale result remains a separate, conditional latency statement. It is incorrect to combine the new capture probability with old logarithmic latency without checking their joint event.

A time-multiplexed banked SRAM design provides a direct implementation: read the target slot, keep the first key and tag, compare subsequent records and overwrite only at batch reset. With lossless serialization it needs H resident records and a small arbitration queue. It uses $\Theta(H(n + \log_2 T))$ bits when exponent tags can be regenerated; "one register" means one full record at each receiver, not one bit. Under a deadline, a separate delivery-failure bound δ gives only $\mathbb{P}(\text{capture by deadline}) \geq (1 - (T-1)/H)p_T - \delta$.

16.3 Imperfect randomness and exact certification

Proposition 16.3 (Robustness at the seed interface)**.** *Suppose the joint physical law of the base-selection randomness, exponent seed and any independent hash seed has total variation distance at most ε from the prescribed ideal law. For any deterministic post-processing pipeline, its success probability changes by at most ε. In particular, the lower bound in Theorem 14.4 becomes $3/10 - \varepsilon$ if all arithmetic and detection remain exact.*

Proof. The event of success is a subset of the seed space. Total variation is the supremum of the probability difference over all such subsets. Equivalently, deterministic maps cannot increase total variation. □

An unbiased histogram of each pbit does not certify this joint-law condition. Perfectly correlated fair bits give a simple counterexample: each marginal is fair but the seed occupies only two words. Likewise, passing a finite statistical test does not prove exact four-wise independence at arbitrary word lengths. A measured entropy source, an appropriate extraction argument, and an explicit dependence model are needed to turn physical sampling into a quantitative theorem.

For a prescribed sequence of J stochastic kernels, if each implemented conditional kernel differs from its ideal one by at most ϵ_j in total variation *uniformly over histories*, sequential coupling bounds the joint trajectory error by $\sum_j \epsilon_j$. Couple each transition while previous states agree; the probability of a first mismatch is at most this sum. Marginal accuracy averaged over an unrelated training distribution does not establish the uniform premise. This is the interface-level error ledger relevant to compiled thermodynamic programs.

Part IV
Order cutoffs and communication networks

17 Order cutoffs, collision probabilities, and hardware tradeoffs

Throughout this section let

$$N = pq, \qquad p < q,$$

where p and q are distinct odd primes. For

$$a \in (\mathbb{Z}/N\mathbb{Z})^\times,$$

write

$$r = \mathrm{ord}_N(a).$$

By the Chinese remainder theorem,

$$(\mathbb{Z}/N\mathbb{Z})^\times \simeq \mathbb{F}_p^\times \times \mathbb{F}_q^\times.$$

If

$$a \longleftrightarrow (x, y),$$

we write

$$d = \mathrm{ord}_p(x), \qquad e = \mathrm{ord}_q(y).$$

Then

$$r = \mathrm{lcm}(d, e).$$

We shall call a *favorable* when the usual order-finding factorization step produces a nontrivial square root of 1 modulo N. In terms of the component orders, this is equivalent to

$$v_2(d) \neq v_2(e),$$

where v_2 denotes the 2-adic valuation, with $v_2(1) = 0$.

17.1 Favorable orders below a prescribed cutoff

We first quantify the probability of obtaining a favorable element whose order does not exceed a prescribed bound.

Theorem 17.1 (A cutoff estimate for favorable orders)**.** *For $B \geq 1$, let*

$$\pi_B(p, q) = \mathbb{P}(\mathrm{ord}_N(a) \leq B \text{ and } a \text{ is favorable}),$$

where a is chosen uniformly from $(\mathbb{Z}/N\mathbb{Z})^\times$. Then

$$\boxed{\pi_B(p, q) = \frac{1}{(p-1)(q-1)} \sum_{\substack{d \mid p-1,\ e \mid q-1 \\ \mathrm{lcm}(d,e) \leq B \\ v_2(d) \neq v_2(e)}} \varphi(d)\varphi(e).}$$

In particular, for

$$B = \sqrt{N},$$

one has the uniform lower bound

$$\boxed{\pi_{\sqrt{N}}(p,q) \geq \frac{1}{q-1}.}$$

Consequently, after k independent choices of the base a,

$$\mathbb{P}(\textit{success in at least one trial}) = 1 - \left(1 - \pi_{\sqrt{N}}(p,q)\right)^k$$

and hence

$$\boxed{\mathbb{P}(\textit{success}) \geq 1 - \exp\left(-\frac{k}{q-1}\right).}$$

In particular,

$$\boxed{k \geq (q-1)\log 100}$$

is sufficient to guarantee success probability at least 0.99*.*

Proof. Since $\mathbb{F}_p^\times$ and $\mathbb{F}_q^\times$ are cyclic, the number of elements of exact order d in $\mathbb{F}_p^\times$ is $\varphi(d)$, and similarly the number of elements of exact order e in $\mathbb{F}_q^\times$ is $\varphi(e)$. Hence the number of elements a with component orders (d, e) equals

$$\varphi(d)\varphi(e).$$

Moreover,

$$\mathrm{ord}_N(a) = \mathrm{lcm}(d, e),$$

and the factorization step is favorable precisely when

$$v_2(d) \neq v_2(e).$$

This proves the exact formula for $\pi_B(p, q)$.

We now take

$$B = \sqrt{N}.$$

For every $x \in \mathbb{F}_p^\times$, let

$$d = \mathrm{ord}_p(x).$$

Choose $y \in \mathbb{F}_q^\times$ by

$$y = \begin{cases} 1, & d \text{ even}, \\ -1, & d \text{ odd}. \end{cases}$$

If d is even, then

$$e = \mathrm{ord}_q(y) = 1,$$

so

$$v_2(d) \neq v_2(e),$$

and

$$\mathrm{ord}_N(a) = d \leq p - 1 < \sqrt{pq}.$$

If d is odd, then

$$e = 2.$$

Since $d \mid p - 1$ and d is odd,

$$2d \mid p - 1.$$

Therefore

$$\mathrm{ord}_N(a) = \mathrm{lcm}(d, 2) = 2d \leq p - 1 < \sqrt{N},$$

and again

$$v_2(d) \neq v_2(e).$$

Thus every $x \in \mathbb{F}_p^\times$ determines a distinct favorable element whose order is smaller than $\sqrt{N}$. Hence there are at least $p-1$ such elements. Since

$$\varphi(N) = (p-1)(q-1),$$

we obtain

$$\pi_{\sqrt{N}}(p,q) \geq \frac{p-1}{(p-1)(q-1)} = \frac{1}{q-1}.$$

Finally,

$$\mathbb{P}(\text{failure after } k \text{ trials}) = (1-\pi_{\sqrt{N}})^k \leq e^{-k\pi_{\sqrt{N}}} \leq \exp\left(-\frac{k}{q-1}\right),$$

which proves the result. □

Remark 17.2. The exact formula applies to any cutoff B. In particular, one may take

$$B = N^\alpha, \qquad 0 < \alpha \leq 1,$$

or more generally any prescribed function $B = B(N)$. The choice $B = \sqrt{N}$ is distinguished by the fact that it gives a simple uniform lower bound without requiring additional information on the arithmetic structure of $p-1$ and $q-1$.

17.2 Hardware cost of the collision search

We now relate the order cutoff to the number of physical cells required by a collision-based implementation.

The following calculation is conditional on the standard *birthday-mixing hypothesis*: once the diffusion process has mixed sufficiently, newly generated endpoints are modeled as approximately independent uniform samples from the cyclic group $\langle a \rangle$ of order r.

Let T_r denote the number of samples required until the first repeated endpoint appears. Then, for $t \leq r$,

$$\mathbb{P}(T_r > t) = \frac{r(r-1)\cdots(r-t+1)}{r^t} = \prod_{j=0}^{t-1}\left(1-\frac{j}{r}\right).$$

Using

$$1 - x \leq e^{-x},$$

we obtain

$$\boxed{\mathbb{P}(T_r \leq t) \geq 1 - \exp\left(-\frac{t(t-1)}{2r}\right).}$$

The classical birthday calculation gives

$$\mathbb{E}[T_r] = \sqrt{\frac{\pi r}{2}} + \frac{2}{3} + O(r^{-1/2}).$$

Since the final sample is the one that repeats an already constructed vertex, the number V_r of distinct vertices constructed before the first collision is

$$V_r = T_r - 1.$$

Consequently,

$$\boxed{\mathbb{E}[V_r] = \sqrt{\frac{\pi r}{2}} - \frac{1}{3} + O(r^{-1/2})}$$

and therefore

$$\boxed{\mathbb{E}[V_r] \sim \sqrt{\frac{\pi}{2}}\sqrt{r} \approx 1.2533\sqrt{r}.}$$

Thus the characteristic hardware cost of reaching a first collision is of order

$$\sqrt{r},$$

rather than r.

If only elements satisfying

$$r \le B$$

are considered, then, at the upper end $r \approx B$,

$$\boxed{\mathbb{E}[V] \lesssim \sqrt{\frac{\pi B}{2}} \approx 1.2533\sqrt{B}.}$$

In particular, for the square-root cutoff

$$B = \sqrt{N},$$

we obtain

$$\boxed{\mathbb{E}[V] \lesssim \sqrt{\frac{\pi}{2}}\, N^{1/4}.}$$

Within this eligible-order and iid-endpoint model, the number of stored records scales from $\sqrt{N}$ to

$$\boxed{N^{1/4}}.$$

One may also ask for a prescribed collision probability. To obtain a collision with probability at least $1-\varepsilon$, it is sufficient that

$$\exp\left(-\frac{t(t-1)}{2r}\right) \le \varepsilon.$$

Equivalently,

$$t(t-1) \ge 2r\log\frac{1}{\varepsilon}.$$

Thus

$$\boxed{t \ge \frac{1+\sqrt{1+8r\log(1/\varepsilon)}}{2}.}$$

Asymptotically,

$$t = \sqrt{2r\log\frac{1}{\varepsilon}} + O(1).$$

For $r \le B$, this gives the characteristic scale

$$\boxed{V_\varepsilon \lesssim \sqrt{2B\log\frac{1}{\varepsilon}}.}$$

For example,

collision probability	$V_\varepsilon/\sqrt{B}$
0.90	2.146…
0.95	2.448…
0.99	3.035…

For $B = \sqrt{N}$,

$$\boxed{V_\varepsilon \lesssim \sqrt{2\log(1/\varepsilon)}\, N^{1/4}.}$$

A single collision need not determine the exact order. In general, the collision relation may produce only a multiple of r, and several independent relations may have to be combined, for example by taking greatest common divisors. For any fixed number of useful collision relations, however, this changes only the multiplicative constant and not the fundamental scale

$$\boxed{O(\sqrt{r})}.$$

17.3 The space–probability tradeoff

The reduction in hardware obtained by imposing an order cutoff has an unavoidable probabilistic cost.

Theorem 17.3 (Space–probability tradeoff)**.** *Let*

$$m = p - 1, \qquad n = q - 1, \qquad g = \gcd(m, n),$$

and let $\tau(s)$ denote the number of positive divisors of s. Then

$$\boxed{\pi_B(p,q) \le \min\left\{1,\ \frac{B\,\gcd(p-1,q-1)\,\tau(p-1)\tau(q-1)}{(p-1)(q-1)}\right\}.}$$

Consequently, if the collision search is restricted to elements of order at most B, then the hardware scale per trial is

$$O(\sqrt{B}),$$

whereas the expected number of independent choices of a before obtaining a favorable element satisfying the cutoff is at least

$$\boxed{\frac{(p-1)(q-1)}{B\,\gcd(p-1,q-1)\,\tau(p-1)\tau(q-1)},}$$

whenever the latter quantity exceeds 1.

Proof. From the exact formula,

$$\pi_B(p,q) = \frac{1}{(p-1)(q-1)} \sum_{\substack{d \mid p-1,\ e \mid q-1 \\ \operatorname{lcm}(d,e) \le B \\ v_2(d) \ne v_2(e)}} \varphi(d)\varphi(e).$$

For every pair occurring in the sum,

$$de = \operatorname{lcm}(d,e)\gcd(d,e).$$

Since

$$\operatorname{lcm}(d,e) \le B$$

and

$$\gcd(d,e) \mid \gcd(p-1,q-1),$$

we have

$$de \le B\,\gcd(p-1,q-1).$$

Moreover,

$$\varphi(d)\varphi(e) \le de.$$

Hence each summand satisfies

$$\varphi(d)\varphi(e) \le B \gcd(p-1, q-1).$$

There are at most

$$\tau(p-1)\tau(q-1)$$

possible pairs (d, e). Therefore

$$\pi_B(p,q) \le \frac{B \gcd(p-1,q-1)\,\tau(p-1)\tau(q-1)}{(p-1)(q-1)}.$$

The remaining assertions follow from the birthday estimate $O(\sqrt{B})$ and from taking the reciprocal of the one-trial success probability. □

Corollary 17.4 (Balanced semiprimes)**.** *Suppose that*

$$p \asymp q \asymp X,$$

so that

$$N = pq \asymp X^2,$$

and assume

$$\gcd(p-1, q-1) = X^{o(1)}.$$

If

$$B = N^\alpha, \qquad 0 < \alpha \le 1,$$

then

$$\boxed{\pi_B(p,q) \le N^{\alpha-1+o(1)}.}$$

Consequently,

$$\boxed{\textit{hardware scale per trial} = O(N^{\alpha/2}),}$$

while

$$\boxed{\textit{expected number of trials} \ge N^{1-\alpha-o(1)}.}$$

Proof. The classical divisor bound

$$\tau(s) = s^{o(1)}$$

gives

$$\tau(p-1)\tau(q-1) = X^{o(1)}.$$

Together with

$$\gcd(p-1, q-1) = X^{o(1)}$$

and

$$(p-1)(q-1) = N^{1+o(1)},$$

Theorem 17.3 gives

$$\pi_B(p,q) \le \frac{B}{N} N^{o(1)}.$$

For $B = N^\alpha$,

$$\pi_B(p,q) \le N^{\alpha-1+o(1)}.$$

The hardware estimate follows from

$$\sqrt{B} = N^{\alpha/2}.$$

□

Corollary 17.5 (Sharp exponent for the square-root cutoff)**.** *Under the hypotheses of the preceding corollary,*

$$\boxed{\pi_{\sqrt{N}}(p,q) = N^{-1/2+o(1)}.}$$

Consequently, the expected number of independent trials is

$$\boxed{N^{1/2+o(1)},}$$

while the hardware scale per trial is

$$\boxed{N^{1/4}.}$$

Proof. The upper bound follows from the preceding corollary with $\alpha = 1/2$:

$$\pi_{\sqrt{N}}(p,q) \le N^{-1/2+o(1)}.$$

On the other hand, Theorem 17.1 gives

$$\pi_{\sqrt{N}}(p,q) \ge \frac{1}{q-1}.$$

Since

$$q \asymp N^{1/2},$$

we have

$$\frac{1}{q-1} = N^{-1/2+o(1)}.$$

Combining the two bounds proves the result. □

Remark 17.6. This gives a simple asymptotic space–probability law. If

$$B = N^{\alpha},$$

then lowering α reduces the hardware requirement according to

$$N^{\alpha/2},$$

but simultaneously reduces the admissible favorable-order probability to at most

$$N^{\alpha-1+o(1)}.$$

Thus an aggressive reduction in physical hardware inevitably incurs a corresponding increase in the number of independent trials.

17.4 Interpretation in the revised architecture

These cutoff theorems remain correct statements about the full order of a uniform base. They are not the mechanism behind Theorem 14.4: that theorem uses partial congruences modulo an unknown factor, and therefore does not condition on the rare event $\mathrm{ord}_N(a) \le \sqrt{N}$. Its constant success probability is proved separately. A count of samples or records is never a count of transistors, pbits or complete arithmetic processors.

The historical billion-cell and one-nanosecond-trial projections have been removed from the active engineering case. They combined an ideal birthday model with an unspecified trial duration and a cell definition that did not include arithmetic, labels or interconnect. Their original text remains in the source archive for provenance; it supplies no current manufacturing or timing claim.

17.5 Logarithmic-depth collision detection by programmed connections

The statistical realization of the dyadic kernel produces records $R_i = (X_i, i)$, where $X_i = b^{E_i} \bmod N$ and i is an identifier from which the exponent E_i can be reconstructed using the common four-wise-independent seed. A collision is *useful* if $X_i = X_j$ and $E_i \neq E_j$.

Theorem 17.7 (Programmable logarithmic-depth collision detection)**.** *Suppose C synchronous processing cells each store one $O(\log N + \log C)$-bit record, and in a single abstract round each cell can take part in at most one exact compare–exchange with an assigned partner. Assume the pairing can be prescribed separately at each round, including nonlocal pairings. Then all repeated endpoint values among $X_1, \dots, X_C$ can be detected in $O(\log C)$ parallel rounds, with $O(C \log C)$ total compare–exchange operations, $O(C(\log N + \log C))$ storage bits, and no table of pairwise comparisons.*

Proof. Use an Ajtai–Komlós–Szemerédi sorting network on the keys X_i, carrying the identifiers along with each key. A sorting network has a fixed, data-independent schedule of disjoint compare–exchange pairs at each level; its depth is $O(\log C)$ and its size is $O(C \log C)$. After sorting, all equal keys occupy consecutive positions. Two adjacent-comparison passes (even–odd and odd–even), or equivalently simultaneous equality comparisons on adjacent output wires, identify all adjacent equalities. If a block of equal keys contains different exponents, at least one of its adjacent pairs has distinct exponents. Regenerate the two exponents from their identifiers and verify $b^{|E_i - E_j|} = 1 \pmod N$. The records have the stated width, and at each level there are at most $C/2$ compare–exchanges. □

Proposition 17.8 (Explicit hypercube schedule)**.** *For $C = 2^d$, a bitonic compare–exchange network detects repeated endpoints in $d(d+1)/2 + O(1)$ rounds using only the programmed partner relations*

$$i \longleftrightarrow i \oplus 2^j, \qquad 0 \le j < d.$$

It requires $dC/2$ undirected hypercube links if every possible partner connection is provided physically, and $Cd(d+1)/4$ compare–exchange operations per batch.

Proof. The standard bitonic sorting construction performs phases indexed by $k = 1, \dots, d$, with comparison subphases $j = k-1, k-2, \dots, 0$. At subphase j, the comparator for processor i uses partner $i \oplus 2^j$, with its min/max orientation fixed by the phase and processor index. Induction on the bitonic merge shows that the final sequence is sorted. The total number of subphases is $\sum_{k=1}^{d} k = d(d+1)/2$, and each employs $C/2$ comparators. For each coordinate j, there are $C/2$ undirected edges in the d-dimensional hypercube. □

Proposition 17.9 (Nearest-neighbor mesh obstruction)**.** *Consider a square mesh of $C = m^2$ physical cells, with one record per cell, and suppose each of the m edges crossing the vertical bisection can transport at most κ whole records per direction per round. If independent, distinct input keys have a uniformly random final sorted rank, then any in-place routing-based sorting implementation needs*

$$\mathbb{E}R \ge \frac{\sqrt{C}}{4\kappa}$$

communication rounds. In particular, such a mesh with fixed κ cannot achieve $O(\log C)$ rounds for all inputs.

Proof. Of the $C/2$ records initially placed on the left side, each has probability $1/2$ of ultimately belonging to the right half of the sorted output. Thus $C/4$ records must cross the vertical bisection from left to right in expectation. As the cut transports at most $m\kappa$ such records per round, the expected number of rounds is at least $(C/4)/(m\kappa) = \sqrt{C}/(4\kappa)$. □

Remark 17.10 (Collision probability and numerical scale). The four-wise independent collision bound of Theorem 13.1 remains valid with the less restrictive choice $Q \geq 128C^2$ (instead of $Q \geq 100C^3$), if one retains its explicit finite bound rather than the asymptotic $1/3 - O(C^{-1})$. For $B = C^2$ this bound tends to $1/3 - 1/256 \approx 0.329427$. For an independent uniform unit base and $N = pq \leq 4C^2$, the factor-extraction success probability is therefore at least one-half of that finite collision bound. In the special case $C = 2^{33}$, $Q = 2^{73} = 128C^2$, the bound is approximately 0.164714 per ideal complete batch, assuming all collisions are detected. The explicit bitonic network uses 561 comparison rounds and 141,733,920,768 possible hypercube links. These are resource counts in an abstract network, not feasibility estimates for a planar integrated circuit.

Remark 17.11 (Analog versus tagged-digital realizations). A scalar analog diffusion cell implements a linear average, but cannot preserve the identities of individual trajectories through mixing. The present sorting construction therefore requires tagged stochastic walkers, or digital packet registers in addition to any analog heat-diffusion substrate. The $O(\log C)$ depth counts atomic word comparisons and nonlocal transfers, not the propagation time, energy, circuit area, nor the cost of modular arithmetic.

18 Alternative network architectures and exactness guarantees

18.1 The distinction between routing, coalescence, and sorting

A tagged endpoint is a pair (X_i, i), where a public polynomial seed reconstructs E_i from i. The verifier rejects any purported equality unless it checks both $X_i = X_j$ and $E_i \neq E_j$. Given those equalities, $b^{|E_i - E_j|} \equiv 1 \pmod{N}$ follows without estimating a small analog signal. In an abstract compare–exchange circuit, AKS sorting provides $O(\log C)$ comparison layers [8]; Batcher's explicit bitonic network gives $d(d+1)/2$ layers when $C = 2^d$ [9]. Both use registers of $\Theta(\log N + \log C)$ bits and potentially non-local wires.

18.2 Butterfly paths and why same-key packets meet

Let a packet start with address $u \in \{0,1\}^D$ and a hash destination $y = h(X) \in \{0,1\}^D$. At stage t, replace bit t of the current address by bit t of y. Thus after stage t the first $t+1$ bits agree with y and the rest with u. At the output its address is y. In particular, equal keys have equal destinations, independent of their origins. This fact establishes *rendezvous correctness*, but says nothing by itself about queue growth or how many rounds are required.

Proposition 18.1 (Hash-controlled occupancy of distinct keys)**.** *Let $B = 2^D$ sources enter a leveled butterfly of D stages, one input packet per source. Endpoints may be arbitrarily dependent. Independently choose a $2D$-wise independent uniform hash on* distinct *endpoint values. Let Y_v be the number of distinct keys whose paths go through a fixed node v. Then*

$$\mathbb{P}\left(\max_v Y_v \geq 2D\right) \leq \frac{B(D+1)}{(2D)!}.$$

Proof. For a node at depth t, at most 2^t input positions can reach it. Take one representative for each distinct key among those inputs. Each such representative reaches the node with probability 2^{-t}, and the corresponding indicators are $2D$-wise independent because the hash is independent of the endpoint list. The $2D$th descending factorial moment of their sum is at most $(2^t)^{2D} 2^{-2Dt} = 1$. On the event that the count is at least $2D$, this factorial is at least $(2D)!$. Markov followed by the union bound across $B(D+1)$ nodes proves the assertion. □

Remark 18.2 (The memory qualification). The proposition controls the number of *distinct* keys using a node across an entire packet batch. It does *not* by itself construct an online collision

merger with constant-size queueing state. In particular, if a node must remember an arbitrary set of keys to recognize all later repeats, its state size grows with the set. The isolated-receiver model in Section 19 avoids this problem by requiring only one successful collision whose bucket is not contaminated by another key.

18.3 An elementary information-theoretic memory lower bound

Proposition 18.3 (Exact one-pass duplicate detection)**.** *Fix a universe of U possible keys. Any deterministic one-pass detector that must answer correctly whether the next key belongs to an arbitrary previously seen set of m distinct keys requires at least $\log_2 \binom{U}{m}$ bits of retained state.*

Proof. If two distinct m-element subsets S, T lead to the same state, take $x \in S \setminus T$. On receiving the same next key x, the detector must respond both "previously seen" and "new", which is impossible. Thus all $\binom{U}{m}$ subsets require different states. □

Remark 18.4. This one-pass lower bound applies to a *single* detector receiving an arbitrary stream. It does not disprove bounded-buffer distributed routing protocols with state spread over a network.

18.4 Planar bisection obstruction

In a $\sqrt{C} \times \sqrt{C}$ nearest-neighbor mesh with κ full-word transmissions per directed cut edge per round, a uniformly random target permutation requires $\mathbb{E}R \geq \sqrt{C}/(4\kappa)$ rounds: $C/4$ words cross the vertical bisection left-to-right in expectation, while only $\kappa\sqrt{C}$ cross per round. A multistage butterfly avoids the *graph theoretic* hypothesis of locality, but demands nonlocal physical wiring and buffers.

18.5 What the general packet-routing theorem actually supplies

The theorem of Leighton–Maggs–Ranade–Rao [10] supplies an online randomized scheduler for a *fixed set of paths* of congestion κ and dilation D on a bounded-degree leveled network. Its $O(\kappa + D + \log P)$ routing bound with constant-size queues is an *external theorem*; it is not a finite-latency specification for our illustrative backpressure simulator. The isolated-collision theorem below provides a sufficient intersection of a collision event with a congestion event before invoking that scheduler. All failures are explicitly separated.

19 Multiscale constant-buffer rendezvous: a physically accountable bridge

19.1 Scope and relation to programmed diffusion

Let $N = pq$ with p, q distinct odd primes and let a be a uniform unit of $(\mathbb{Z}/N\mathbb{Z})^\times$. As in the main manuscript, put

$$b = a^{2^M} \pmod{N}, \qquad 2^M > N, \qquad r = \mathrm{ord}_N(b).$$

The programmed kernel introduced in the main manuscript is the autocorrelation of the probability measure associated with independent uniform exponents in $\{0, \ldots, Q-1\}$ for $Q = 2^K$. In this section we do not attempt a physical analog readout of the return mass $h_K(e)$. Instead, tagged exponent samples produce endpoints $X_i = b^{E_i} \pmod{N}$. Two different exponents with the same

endpoint give a nonzero relation divisible by r, which can be fed into the exact factor-extraction theorem in the main manuscript.

We emphasize that a sampling/rendezvous implementation is not an analog RC realization of the B_t operators. Its potentially thermodynamic part is a physical source of random samples; modular exponentiation, packet comparison, and factor verification remain exact digital tasks. The contributions below are explicit positive-probability network bounds, not an unqualified polylogarithmic-time factorization algorithm.

19.2 A uniform multiscale source of relations

We use $C = 2^d$ as an *abstract total sampling budget*; C is not a count of transistors or physical switches. Set aside independent sampling lanes of sizes

$$m_j = 2^j, \qquad 1 \le j \le d-1, \qquad \sum_{j=1}^{d-1} m_j = C - 2.$$

Each lane draws m_j four-wise independent uniform exponent samples from $\{0, \dots, Q-1\}$, where $Q = 2^K \ge 128C^2$ is shared. For example, evaluate a random polynomial of degree at most three at distinct points of $\mathbb{F}_Q$ and identify field elements with K-bit integers. A lane is *appropriate* if it is the smallest of these powers of two for which $m^2 \ge r$. For $r > 4$ this lane satisfies

$$\frac{m^2}{4} < r \le m^2. \tag{19.1}$$

For $r \le 4$, use the lane $m = 2$.

Recall the four-wise birthday estimate from the preceding collision analysis: if $r \le m^2$ and $Q \ge 128m^2$, then

$$\mathbb{P}(\exists i < j : X_i = X_j,\ E_i \ne E_j) \ge \frac{1}{4 + 1/192} - \frac{1}{256} > 0.245. \tag{19.2}$$

The allocation is oblivious to the unknown order. Provided $r \le C^2/4$, at least one lane is appropriate and every lane is executed.

19.3 A one-register receiver rather than an in-network merger

The main new architectural choice is to use $B = 4m$ receivers for a lane with m input packets. A hash function $h : \langle b \rangle \to \{0, \dots, B-1\}$ determines the receiver of each endpoint X_i. At each receiver, the first arriving packet's pair (X_i, i) is retained. Every later incoming pair (X_j, j) is compared with that first pair. If the keys match, the receiver reports (i, j, X_i) to the verifier; otherwise it discards the incoming packet. The verifier reconstructs E_i, E_j from their sampling seed and checks $E_i \ne E_j$ and $b^{|E_i - E_j|} \equiv 1 \pmod N$.

This receiver detects any *isolated useful pair*: indices $i < j$ for which $X_i = X_j$, $E_i \ne E_j$, and no other packet has their hashed receiver address. It may miss collisions at busy receivers. That is allowed: one correct relation suffices, and no claimed relation is accepted without exact modular verification. Each receiver holds only one resident record and one input word (two complete-record slots), regardless of the number of arrivals over time.

Theorem 19.1 (Uniform probability of an isolated useful collision)**.** *Let $m = 2^j \ge 16$, $m^2/4 < r \le m^2$, and $Q \ge 128m^2$. Let $E_1, \dots, E_m$ be four-wise independent and uniform in $\{0, \dots, Q-1\}$, and set $X_i = b^{E_i}$. Choose, independently of these exponents, a pairwise-independent hash function on distinct endpoints, with uniform values in $B = 4m$ receivers. Then*

$$\mathbb{P}(\textit{at least one isolated useful pair}) \ge \frac{1}{8}\left[\frac{127}{256}\frac{15}{16}\frac{49}{100}\right]^2 > 0.0064. \tag{19.3}$$

The bound is independent of m, r, and N.

Proof. Let $Z = \sum_{i<j} \mathbf{1}_{\{X_i = X_j\}}$ count all equal-endpoint pairs, including those caused by equal exponents. Write $A = \binom{m}{2}$, $U = 6\binom{m}{3}$ and $V = 6\binom{m}{4}$. With $s_2 = \sum_x \mathbb{P}(X_i = x)^2$ and $s_3 = \sum_x \mathbb{P}(X_i = x)^3$, four-wise independence gives the exact identities

$$\mathbb{E}Z = As_2, \qquad \mathbb{E}Z^2 = As_2 + Us_3 + Vs_2^2.$$

Since each residue modulo r occurs either $\lfloor Q/r \rfloor$ or $\lceil Q/r \rceil$ times among the possible exponents,

$$\frac{1}{r} \le s_2 \le \frac{1}{r} + \frac{1}{Q}, \qquad s_3 \le \left(\frac{1}{r} + \frac{1}{Q}\right)^2.$$

Put $\alpha = 4 + 1/128$. Using $r > m^2/4$ and $Q \ge 128m^2$,

$$\begin{aligned} \mathbb{E}Z^2 &\le \frac{\alpha}{2} + \frac{\alpha^2}{m} + \frac{\alpha^2}{4} \\ &\le \frac{\alpha}{2} + \frac{\alpha^2}{16} + \frac{\alpha^2}{4} < 8. \end{aligned}$$

Let W count unordered pairs (i, j) that are useful and isolated, namely $X_i = X_j$, $E_i \ne E_j$, with all other $m - 2$ packets mapped to different receivers. The probability of a useful collision for a fixed pair is

$$\mathbb{P}(X_i = X_j, E_i \ne E_j) = s_2 - Q^{-1} \ge \frac{1}{m^2} - \frac{1}{Q} \ge \frac{127}{128m^2}.$$

Conditional on E_i, E_j and this event, each third exponent E_k remains uniform, by three-wise independence. Therefore $\mathbb{P}(X_k = X_i \mid E_i, E_j) \le 1/r + 1/Q$. Hashing is independent of the exponents, and for different endpoints a pairwise-independent hash has collision probability $1/B$. Hence

$$\mathbb{P}\big(h(X_k) = h(X_i) \mid E_i, E_j\big) \le \frac{1}{B} + \frac{1}{r} + \frac{1}{Q}.$$

A union bound over $k \notin \{i, j\}$ gives, for $m \ge 16$,

$$\begin{aligned} \mathbb{P}(\text{pair } (i,j) \text{ isolated} \mid X_i = X_j, E_i \ne E_j) &\ge 1 - (m-2)\left(\frac{1}{4m} + \frac{4}{m^2} + \frac{1}{128m^2}\right) \\ &> \frac{49}{100}. \end{aligned}$$

Summing over pairs,

$$\mathbb{E}W \ge \binom{m}{2} \frac{127}{128m^2} \frac{49}{100} \ge \frac{127}{256} \frac{15}{16} \frac{49}{100}.$$

Since $0 \le W \le Z$, we have $\mathbb{E}W^2 \le \mathbb{E}Z^2 < 8$. Cauchy–Schwarz therefore yields $\mathbb{P}(W > 0) \ge (\mathbb{E}W)^2 / \mathbb{E}W^2$ and proves (19.3). □

Remark 19.2 (Why the separate hash is needed). A receiver may see multiple unrelated residues whose hash values happen to coincide. Storing only its first key can miss such collisions. The probability bound above avoids this logical gap by requiring that one useful pair occupy a receiver *alone*. This is stronger than merely having an equal-endpoint pair somewhere in the network, but it still occurs with constant probability.

19.4 Bounded congestion on a butterfly

Let $B = 4m = 2^D$. Use a leveled butterfly with $D = \log_2 B$ stages and B nominal sources; only m sources are occupied. A packet at (t, u) traverses one of the two links to $(t+1, u)$ or $(t+1, u \oplus 2^t)$, selected by the t-th bit of its hashed receiver. Each route has D switching links. For a one-word-per-cycle receiver, append one unit-capacity service link to each output; the total packet path length is then $D + 1$.

For a uniform network-wide congestion bound, choose h from a family which is $2D$-wise independent over distinct endpoint keys. For example, embed residues injectively into $\mathbb{F}_{2^T}$, where $T \geq \max(\lceil \log_2 N \rceil, D)$, evaluate a uniformly random degree $2D-1$ polynomial, and retain a fixed D-bit linear projection of its values. It takes $2DT$ ideal seed bits. Independence across distinct keys remains valid even though equal keys necessarily hash identically.

Lemma 19.3 (Edge congestion with few duplicate pairs)**.** *Let κ be the maximum number of packet routes using a directed link of the butterfly. With Z as in Theorem 19.1,*

$$\mathbb{P}(\kappa \geq 2D + 65) \leq \frac{8}{65^2} + \frac{2BD + B}{(2D)!}. \tag{19.4}$$

For $m \geq 16$, the right-hand side is less than 0.001896*.*

Proof. Fix an edge at stage t. At most 2^{t+1} source positions can use that edge. From each *distinct* endpoint among these positions select one representative source. Conditional on the complete list of endpoints, these representatives form a deterministic set of distinct keys; each uses the edge with probability $2^{-(t+1)}$. With Y_e their count, the $2D$-wise independence of hash values gives

$$\mathbb{E}(Y_e)_{2D} \leq (2^{t+1})^{2D} 2^{-(t+1)2D} = 1, \qquad \mathbb{P}(Y_e \geq 2D) \leq \frac{1}{(2D)!}.$$

Including one serialized service edge per receiver, there are at most $2BD + B$ directed edges. By the union bound, all edges satisfy $Y_e \leq 2D - 1$ outside an event of probability at most $(2BD + B)/(2D)!$. For each key, the number of additional packets beyond the first is at most its number of unordered equal-endpoint pairs. Thus the total number of extra copies over all keys is at most Z, and the actual congestion of any edge is at most $Y_e + Z$. By $\mathbb{E}Z^2 < 8$,

$$\mathbb{P}(Z \geq 65) \leq \frac{\mathbb{E}Z^2}{65^2} < \frac{8}{65^2}.$$

Outside both exceptional events $\kappa \leq (2D-1) + 64 < 2D + 65$. For $D \geq 6$, the factorial term is at most $64(2 \cdot 6 + 1)/12! < 1.737 \times 10^{-6}$. □

Theorem 19.4 (Positive-probability logarithmic rendezvous with constant buffers)**.** *Assume the hypotheses of Theorem 19.1; route all packets* without in-network fusion *along their hashed butterfly paths. There exists a randomized store-and-forward scheduling protocol, using a constant number of complete-record buffers at each internal node and one complete-record transmission per directed edge per cycle, such that*

$$T_m = O(\log m) \tag{19.5}$$

with probability at least $0.0045 - \varepsilon_m$, where $\varepsilon_m \to 0$ is the scheduling failure probability for the fixed paths. On this event an isolated useful collision is detected by a one-resident-record receiver. The constants hidden in $O(\log m)$ and in the fixed buffer capacity are those supplied by the cited general randomized-routing theorem; no realistic transistor count or clock period follows from this existence result.

Proof. Let $\mathcal{I}$ be the event of an isolated useful pair. By Theorem 19.1, $\mathbb{P}(\mathcal{I}) > 0.0064$. By Lemma 19.3, the congestion is at most $2D + 64$ except on an event with probability less than 0.001896. Consequently the joint event has probability greater than $0.0064 - 0.001896 > 0.0045$. Conditional on any realization of this event, the packet routes are fixed, have dilation $D + 1 = \log_2 B + 1 = O(\log m)$ and congestion $\kappa = O(D)$, and lie in a bounded-degree leveled network with $O(BD)$ nodes. The randomized *online* scheduling theorem of Leighton, Maggs, Ranade and Rao [10] routes *any such fixed set of paths* in $O(\kappa + D + \log m) = O(\log m)$ steps with constant-size queues, with high probability over its independent scheduling randomness. The packets of an isolated useful pair reach the same receiver, which holds the first and compares it against every subsequent packet. No other packet reaches that receiver, so the pair is detected. Exact verification of the distinct exponents produces a nonzero multiple of r. Subtracting the scheduling failure probability finishes the proof. □

Remark 19.5 (Important qualification on high probability). The theorem asserts a uniform *positive* chance of successful collision detection within logarithmic rounds, not a high-probability collision event. The high-probability statement applies to transport *conditioned on* a favorable path set. Moreover, ε_m is an asymptotic failure term in an external randomized scheduling theorem: this paper does not specify its finite-size constants, so no numerical claim such as "33 nanoseconds" can be deduced. For the finitely many small sizes below the regime where the scheduling bound is effective, a standard constant-buffer bitonic sorting network is a deterministic fallback, with depth $D(D+1)/2$; since that regime has bounded D, this is consistent with an asymptotic logarithmic upper bound.

Corollary 19.6 (Multiscale order-oblivious search)**.** *Let $C = 2^d$ and $r \le C^2/4$. Run the above architecture at every power-of-two sample size $m_j = 2^j$ $(1 \le j < d)$, using a sufficiently large common exponent interval $Q \ge 128C^2$. For the appropriate lane, use Theorem 19.4 if $m_j \ge 16$ and a fixed constant-size deterministic sorting fallback if $m_j < 16$. Then, with a probability bounded below by a positive absolute constant for all sufficiently large C, one of the lanes outputs a verified nonzero relation in $O(\log C)$ parallel communication rounds. For odd semiprimes $N \le C^2$, choosing the base uniformly at each independent trial and using the CRT factor-extraction test gives the same positive-probability statement for obtaining a proper factor.*

Proof. If $r \le 4$, the $m = 2$ lane qualifies and direct comparison succeeds with probability at least $1/r - 1/Q \ge 1/4 - 1/Q > 0.245$; no four-sample theorem is invoked for this two-sample lane. If $r > 4$, let m be the first lane with $r \le m^2$. It satisfies (19.1). If $m \ge 16$, apply Theorem 19.4; if $m = 4$ or 8, use deterministic bitonic sorting and the 0.245 birthday estimate. Every lane executes concurrently, and $\log m \le \log C$. For $N = pq$ with distinct odd primes, $\mathrm{ord}_N(b) \le (p-1)(q-1)/4 < N/4$, so $N \le C^2$ implies $r < C^2/4$. Finally, the odd-order projection $b = a^{2^M}$ is independent of the two-primary favorable-base event for uniform a (proved in the preceding section). This event has probability at least $1/2$. □

19.5 Physical resource ledger and prototype

To distinguish mathematical parallelism from electronic feasibility, let H denote the number of *physical switch/processing sites*, not the number of mathematical exponent samples. A fully laid-out B-input, D-level butterfly contains $B(D+1)$ logical node positions and up to $2BD$ directed wires. Its *symbolic* total storage with q complete-record slots per intermediate node is

$$M_{\text{words}} = \Theta(qBD), \qquad M_{\text{bits}} = \Theta\big(qBD(\log N + \log C)\big).$$

The total across multiscale lanes is $\Theta(C \log C)$ logical network sites, not merely C physical components. Consequently, from an overall budget H of such logical sites one should expect

at most $C = O(H/\log H)$ sample origins in a fully unfolded implementation; the ideal bound $N = O(C^2)$ becomes, at best, a conditional scale $N = O(H^2/(\log H)^2)$ before accounting for gate-level width, wiring area, arithmetic engines and yield.

m	$B = 4m$	$D = \log_2 B$	$B(D+1)$ node positions	$2BD$ directed links
64	256	8	2 304	4 096
256	1 024	10	11 264	20 480
1 024	4 096	12	53 248	98 304
4 096	16 384	14	245 760	458 752

A word carries $\lceil \log_2 N \rceil$ residue bits plus a lane-local identifier of $\lceil \log_2 m \rceil$ bits and, where needed, the D-bit receiver address. "One word per link per clock" is therefore not "one bit per link per clock". Constant queue size means a fixed number of *words*, not constant bit storage independent of N.

We recommend a phased demonstrator:

1. *CPU reference:* exact modular endpoint generation and independent end-to-end relation/factor verification; baseline against Pollard–Brent.
2. *FPGA transport:* for $m = 64, 256, 1024$, instantiate bounded queues, fixed butterfly links, independent collision-hash generation, one-resident-record receivers, and telemetry of actual congestion, throughput, dropped packets, and energy.
3. *Thermodynamic sampler:* replace only the generation of prescribed Bernoulli inputs or random seeds with a measured probabilistic primitive. Compare energy *per verified factor* against the same all-digital circuit, counting data movement and seeding overhead.

An RC switch network realizing the averaging operators B_t could be tested *separately*, but it cannot substitute for an exact modular multiplication engine or an equality detector. Extropic's published architectures emphasize programmable sampling primitives and sparse local couplings, rather than generic 100-bit packet routers [12].

19.6 Reproducibility, open issues, and publication boundaries

The accompanying `simulate_bounded_butterfly.py` provides a finite-size *illustrative* store-and-forward schedule with queue capacity $q = 2$, one record per directed edge per cycle and a single-word streaming receiver. It samples fully independent exponents (a stronger condition than four-wise independence) in an *abstract cyclic group* represented by integer reduction modulo r. It does not implement the Leighton–Maggs–Ranade–Rao scheduler and hence cannot validate that scheduler's finite-size constants. It also does not emulate thermal physics.

No claim is made of an asymptotic improvement over classical factoring algorithms: the number of physically laid-out routers and the modular arithmetic work per trial can dominate. A candidate contribution for peer review, subject to a priority search, is the particular *isolated collision theorem plus its multiscale composition and resource accounting*, carefully distinguished from classical birthday paradoxes, pre-existing constant-buffer packet routing [10, 11], and parallel collision search [5].

The next non-asymptotic milestone is a clocked RTL prototype with synthesis-reported area and timing, a measured random-seed error bound, and a reproducible end-to-end benchmark. Only after these gates are met should a device-scale forecast or an external hardware proposal quote factorization throughput.

19.7 Illustrative finite-size routing trace

For a smoke test of the stated data-flow rules, a separate event simulator used $q = 2$ slots per internal node, each directed edge carrying at most one word per cycle, $B = 4m$ destinations, and a one-resident-word comparator at each destination. The cyclic endpoints were represented exactly as $E_i \bmod r$ with $r = m^2/2$, and the test used *fully independent* uniform exponents and independent random hash labels for distinct keys. The transport scheduler here is a simple backpressure priority rule, *not* the cited optimized randomized scheduling algorithm. Trials were performed with independent integer seeds.

m	Trials	Mean rounds	Max rounds	Isolated-pair rate	Detected-relation rate
16	400	7.86	10	0.535	0.605
64	400	10.71	13	0.545	0.608
256	200	13.52	16	0.535	0.610
1 024	80	16.35	18	0.625	0.688

No packet was dropped in the reported runs, the largest observed internal queue occupied two record slots, and each verified detection implied a genuine nonzero exponent relation in the synthetic cyclic group. These statistics describe a small *software model*, not an FPGA timing measurement, thermal sample, or factorization benchmark. They should not be extrapolated to $m \sim 10^{10}$.

Part V
Physical models and reproducible experiments

20 What a hardware realization must implement

The programmed theorem concerns a vector-valued averaging primitive. A passive resistor network implements a continuous semigroup. A digital collision processor generates labeled samples. These are three different realizations with different resource accounts.

20.1 A layout that does not assume the unknown order

Building a cycle indexed by $0, \dots, r-1$ assumes the quantity to be found. Enumerating the orbit to build such a cycle can already require r modular multiplications. A direct construction that avoids this circularity uses residue addresses $x \in \{0, \dots, N-1\}$ and the permutations

$$x \longmapsto x b^{\pm 2^t} \pmod{N}.$$

Because b is a unit, these maps are permutations of all residue addresses. Initialize only address 1. The nonzero state remains on its orbit G, which has unknown size r, although the device supplies N addresses. The files accompanying this article use this construction for the small full-vector prototype.

A sparse implementation must merge equal residue labels and preserve all their accumulated mass. Once a probability vector is within $1/(2r)$ of uniform at every point, every one of the r coordinates is positive. Therefore an exact explicitly stored sparse vector can no longer contain only $O(\sqrt{r})$ entries at that stage. A local rule for computing a neighbor is a succinct specification of an edge; it is not a guarantee of geometrically local wiring or small physical state.

20.2 Continuous-time RC diffusion: an exact analysis

Attach a capacitance C_0 from each node to ground. For symmetric conductances g_{xy}, Kirchhoff's law gives

$$C_0 \dot{V} = -\mathcal{L} V, \qquad V(t) = e^{-t\mathcal{L}/C_0} V(0), \tag{20.1}$$

where $\mathcal{L}$ is the conductance Laplacian. With conductance g per signed-generator occurrence, the orbit Laplacian is

$$\mathcal{L} = g \sum_{j=0}^{M} (2I - T^{2^j} - T^{-2^j}).$$

Repeated edges are combined by adding conductances; identity shifts contribute zero. Its nonzero eigenvalues are

$$\ell_k = 2g \sum_{j=0}^{M} (1 - \cos(2\pi k 2^j / r)) \geq 2g,$$

by Lemma 9.1. Thus for $r > 1$, $V(0) = V_0 \delta_e$, and normalized return $v_e(t) = V_e(t)/V_0$,

$$0 \leq v_e(t) - 1/r \leq e^{-2gt/C_0}. \tag{20.2}$$

This follows directly from the character expansion of (20.1); no Euler-discretization approximation is needed.

For a target deterministic error $1/(8B^2)$, a sufficient physical relaxation time is

$$t \geq \frac{C_0}{2g} \ln(8B^2).$$

If g per edge is held fixed, this is $O(\log B)$ in units C_0/g, while the nominal total incident conductance grows as $2g(M+1)$. If instead that conductance budget is fixed at G_Σ, putting $g = G_\Sigma/[2(M+1)]$ gives

$$t \geq \frac{C_0(M+1)}{G_\Sigma} \ln(8B^2).$$

The apparent difference between logarithmic and quadratic logarithmic time is partly a conductance-normalization issue. Wiring parasitics, charge injection, and detection time are not covered by either ideal bound.

Sampling a passive circuit produces $e^{-\Delta t \mathcal{L}/C_0}$, not the finite averaging matrix B_t or W. In general, using its first-order Taylor expansion at a fixed non-infinitesimal step does not implement the exact discrete operator. The $O(\log N)$ programmed theorem requires a switched averaging realization or an explicitly analyzed replacement schedule.

20.3 Conductance tolerance versus capacitance mismatch

Symmetric positive conductance changes preserve the equal-voltage equilibrium on a connected component when capacitances are equal. They alter relaxation rates. Capacitance mismatch changes the value used for order recovery itself.

Proposition 20.1 (Equilibrium bias from capacitances)**.** *Let a connected component have r nodes with positive capacitances C_x, symmetric conductances, and initial condition $V_e(0) = V_0$, $V_x(0) = 0$ for $x \neq e$. Then*

$$\frac{V_e(\infty)}{V_0} = \frac{C_e}{\sum_x C_x}.$$

Writing $C_x = C_0(1+\epsilon_x)$ and $\bar{\epsilon} = r^{-1}\sum_x \epsilon_x$, the reciprocal equilibrium reading equals

$$\frac{V_0}{V_e(\infty)} = r\frac{1+\bar{\epsilon}}{1+\epsilon_e}.$$

For rounding this exact equilibrium reading to r, the strict condition

$$|\bar{\epsilon} - \epsilon_e| < \frac{1+\epsilon_e}{2r}$$

is sufficient, and apart from rounding ties it is also necessary.

Proof. The equations are $C_x\dot{V}_x = -\sum_y g_{xy}(V_x - V_y)$. Summing them shows that $\sum_x C_x V_x$ is conserved. The symmetric connected Laplacian, after conjugation by $\mathrm{diag}(C_x)^{-1/2}$, has one zero eigenvalue and all others positive. The voltages therefore tend to a common value. Conservation of charge identifies it as $C_e V_0/\sum_x C_x$. Substitution and subtraction of r prove the remaining statements. □

Thus uncalibrated relative mismatch generally has to decrease like $1/r$ for this readout protocol. Uniform component mismatch cancels, but the injection capacitance relative to the component average matters. A calibration scheme must include its own cost and cannot simply assume that an unknown orbit contains perfectly matched cells. Uniform leakage to ground introduces a common decay; nonuniform leakage can also change the normalized profile.

20.4 Resolution and a model-dependent noise budget

Adjacent equilibrium return values differ by

$$\frac{1}{r} - \frac{1}{r+1} = \frac{1}{r(r+1)}.$$

Resolving orders near B from this scalar therefore requires absolute precision of order B^{-2} on a normalized full scale, or about $2\log_2 B$ binary digits plus guard bits. The fact that the digit count is logarithmic does not make a physical precision of B^{-2} inexpensive.

For a concrete illustration, suppose an independent sample-and-hold readout contributes voltage variance $k_{\mathrm{B}}T_{\mathrm{bath}}/C_{\mathrm{read}}$ per sample. This is the usual kT/C circuit model [7]. Averaging J independent samples gives rms error

$$\sigma_{\mathrm{mean}} = \sqrt{\frac{k_{\mathrm{B}}T_{\mathrm{bath}}}{JC_{\mathrm{read}}}}.$$

Requiring an rms error no larger than $V_0/(8B^2)$ imposes

$$JC_{\mathrm{read}} \geq \frac{64k_{\mathrm{B}}T_{\mathrm{bath}}B^4}{V_0^2}. \tag{20.3}$$

For a Gaussian confidence requirement $z\sigma_{\mathrm{mean}} \leq V_0/(8B^2)$, multiply the right-hand side by z^2. Correlated samples provide fewer independent observations and cannot be counted at the raw ADC clock rate.

At $T_{\mathrm{bath}} = 300$ K, $V_0 = 1$ V, and $C_{\mathrm{read}} = 1$ pF, the rms condition gives the illustrative numbers

Order bound B	Required J (before confidence margin)
10^3	2.65×10^5
10^6	2.65×10^{17}
10^9	2.65×10^{29}

These are consequences of a stated simple readout model, not universal lower bounds on all conceivable analog architectures. They show why a large cell count alone is not a feasibility argument. Recent thermodynamic hardware work demonstrates useful sampling and linear-algebra primitives [6]; it does not establish the extreme scalar precision required by this order-recovery protocol.

20.5 Why sampling a return indicator is a different resource

If the interface returns only an independent Bernoulli indicator of a return, with probability $p = 1/r$, its empirical mean has standard deviation approximately $1/\sqrt{Jr}$. Achieving error of order $1/r^2$ requires order r^3 independent samples.

This exponent also has a testing lower bound. For $r \geq 2$, let $P = \mathrm{Bernoulli}(1/r)$ and $Q = \mathrm{Bernoulli}(1/(r+1))$. The elementary relative-entropy bound

$$D(P\|Q) \leq \frac{(1/r - 1/(r+1))^2}{(1/(r+1))(1-1/(r+1))} = \frac{1}{r^3}$$

follows from $\ln x \leq x - 1$. Relative entropy adds for independent observations, and Pinsker's inequality gives

$$\|P^{\otimes J} - Q^{\otimes J}\|_{\mathrm{TV}} \leq \sqrt{J/(2r^3)}.$$

Any test with both error probabilities at most $1/3$ requires total variation at least $1/3$, hence $J \geq 2r^3/9$. Standard Bernoulli concentration supplies a matching order upper bound for fixed confidence at error scale r^{-2}. This statement is specific to that readout channel. It is not a lower bound on using all endpoint labels, on modular arithmetic, or on factoring in general.

20.6 A concrete small programmable prototype

A useful first digital prototype stores two arrays, indexed by all residues, and performs the integer update

$$c_{t+1}(x) = 2c_t(x) + c_t(xb^{2^t}) + c_t(xb^{-2^t}), \qquad c_0 = \delta_1. \tag{20.4}$$

The normalized state is exactly $c_t/4^t$. There is no numerical rounding. Since $\sum_x c_t(x) = 4^t$, a width of $2L+1$ unsigned bits suffices for all counters up to layer L. The return readout and stopping candidate can be computed as integers:

$$z = \frac{c_t(1)}{4^t}, \qquad c = \left\lceil \frac{4^t}{c_t(1)} \right\rceil.$$

Check $b^c \equiv 1 \pmod N$ as in Proposition 10.3.

For a small FPGA, a time-multiplexed implementation can compute the neighbor addresses digitally and update a few coordinates per clock. It uses $O(NL)$ stored bits and $O(NL)$ coordinate updates, so it is a verification platform, not a fast large-integer factoring machine. Fully spatial layers require parallel routing of the modular-permutation connections. A proposed switched-capacitor version would additionally need nonoverlapping clock phases, charge-conserving averaging, controlled loading, and calibrated readout. Merely connecting every node simultaneously to its neighbors does not implement (10.1).

The supplied `core.py` implements (20.4) without any order oracle. The supplied RTL file `dyadic_layer.sv` expresses one fixed-route arithmetic layer; it is an RTL specification, not a synthesized timing or area result. A complete configurable system still needs scheduling, state buffers, routing selection, and host-side modular verification.

For a passive experiment, the supplied `rc_N21.cir` uses all 21 residue addresses, $b = 4$, $C_0 = 1$ nF, and $g = 1\,\mu$S per occurrence. The initialized component happens to have three vertices. Its ideal return is

$$V_1(t)/V_0 = \frac{1}{3} + \frac{2}{3}e^{-18gt/C_0}.$$

This has been checked against the matrix-exponential circuit model, not against a fabricated device or a SPICE engine. Measuring its decay and its sensitivity to a deliberately changed injection capacitor would test the central physical assumptions on a manageable scale.

20.7 Implementation priorities and acceptance criteria

1. Reproduce the exact integer-array experiments at $N = 21, 143, 299$, with the construction using only N, b. Verify the modular stopping certificate on the host.

2. Measure a small RC network: conservation of total charge within uncertainty, equilibrium value, fitted gap, settling time, and readout noise. Repeat with a known capacitance mismatch and compare with Proposition 20.1.

3. If pursuing programmable averaging, characterize one three-tap switched stage, then a sequence of stages. Quantify η_{init}, each effective η_t, and ρ rather than inferring them from nominal component specifications.

4. For the scalable digital path, benchmark batched modular exponentiation and a shared collision table on FPGA or GPU. Record products per second, sustained table throughput, bits per record, power, and verified factors per joule. Include Pollard–Brent and suitable established methods on the same hardware.

For a 40-decimal-digit modulus, $\lceil \log_2 N \rceil$ is about 133. With $K = 2\,\mathrm{bitlength}(N) + 8$, a bare endpoint–exponent record has about 407 bits. A billion such records already require roughly 51 GB before hash-table overhead. These memory slots cannot be identified with individual analog cells or single transistors.

21 A thermodynamic interface with explicit arithmetic certificates

21.1 What current public descriptions support

Extropic describes XTR-0 as an experimental platform coupling its probabilistic circuits to a conventional processor. Its hardware page lists Z1 sticks and cards for early access in 2027, rather than confirming general availability in 2026 [18]. The summer 2026 announcement describes Z1 as a sparse Ising sampler with roughly 269,000 pbits, sixteen-neighbor connectivity, a sampling rate above 50 MHz, and power below one watt. These are the company's published specifications and plans, not measurements made for this project [12].

The same announcement presents Torx as a framework of stochastic transition kernels and Thermalizers as a compilation approach targeting energy-based models. The associated paper studies approximate factor compilation and error accumulation [17]. These developments motivate an interface-level experiment. They do not establish an efficient physical implementation of modular multiplication, a product tree, or the exact dyadic averaging operator in this work.

21.2 Two distinct experimental roles

The lowest-risk experiment uses thermodynamic hardware to supply the seed to an otherwise unchanged digital pipeline. This directly tests energy and latency of sampling. Since only $4K$ random bits specify T four-wise-independent exponents, randomness generation may already be a small part of the total energy. A fast source alone can therefore have little end-to-end effect.

The more substantive experiment is a *conditional arithmetic proposal module.* For a known residue x, a fixed multiplier c and modulus N, the thermodynamic module proposes a quotient–remainder pair (q, y) satisfying

$$xc = qN + y, \qquad 0 \le y < N, \qquad 0 \le q < N. \tag{21.1}$$

The digital controller supplies $c = b^{2^t}$, or a fixed-window table value, and retains the exponent choice. The candidate is checked before its result is used, or its errors are included in a quantified trajectory error budget. This turns approximate dynamics into proposals for exact algebra, a general pattern with substantial antecedents in invertible probabilistic logic [15, 16]. No novelty is claimed for pbits, reverse multiplication, or exact verification as general concepts.

The module is useful only if proposal generation plus checking and any fallback costs less than the exact digital operation it replaces. Recomputing xc with the same full-width multiplier may make the checker as expensive as the original task. That cost must be measured, not absorbed into a free verification oracle.

21.3 An explicit local-constraint energy construction

A finite Boolean circuit can check (21.1): compute xc, compute $qN + y$, compare them, and check $y < N$. Schoolbook multiplication and addition use $O(n^2)$ Boolean gates, where $n = \lceil \log_2 N \rceil$. The quotient and remainder are the variable inputs; x, c, N are clamped. Auxiliary bits record gate values and carries. Overflow is prevented by working at the full $2n + 1$-bit width before testing equality.

For an AND gate $z = uv$, the quadratic penalty

$$P_{\mathrm{AND}}(u, v, z) = uv - 2uz - 2vz + 3z$$

is nonnegative on binary assignments and vanishes exactly when the gate is satisfied. For a NOT gate $z = 1 - u$, use $(u + z - 1)^2$. For a full adder with bits u, v, w, sum bit s and carry bit k, use

$$P_{\mathrm{add}} = (u + v + w - s - 2k)^2.$$

This is quadratic after using $z^2 = z$ for each binary variable. It vanishes exactly for a correct full-adder relation; the binary representation of the sum in $\{0, 1, 2, 3\}$ is unique. A circuit of AND and NOT gates alone is already universal, so no unproved gate primitive is needed for existence of the construction.

Sum the gate penalties and the penalties pinning the output checks to one. If necessary, duplicate variables connected by equality penalties to respect a chosen maximum logical degree. Mapping this logical graph to an actual chip may require additional copies and couplings, and can change the effective energy gap; the $O(n^2)$ logical gate count is not a physical pbit count.

Proposition 21.1 (Ground states and the precision of the claim)**.** *For fixed $0 \le x, c < N$, a constraint energy constructed as above is nonnegative, and every zero-energy state encodes the unique quotient–remainder pair in (21.1). Conversely that pair admits a zero-energy completion. If the circuit is deterministic once its variable inputs are supplied, all auxiliary wires in a valid completion are fixed as well.*

Proof. Every local penalty is nonnegative and zero precisely when its gate or pinning condition holds. Thus a zero sum satisfies the whole circuit, which checks the integer identity and range constraints. Euclidean division gives a unique pair (q, y), and $xc < (N - 1 + 1)^2 = N^2$ ensures $q < N$. Evaluating the circuit on that pair assigns every intermediate wire and satisfies every penalty. Determinism of the gates proves the last assertion. □

Proposition 21.2 (A sufficient equilibrium bound, not a mixing theorem)**.** *Suppose an implementation has v free binary variables, at least one zero-energy state and every invalid state has energy at least $\Delta > 0$. At an* exact *Gibbs equilibrium with inverse temperature β,*

$$\mathbb{P}(\textit{invalid}) \le (2^v - 1)e^{-\beta\Delta}.$$

Thus $\beta\Delta \ge v \ln 2 + \ln(1/\epsilon)$ suffices for invalid probability at most ϵ.

Proof. The total Boltzmann weight of invalid states is at most $(2^v - 1)e^{-\beta\Delta}$. The partition function is at least one because a zero-energy state exists. Divide by this lower bound. □

This deliberately conservative estimate illustrates the missing engineering questions: coupling dynamic range, temperature, calibration, embedding, and equilibration time. A correct ground-state encoding *does not prove* rapid Gibbs mixing. If the actual distribution is within δ of Gibbs equilibrium in total variation, add δ to the error bound. Neither the needed δ nor the mixing time is established by the logical construction.

21.4 Checking a proposed quotient without full-width multiplication

A low-width modular check can be cheaper than recomputing the entire product, at the price of a quantified false-accept probability.

Proposition 21.3 (Fresh-prime checking)**.** *Let $N < 2^n$ and let a frozen candidate have $0 \le x, c, q, y < N$. Choose a prime ℓ uniformly from a fixed set $\mathcal{P}$ of S distinct primes, all at least 2^w. If $xc \neq qN + y$, then the chance it nevertheless satisfies*

$$xc \equiv qN + y \pmod{\ell}$$

is at most

$$\min\{1, \lfloor 2n/w \rfloor / S\}.$$

With k independent fresh challenges it is at most the kth power of this bound, provided the same candidate remains fixed.

Proof. Put $D = xc - qN - y$. The range checks imply $0 < |D| < N^2 < 2^{2n}$. If h distinct members of $\mathcal{P}$ divide D, their product is at least 2^{wh} and divides $|D|$. Therefore $wh < 2n$, so in particular $h \le \lfloor 2n/w \rfloor$. At most that many prime challenges accept an incorrect candidate. Independent choices with replacement multiply the conditional acceptance probabilities. □

The challenges must be generated after the candidate is committed, or be independent of the mechanism producing its error. A device that knows fixed checking primes can produce errors divisible by their product. Reduction of wide inputs modulo each challenge prime, control, and random prime selection also have a cost. This checker is an optional measured tradeoff, not an asserted universal improvement over digital arithmetic.

21.5 Repair without biasing the exponent distribution

For a given exponent seed, retain each requested arithmetic operation while making at most R thermodynamic proposals. On rejection, retry the *same* operation; after R rejections use a digital fallback. Do not discard the exponent and draw another one merely because its endpoint was expensive to realize. That practice could bias the sample distribution and invalidate the four-wise proof.

With an exact checker and exact fallback, this fixed-input repair produces exactly the ideal endpoint for every exponent, regardless of proposal quality. With a probabilistic checker having conditional false-accept probability at most η for each of at most J proposed operations, a union bound gives at most $J\eta$ chance of a corrupted computation. On the complement, the ideal sampling theorem applies. Adding seed deviation ϵ and independently quantified delivery or detector failure δ yields the conservative bound

$$\mathbb{P}(\text{verified factor}) \ge 3/10 - \epsilon - J\eta - \delta.$$

The final divisor test remains exact even if an internal check fails; therefore internal noise can destroy success or waste work, but cannot make an invalid divisor pass a correct final verifier.

21.6 Energy break-even and the proper experimental objective

Let $E_{\text{prop}}, E_{\text{check}}, E_{\text{dig}}$ be measured energies per proposal, check and digital fallback. Under an *explicit iid* proposal-success model with probability s and an exact checker, truncation after R proposals gives expected energy

$$E_{\text{hybrid}} = (E_{\text{prop}} + E_{\text{check}}) \frac{1 - (1-s)^R}{s} + (1-s)^R E_{\text{dig}} + E_{\text{interface}}. \tag{21.2}$$

For $s = 0$ interpret the fraction as R. This is the sum of the probabilities of reaching proposal numbers $1, \ldots, R$, plus the fallback probability. Correlated proposal errors require measuring the

actual survival probabilities instead of replacing them by $(1-s)^j$. A useful module must beat E_{dig} after accounting for interface and amortized programming energy.

At system level, if sampling constitutes fraction f of a digital pipeline's energy and is accelerated by factor s_E, the best possible energy ratio from replacing only that component is

$$\frac{E_{\mathrm{old}}}{E_{\mathrm{new}}} \leq \frac{1}{(1-f)+f/s_E},$$

before new interface overhead. For $f = 0.01$, even eliminating that component entirely improves total energy by at most about one percent. This arithmetic is why a modular-channel experiment is scientifically more informative than quoting the energy of isolated random bits.

The final metric is joules per verified factor, alongside time to factor and silicon resources, at matched success probability. Under independent identically distributed batches with mean energy $\overline{E}$ and success probability p, the expected energy to the first success is $\overline{E}/p$. This remains true even if energy and success within a batch are correlated: the expected total satisfies $V = \overline{E} + (1-p)V$ by restarting with an independent batch.

22 Reproducible checks and small computational experiments

All new numerical results in this section are generated by `experiments/run_experiments.py`. The factoring routines take no factorization or order as input. A separate trial-division order oracle is used only for verification and to label diagnostic plots. All data are from software or ideal circuit equations; none are measurements of a physical implementation.

22.1 Exact algebraic and arithmetic checks

The script performed the following checks without discrepancies:

- 3,328 exact kernel cases: every $1 \leq r \leq 256$ and $0 \leq L \leq 12$. Integer numerator vectors verify the return formula, the pointwise error bound, and conservation of mass.
- Exact enumeration of uniform units for all 78 odd integers $N \leq 300$ with at least two distinct prime factors, verifying (11.1).
- 45,400 extraction cases using five distinct relation multipliers for every unit in that enumeration. A proper factor is returned exactly for the predicted favorable two-primary patterns.
- 1,438 full-residue-array order recoveries, for every unit modulo odd $3 \leq N \leq 83$, using the one-sided certificate without supplying the order to the recovery routine.
- Four exact nonuniform second-moment checks, for sample sizes $2, 3, 4, 5$, by complete enumeration with rational probabilities.

Finite tests do not replace the proofs, but they cover parity, coincident shifts, $r = 1$, and nonzero-relation cases that are easy to mishandle in implementations.

22.2 Fixed versus programmed evolution

Table 1 compares the original sufficient bound, observed fixed-walk rounding, the programmed guarantee including an error budget, and the first exact programmed certificate. The fixed-walk observed column uses double-precision spectral evaluation against the known diagnostic order; it is not itself an interval certificate. The programmed certificate uses exact integer arithmetic.

N	b	r	Fixed bound	Fixed observed	Safe L	First certificate
21	4	3	154	2	8	2
143	16	15	330	10	12	4
299	3	33	409	17	13	6
1022117	576	5313	1845	46	31	18

Table 1: Operator counts. "Safe L" is the least integer with $4^L \geq 2N^3$. Fixed and programmed operators have different fan-in and routing requirements; these counts are not measured speedup factors.

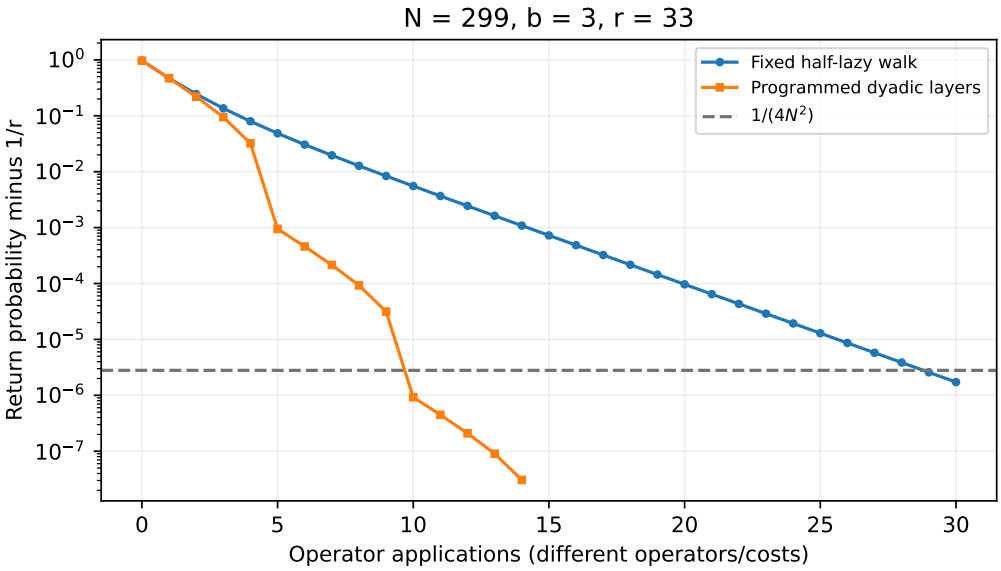


Figure 7: Error at the identity for the original $N = 299$, $b = 3$ example. The scheduled errors are computed from the exact formula.

For $N = 299$, the old sufficient bound is 409 fixed steps, although rounding actually succeeds after 17 steps. The programmed construction has a 13-layer error-budget guarantee and an exact certificate after six layers on this instance. Comparing 409 with six alone would confound a worst-case bound and an observed early stop; all four columns are therefore reported.

22.3 First-relation extraction on the original examples

The first displayed relation in the original Fermat-number example already suffices:

$$N = 4294967297, \quad a = 3945765912, \quad D = 314919552.$$

Direct modular verification gives $a^D = 1 \pmod N$. Removing only the power of two from D and using Theorem 11.1 returns 6700417, whose complementary factor is 641. The later eight-hit gcd-stability criterion is unnecessary for this factorization.

Likewise, for $N = 8219999$, $a = 7081686$, the first displayed relation $D = 12962750$ is already valid and yields the factor 32749, with complementary factor 251. These are exact checks of recorded relations, not attempts to reproduce the original wall-clock timings.

22.4 A controlled software comparison

The new benchmark uses seeds 1729, 1730, and 1731 for each input. The word sampler uses 2,000 independent signed dyadic increments per endpoint. The binary sampler uses $K = 2\,\mathrm{bitlength}(N) + 8$. Both relation samplers use the same oddified base for each input and stop at the first useful relation. The selected original-style bases are favorable; the experiment measures per-base work, not the success distribution of unrestricted random-base factoring. A conventional Pollard–Brent implementation is included as a diagnostic baseline.

N	Words: products	Binary: products	Brent: products	Binary seconds
1022117	50,000	964	93	0.0002
8219999	792,000	18,467	93	0.0027
4294967297	792,000	9,147	189	0.0023

Table 2: Medians of three runs. Relation-method product counts cover endpoint generation, excluding shared table setup, oddification, and extraction. Brent counts its iteration and product-accumulator products. Runtime includes routine setup and extraction. The counts are transparent diagnostics, not a full hardware cost model.

On these inputs, binary sampling uses about 43–87 times fewer endpoint products than the 2,000-increment comparator. This is a direct improvement to the supplied digital strategy. Nevertheless, Pollard–Brent uses fewer products and less time in every displayed case. Three seeds on small, deliberately selected inputs cannot establish average-case superiority, cryptographic scaling, or a practical factoring record. The raw run data, interpreter version, platform string, and counts are in `results/benchmark_runs.csv` and `results/verification.json`.

22.5 Ideal RC-model check

The equal-capacitance matrix-exponential calculation agrees with the analytic three-node return formula to within 10^{-12} at all evaluated times. The supplied SPICE netlist is generated from N, b and the modular permutations, without an order input. It has not been executed in a SPICE simulator in this review. Likewise, the RTL layer has not been synthesized or simulated with an HDL toolchain. The Python integer-array specification is the verified reference for its arithmetic.

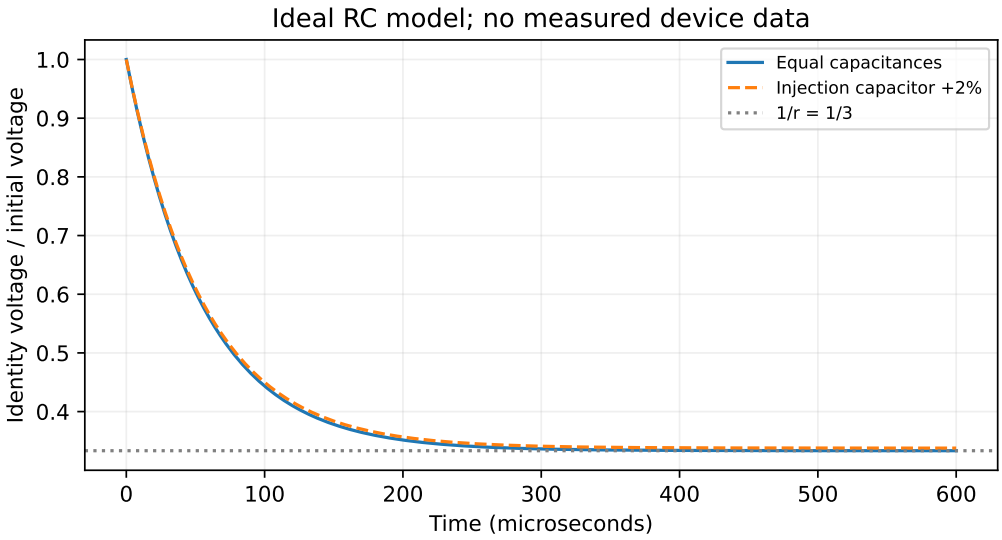


Figure 8: Continuous-time model for the 21-address network. Increasing only the injection capacitance by 2 percent moves the limiting normalized value from 1/3 to 1.02/3.02. No physical measurements are represented.

23 Research directions and limits of the present results

The strongest mathematical development is a constructive logarithmic-depth averaging schedule with an exact finite-size error, an arithmetic stopping certificate, and an extension to finite abelian subgroup cardinalities. The most immediate digital improvements are controlled binary sampling, first-relation extraction, and restarting an unfavorable base without redundant gcd stabilization.

Several further questions are concrete enough to guide subsequent work:

1. Determine the least routing and storage resources needed to implement the modular permutation sequence while retaining the programmed-depth benefit. A short algebraic description of the routes is not a physical layout.

2. Optimize the radix and schedule subject to a fixed conductance, fan-in, switching-energy, and wire-length budget. A lower layer count alone does not select the best physical design.

3. Develop an experimentally validated interval readout, with capacitance calibration and correlated-noise accounting, that can satisfy Theorem 10.4 at increasing orbit sizes.

4. Analyze a specific shared distinguished-point transition rule with integer-exponent labels, instead of transferring iid birthday estimates to a correlated process without proof.

5. Extend comparative software benchmarks to random balanced semiprimes and several size ranges, reporting distributions, failed trials, arithmetic work, memory traffic, and energy. Established factoring algorithms must remain the primary baselines.

The exact kernel does not remove the potentially exponential number of represented coordinates, and the scalar readout retains an order-dependent precision requirement. Consequently the present results establish improved depth in an explicitly strengthened diffusion model and improved implementations of the relation strategy. They do not establish polynomial-resource classical factoring, a competitive large-scale physical machine, or an advantage over quantum computation.

24 New computational verification and reproducible comparison

24.1 What was executed

The entire legacy verification script was rerun: it checks 3,328 exact kernel instances, 78 CRT success-law enumerations, 45,400 relation extractions, 1,438 exact all-residue order recoveries and four rational second-moment identities. Its generated results and figures are included. The independent revision checks add the following.

Cases	Check
65,536	All cubic coefficient vectors over $\mathbb{F}_{16}$ give distinct four-point output vectors, checking the finite Vandermonde construction
52	Translation-invariant partial-congruence probabilities and overlapping-pair moments, with exact fractions
480	Product/remainder evaluation of F' against direct products of all differences over composite rings
1,110	Fixed-window endpoints against independent built-in modular exponentiation
1,728	Full detector results against all-pairs gcd and relation-extraction enumeration
512	All small receiver hash assignments for a fixed useful key and two contaminating keys
1,968	All candidate quotient–remainder pairs for fifteen small clamped arithmetic circuits, with consistent auxiliary-wire evaluation

The rational constants in Theorem 14.4 were checked without floating-point rounding. A dedicated saturation test recovers 3 or 5 from the distinct-key example $(1, 4, 6)$ modulo 15. The logical AND penalty was exhaustively checked on its eight possible assignments. The circuit proof, rather than enumeration of all auxiliary-bit strings, establishes correctness for arbitrary inconsistent wire assignments. None of these finite tests substitutes for the mathematical proofs.

The factor routine receives only N, a seed and optional budget/base parameters. Trial factorization and multiplicative-order computation appear solely in the independent diagnostic code. The sampler derives all its constants from N, and the product detector works in $\mathbb{Z}/N\mathbb{Z}$. This separation prevents known factors from being used to construct the computation that purports to find them.

24.2 Small random semiprimes

A fixed diagnostic schedule sampled 300 pairs of distinct primes from the primes between 11 and 500 and ran one full batch on each product. All 300 batches returned proper factors, with the following routes:

Route	Batches
Nonunit base gcd	7
Useful full endpoint equality	104
Proper derivative-product gcd	149
Saturated product resolved by splitting	40

These are small, favorable finite observations under deterministic software pseudorandom seeds. They do not prove a 100 percent general success rate, ideal physical randomness, or cryptographic scaling. The theorem's conservative 3/10 bound is an analytical worst-case statement under its specified probability model. Every input, seed, result and route is retained in the accompanying JSON file.

24.3 Matched endpoint lists and a conventional baseline

For each of five semiprimes, seeds 1729–1733 were run without selecting favorable bases. An equality-only control uses exactly the same base and endpoint list as the combined detector. Pollard–Brent is a separate standard-Python reference with the same numeric seeds, not an optimized external library. The table reports all success counts and medians; "products" refers to modular ring products, not finite-field operations, gcd costs, bit operations or joules.

N	Records	Combined	Equality	Products	Brent prod.	Time (ms)	Brent (ms)
66,013	34	5/5	0/5	2,216	45	0.708	0.025
1,022,117	64	5/5	2/5	7,088	93	2.047	0.026
16,016,003	128	5/5	2/5	26,614	93	5.606	0.025
100,160,063	202	5/5	0/5	64,682	189	16.423	0.035
4,292,870,399	512	5/5	0/5	403,026	381	129.248	0.096

Table 3: Five runs per input. Combined products include table setup, endpoint generation, schoolbook polynomial arithmetic and verifier products. Full finite-field, gcd and timing data are in the CSV. No hardware measurements are represented.

The added partial-congruence route finds factors on these same short lists even when full equality fails. However, the schoolbook polynomial implementation consumes more work and time than Pollard–Brent on every listed median. This result limits the practical claim: the revised sampling theorem is stronger than the former equality-only pipeline, but the present Python implementation is not a competitive factoring engine. A fast polynomial backend and hardware study are still needed, and may or may not change that comparison.

24.4 The separately identifiable window improvement

The following comparison counts complete endpoint setup and generation for the same lists, excluding detector work in both columns. Binary sampling uses shared powers b^{2^j}; the window implementation uses $w = 4$, including all window-table filling. Thus the improvement is not obtained by comparing an optimized setup with an untimed one.

N	Binary products	Window products	Ratio
66,013	424	278	1.53
1,022,117	890	512	1.74
16,016,003	2,022	1,069	1.89
100,160,063	3,452	1,787	1.93
4,292,870,399	9,990	4,907	2.04

The theoretical $w \asymp \log T$ choice in Proposition 16.1 minimizes a different abstract objective. The conservative four-bit window here demonstrates a concrete arithmetic saving without claiming an optimal FPGA memory configuration.

24.5 Sample scales and their interpretation

24.6 How to reproduce the evidence

From the project directory, use a standard Python 3 installation:

```
python3 experiments/verify_revision.py
python3 experiments/divisor_sensitive.py 1022117 --seed 1729
python3 experiments/thermodynamic_constraints.py --verify
python3 experiments/run_experiments.py
```

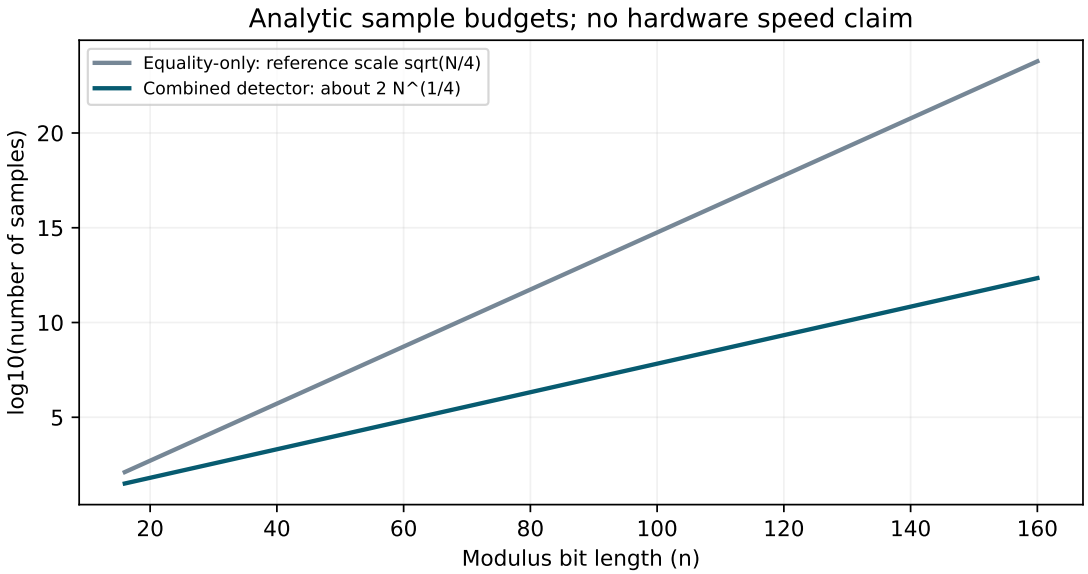


Figure 9: Analytic sample counts, not device sizes or timings. The older worst-order equality scale is represented by $\sqrt{N/4}$; the new theorem uses $2\lceil N^{1/4} \rceil$. Constants, actual local orders, repeated batches and detector cost matter. The same diagram cannot be read as a plot of speedup against established factoring.

The first three commands use only the standard library. The legacy plot/model script uses NumPy, SciPy and Matplotlib. The source and verification records state the interpreter and platform actually used. Run the plotting helper to regenerate the new analytic sample-scale figure. The package includes raw CSV/JSON rather than only rounded tables.

The absence of an HDL or SPICE toolchain in this execution environment means that the RTL and netlist are provided as design specifications. No HDL simulation, synthesis, timing closure, place-and-route, chip measurement or thermal-sampling experiment is claimed. The logical QUBO file is a reproducible constraint instance; it has not been embedded on Extropic hardware or used to infer a sampling rate.

Part VI
Industrial evaluation and technical disclosure

25 Technical evaluation dossier for Extropic

25.1 The proposed collaboration and its first decision

The proposed collaboration is an evaluation of a *certified hybrid arithmetic pipeline* for integer factorization. The mathematical core is the dyadic exponent distribution and the divisor-sensitive detector. The industrial question is whether programmable stochastic primitives can implement a material part of that pipeline at lower energy or latency than a carefully matched digital implementation.

The initial request should be a limited feasibility study with access to interface specifications, a simulator and, if mutually available, a small experimental platform. It should not be framed as a request to manufacture a billion-cell factoring machine. No partnership, financing, production availability or performance commitment is assumed here. The full technical annex should be shared only after an appropriate confidentiality and intellectual-property process has been agreed.

There are two concrete deliverables for the first joint review: a working exact reference that anyone can reproduce, and a measured cost decomposition identifying which operations dominate. If seed production is negligible compared with exact arithmetic, replacing only that source has little industrial value. The conditional modular channel in Section 21 is therefore a separate research candidate, with a clear rejection criterion if proposal, verification and repair cost exceed digital execution.

25.2 Reference block architecture

25.3 Module contract and reproducible messages

A batch descriptor contains N, base a, batch identifier, exponent bit width K, count T, finite-field modulus, four seed coefficients, window width and a detector mode. A seed commitment precedes any sampling-dependent admission rule. An endpoint record contains its batch identifier, sample index and exact residue. The exponent is reconstructed from the index and seed when needed.

An arithmetic proposal request contains the batch identifier, sample index, stage index, x, c, N and a fixed retry limit. Its response is (q, y) with range flags, a valid/ready handshake and an error status. A challenge-based checker freezes that response before choosing its fresh check moduli. Repair preserves the same requested transition. A witness contains either two different exponents for the same endpoint, or a gcd/product-tree certificate. The host verifies divisibility independently of the stochastic component.

A timeout is a failed trial, recorded with its consumed work and energy. Dropping packets and then reporting success only on completed batches would bias throughput measurements. Software exceptions, nonconvergent thermal trajectories, malformed candidates and saturated products are distinct telemetry events; only saturation has the specific exact recovery procedure proved above.

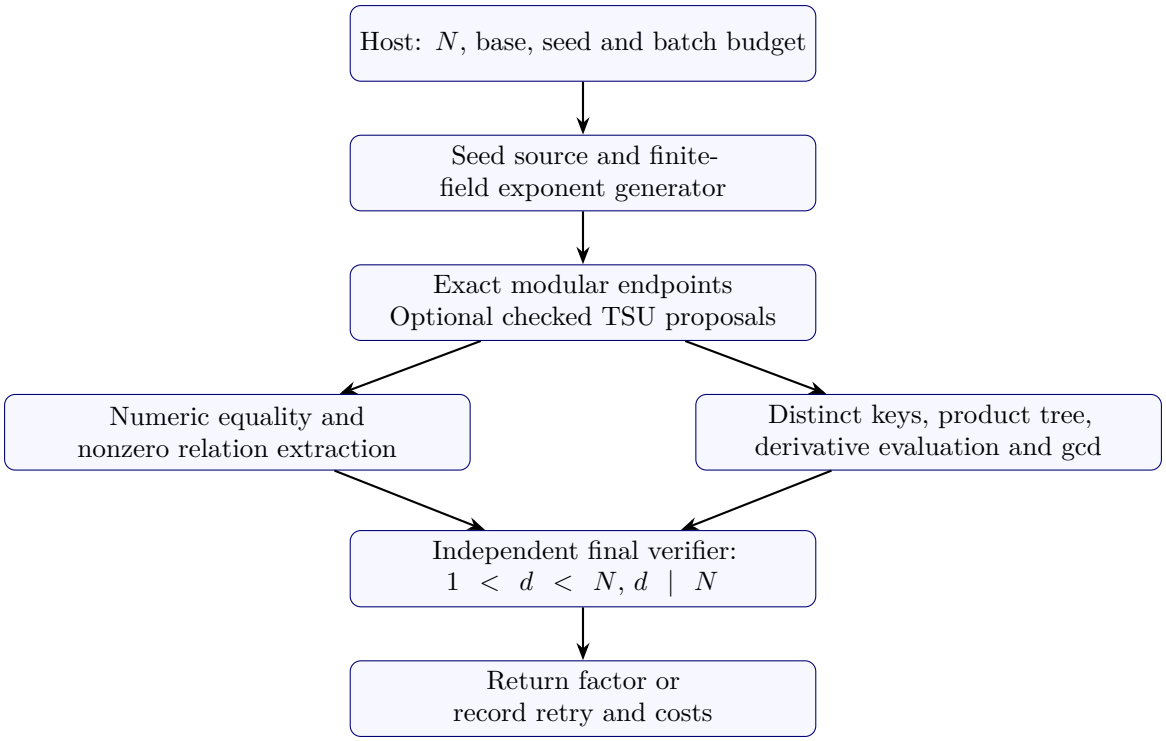


Figure 10: Proposed functional architecture. Only the explicitly marked proposal or seed block is a candidate thermodynamic component; the remaining blocks have exact digital reference semantics. Physical placement, buses and clock domains are not inferred from this diagram.

25.4 Implementation stages and gates

Stage	Deliverable	Acceptance gate
Reference	Exact exponents, endpoints, derivative detector and independent verifier	All finite tests pass; every reported divisor divides its input
Digital tile	Banked endpoint and coefficient memory; modular arithmetic and gcd engines	Synthesized area, timing, memory traffic and full-batch energy; no omitted setup
Seed interface	Matched digital and thermal seed experiments	Quantified dependence/entropy model and measured whole-system impact
Arithmetic module	Small conditional quotient–remainder kernels	Accuracy, autocorrelation, verification cost and fallback survival measured
Integrated study	Same algorithm on two hardware configurations	Lower joules or time per verified factor at matched confidence and accounting scope

The digital stage may begin with $n = 16, 24, 32$ and $T = 32, 64, 128, 256$, using a programmable smaller batch than the uniform theorem where appropriate. The theorem budget is a conservative reference; measured smaller-budget success must be reported separately. The thermal arithmetic stage should begin at four, eight and sixteen bits, not by extrapolating a pbit count into a 133-bit multiplier.

The supplied constraint generator constructs a logical QUBO for $N = 15, x = 7, c = 11$, and finds the certificate $77 = 5 \cdot 15 + 2$ when evaluated on consistent Boolean wires. Its deliberately unoptimized AND/NOT circuit has 1,698 logical variables. That concrete count is useful: even a tiny arithmetic task can require far more than one pbit per answer bit. Gate simplification and physical embedding are future work, not included in that count. No Gibbs trajectory or real chip has been simulated by this Boolean check.

25.5 Cost ledger with quantities that can actually be measured

Let E_{setup} include base preparation, field setup, table fill, parameter transfer and reset. Let E_{end}, E_{det}, E_{check} and E_{move} denote endpoint generation, detection, verification and movement. Then

$$E_{\mathrm{batch}} = E_{\mathrm{setup}} + E_{\mathrm{end}} + E_{\mathrm{det}} + E_{\mathrm{check}} + E_{\mathrm{move}}.$$

If the terms overlap physically, measure total board energy directly and use the decomposition only as diagnostic attribution. The same principle applies to wall time: summing stage latencies is valid for a serialized pipeline, while an overlapped pipeline needs startup, steady-state throughput and drain times separately.

For P arithmetic engines sustaining ν modular products per second each, a workload of W such products requires at least $W/(P\nu)$ seconds of service. For B_{moved} bytes and sustainable bandwidth $\mathcal{B}$, movement requires at least $B_{\mathrm{moved}}/\mathcal{B}$. A lower bound on elapsed time is the maximum of applicable resource service bounds; dependencies can make it larger. Neither a pbit update frequency nor the depth of a sorting network gives an independent full-batch clock rate.

Memory should be reported as residue registers, exponent/identifier registers, coefficient levels, queues and instruction/seed storage. A stored polynomial tree uses $O(T \log T)$ residue coefficients in the straightforward design. A recompute or checkpoint design can reduce memory but pays extra work; its schedule must be specified. Word width grows with $\log_2 N$ in every exact block.

25.6 Benchmark protocol and decision thresholds

Use randomly selected balanced semiprimes from declared bit ranges, retain every seed and every unsuccessful attempt, and reserve a held-out set of inputs before tuning window size, queue thresholds or thermal parameters. Include uneven semiprimes, smooth local orders and unfavorable two-primary patterns as separate diagnostic families. Keep prime and prime-power cases as interface tests rather than counting them as successes under the semiprime theorem.

The comparisons should include: the same reference with digital seeds; equality-only sampling on identical endpoint lists; the divisor-sensitive pipeline with the same arithmetic backend; Pollard–Brent; and appropriate ECM, sieve or other established implementations at larger sizes. The small Python benchmark supplied here is not a comparison with optimized libraries and cannot establish industrial competitiveness.

For h successes in J complete independent trials, report the raw fraction and a binomial interval, for example the Wilson interval

$$\frac{\widehat{p} + z^2/(2J) \;\pm\; z\sqrt{\widehat{p}(1-\widehat{p})/J + z^2/(4J^2)}}{1 + z^2/J}, \qquad \widehat{p} = h/J.$$

This interval presupposes a common trial probability and independence; heterogeneous sizes should be stratified, and correlated thermal runs need an appropriate design. For times and energies, retain distributions and timeouts, not merely the best run. Count board/controller power, communication, programming and reset under an agreed system boundary.

Proceed to larger hardware only if the measured improvement persists with uncertainty margins and matched correctness. If the thermal seed interface saves negligible total energy, retain the mathematical results and stop that hardware track. If a checked arithmetic module beats its digital control, integrate it gradually while monitoring the error ledger of Proposition 16.3. Negative results are informative engineering outcomes and must remain in the dossier.

25.7 Questions for an engineering discussion

The first technical discussion should establish which conditional kernels can be pinned and sampled, the measured parameter-programming cost, the available readout bandwidth, whether

intermediate states can be retained between stages, and the dependence structure of repeated samples. For a QUBO embedding, it must also establish coupling precision, connectivity constraints, variable-copy costs, effective energy scales and the observable available for checking thermalization.

These are specific interface questions, not requests for proprietary circuit designs. A simulator experiment should be described as such. Access to Torx or THRML alone does not imply access to Z1 hardware or validate a chip-level energy model for this workload.

26 Technical invention disclosure and preliminary prior-art map

26.1 Purpose and limits of this annex

This annex organizes a technical disclosure for discussion with patent counsel. It is not a filed application, a patentability opinion, an inventorship determination, or a freedom-to-operate conclusion. The prior-art check below is a targeted public-source screen dated 16 October 2026, not an exhaustive search of claims, patent families or unpublished applications.

The technical subject is a device workflow that preserves tagged exponents, produces modular endpoints, detects either exact relations or partial congruences, and returns only independently verified factors. An optional stochastic arithmetic module uses a quotient–remainder certificate and bounded retry/fallback without resampling the exponent. Potentially protectable subject matter, if any, would need to be assessed as a concrete technical implementation in its jurisdiction; the mathematical identities alone do not settle that assessment.

26.2 Technical problem and proposed mechanism

An analog order readout encodes the unknown cardinality in a voltage near $1/r$. Its absolute resolution requirement becomes of order r^{-2}. Tagged sampling avoids that scalar precision problem, but an equality-only detector may need order $\sqrt{r}$ samples and a large packet network. The revised design retains exact labels and adds divisor-sensitive arithmetic, which yields the finite sampling guarantee in Theorem 14.4.

The proposed apparatus comprises a host controller, exponent-seed register, finite-field evaluator, shared window memory, modular arithmetic units, numeric-key grouping, a polynomial product/remainder engine, a gcd engine and an independent output verifier. A stochastic proposal module may replace specified arithmetic operations while the supervisor retains the same sampled input and uses checked repair. The apparatus exposes counters for unsuccessful proposals, arithmetic operations, memory movement, saturation resolution and batch retries.

The corresponding technical effects to investigate are reduced endpoint work from shared tables, increased success from partial congruences, reduced receiver state when equality-only mode is used, and lower energy from a successful stochastic arithmetic substitution. Only the first three have mathematical or reference-software support here; the last remains an experimental hypothesis.

26.3 Prior-art map: known building blocks and unresolved differences

Source	Relevant disclosed subject	Consequence for this proposal
Pollard (1975), Brent (1980)	Collisions modulo an unknown factor, gcd-based factor extraction and batching	Do not claim gcd detection or partial collisions as new
Bostan–Gaudry–Schost (2007)	Fast polynomial arithmetic and the Pollard–Strassen factoring tradition	The $N^{1/4}$ scale and product-tree ingredients are established antecedents
van Oorschot–Wiener (1999)	Parallel collision search and distributed record strategies	Distributed collision search alone is not a distinct invention
Camsari et al. (2017)	Invertible stochastic logic based on pbits	A generic pbit arithmetic or reverse-computation claim is too broad
Extropic public stack (2026)	Programmable stochastic kernels and approximate thermodynamic compilation	Identify a concrete workload mapping and measured effect, not a claim to its substrate
US20230024744A1	Problem solving with probabilistic-bit circuits, including reverse computation	Compare actual independent claims and families with any proposed device claims
US20250094524A1	Electronic factorization device with energy calculation and candidate-bit updates	Energy-based factoring and a final decision circuit already appear in published applications

The two patent publications are cited for their disclosed text, not for an assertion that their claims are valid, enforceable, granted or applicable in a particular country [19, 20]. The 2025 publication's first claim concerns an energy-calculation circuit updating candidate-factor bits; additional claims include a factorization decision circuit. The present proposed relation/derivative pipeline uses different intermediate objects, but that observation is not a legal novelty or noninfringement conclusion. A claim-by-claim comparison and family review remain necessary.

26.4 Illustrative claim structure for counsel to evaluate

The following language records a candidate combination of technical features. It is intentionally not represented as a legally sufficient or novel claim set.

Apparatus combination. An apparatus having a controller that receives an integer modulus; a generator that associates reproducible exponent identifiers with sampled exponents; an arithmetic subsystem that produces modular endpoint records; a detector that groups equal numeric endpoint records and also forms a polynomial from distinct endpoint residues; an evaluation subsystem that computes derivative values at those residues; a greatest-common-divisor subsystem that tests those values against the modulus and splits a saturated product along stored subproducts; and an independent output subsystem that releases a candidate divisor only after verifying its proper-divisor conditions.

Optional dependent features. Candidate dependent features include: cubic finite-field seed expansion; base projection eliminating the two-primary order component; shared fixed-window tables; reconstruction of exponents from identifiers; use of sibling subtrees to exclude the derivative's self-factor; a lossless one-resident-record receiver for exact-equality mode; and telemetry that records failed trials and repair costs. Each feature is an implementation option, not an assertion of novelty.

Stochastic arithmetic embodiment. A conditional stochastic proposal subsystem supplies quotient and remainder candidates for a fixed modular multiplication request. A checking subsystem freezes the candidate before issuing independent residue challenges, or checks it exactly. A bounded retry controller preserves the original exponent request and uses an exact fallback when the retry budget is exhausted. The final factor verifier remains independent of the proposal subsystem.

Method counterpart. A method claim, if appropriate in the chosen jurisdiction, could track the same concrete data flow and certificate handling. It should state which operations execute on which hardware and what technical effect is demonstrated. Substituting a mathematical algorithm's name for a physical implementation does not supply those missing details.

26.5 Enablement material already supplied

This package contains the full proofs, concrete pseudocode, a standard- Python implementation that accepts only N and randomness, a finite- field sampler, exact window endpoint generation, the derivative and saturation detector, raw tests and timings, a logical QUBO generator, a small RC netlist and receiver/layer RTL specifications. The Python and Boolean-energy checks have been executed. The RTL has not been compiled or synthesized, the RC netlist has not been run in SPICE during this revision, and no physical device or chip embedding has been tested. Those limitations delimit the present technical record.

Before any filing, the inventor and counsel should decide which embodiment is sufficiently described, whether an actual improvement is supported, what alternatives are enabled and which proposed features must remain research questions. The document must not transform aspirational performance into a description of an experiment that never occurred.

26.6 Disclosure history, ownership and next review

WIPO advises avoiding nonconfidential disclosure before filing and using an appropriate confidentiality agreement if disclosure to a potential partner is necessary [21, 22]. The author should have counsel review the actual disclosure history, including any earlier public article, preprint, lecture, circulated manuscript or online code. A confidentiality label added now cannot undo an earlier public disclosure, and a journal manuscript is not itself a patent filing.

The source names J. D. Vélez as author; that metadata is preserved. It does not establish the inventor list or ownership of every proposed embodiment. Human contributions, dates of conception and development, collaborator agreements and any institutional obligations should be established from actual records. No dates, inventors, assignees or priority claims have been invented in this revision.

The review sequence is concrete: identify the technical combination worth pursuing; search its claims and nearby patent families; assess jurisdictions and prior disclosures; obtain an appropriate confidential review route; and decide whether to file before distributing the full technical package. This document has not been sent to Extropic or to any patent office as part of the present work.

27 Final result ledger and remaining research decisions

Result	Evidence	Practical implication
Exact dyadic return kernel	Proof and exact tests	Logarithmic layers; scalar precision and state size remain costly
Any nonzero oddified relation	CRT proof and enumeration	Exact-order recovery and gcd stabilization are unnecessary for this extraction
Quarter-power combined batch	Finite second-moment proof	At least 3/10 success for the stated semiprime model with a complete detector
Derivative-product detector	Proof and exact implementation	Partial congruences can be found without all-pairs gcds; fast polynomial backend remains to be engineered
Shared fixed-base windows	Exact work count and software	Fewer endpoint modular products with an explicit table-memory tradeoff
One-record receiver capture	Conditional probability proof	Stronger capture guarantee; transport and queue latency are separate
Constant-buffer butterfly	Prior routing theorem plus supplied bounds	Asymptotic network possibility, without finite chip timing constants
Conditional stochastic arithmetic	Constraint proof and QUBO artifact	A concrete experiment; no mixing-time or energy advantage established
Patent candidate combination	Technical disclosure and limited prior-art screen	Counsel and further search must assess novelty, ownership and filing strategy

The first decision is whether the fast polynomial detector is competitive on a digital prototype at useful input sizes. The second is whether a stochastic component can improve a dominant cost while preserving the seed and verification contracts. A positive answer to the first does not imply a positive answer to the second. Conversely, a negative hardware result would not invalidate the mathematical identities or sampling bounds.

Before publication as a mathematical paper, the integrated finite semiprime theorem and detector should receive an independent proof and priority review. Before an industrial claim, synthesis and matched measurements are required. Before a patent filing or nonconfidential partner disclosure, the actual technical contribution and disclosure history need a specialist review. These are different decisions with different evidence, and this monograph supplies a concrete basis for each of them.

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